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Asymptotics of Partition Functions of Coulomb Gases

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"En remontant chez moi pour y passer la soirée à travailler de mon mieux,
je me disais que le monde n'est pas construit pour l'équilibre.
Le monde est désordre.
L'équilibre n'est pas la règle, c'est l'exception.
Et je faisais le serment de travailler pour l'ordre et l'équilibre.
Que de serments! Tu vas sourire."
G.Duhamel, Maitres, 1937

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For all of those who are or were:
Your courage, patience, daring, and caring has brought me here.
To hoping that I become deserving of such act, for I feel loved and dear; of immense luck.
For all of those who will be:
To promise that I can for you as it was done for me.

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"João, é bom ver que você aprendeu e fez tantas coisas bonitas!
Continue assim, muito feliz! Brinque e aprenda muito."
Te amo muito, Papai.

RESUMO

JOÃO V. A. PIMENTA. **Assintóticas para funções partição de gases de Coulomb**. 2026. 130 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

A Teoria de Matrizes Aleatórias (RMT) tem sido uma teoria probabilística bem sucedida no estudo de sistemas complexos. Na década de 1950, Wigner propôs que as estatísticas locais dos níveis de ressonância de espalhamento de nêutrons em núcleos pesados poderiam ser modeladas pelo comportamento estatístico dos autovalores de grandes matrizes hermitianas com invariância ortogonal, unitária ou simplética. Desde então, físicos e matemáticos perceberam que poderiam usar as estatísticas de grandes matrizes para modelar as estatísticas de sistemas anteriormente inacessíveis, como sistemas muito grandes ou complexos. Embora na física a aplicação para o espalhamento de núcleos atômicos fosse direta, os estatísticos encontraram muitas aplicações para a teoria como no estudo de matrizes de correlação, e matemáticos fizeram grandes avanços ao conectar a RMT a sistemas integráveis. Focaremos nas distribuições de autovalores cujas funções de correlação podem ser expressas como o determinante de uma função matricial de polinomios ortogonais. A função partição é uma soma sobre os microestados deste sistema de partículas e, como tal, muitas das macroquantidades, como entropia e energia, são funções da mesma. Além disso, pode ser expressa em termos dos polinomios ortogonais e, por isso, frequentemente, as macroquantidades do sistema estão diretamente relacionadas com a assintótica dos mesmos polinomios ortogonais. Tal assintótica pode ser reduzida à técnica de Laplace se o polinomio ortogonal possuir a representação integral apropriada. Neste manuscrito, contextualizaremos e replicaremos uma importante e recente técnica para derivar a assintótica da chamada função partição de uma classe de conjuntos invariantes de matrizes aleatórias com autovalores complexos.

Palavras-chave: Teoria de Matrizes Aleatórias, Ensembles Invariante de Matrizes, Polinômios Ortogonais, Processos Pontuais, Assintótica.

ABSTRACT

JOÃO V. A. PIMENTA. **Asymptotics of Partition Functions of Coulomb Gases** . 2026. 130 p. Dissertação (Mestrado em Ciências – Matemática) – Instituto de Ciências Matemáticas e de Computação, Universidade de São Paulo, São Carlos – SP, 2026.

Random Matrix Theory (RMT) has been a successful probabilistic theory in the study of complex systems. In the 1950s, Wigner proposed that local the statistics of scattering resonance levels for neutrons off heavy nuclei could be modeled by the statistical behavior of eigenvalues of large hermitian matrices with orthogonal, unitary or symplectic invariance. Thenceforth physicist and mathematicians realized they could make use of the statistics of large matrices to model statistics of previously inaccessible systems, e.g. very large or complicated. Although in physics the application for atomic nuclei scattering was straightforward, statisticians found many uses of the theory in the study of correlation matrices and mathematicians made large advances linking RMT to integrable systems. We focus on eigenvalue distributions which correlation functions can be expressed as the determinant of a matricial function of orthogonal polynomials. The partition function is a summation over the micro-states of this particle system and as such, many of the macro-quantities, i.e. as entropy an energy, are a function of it. Moreover it can be expressed in terms of the orthogonal polynomials and therefore, oftentimes, macro-quantities of the system are directly related the asymptotic of orthogonal polynomials. Such asymptotics can be reduced to the Laplace technique if the orthogonal polynomial has a appropriate integral representation. In this manuscript we contextualize and replicate a important recent technique for deriving the asymptotics for the so-called partition function of a class of invariant ensembles of random matrices with complex eigenvalues.

Keywords: Random Matrix Theory, Invariant Matrix Ensembles, Orthogonal Polynomials, Point Processes, Asymptotics.

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INTRODUCTION

CHAPTER

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RANDOM MATRIX THEORY

There are many instances where one might want to describe the behavior of a set of random points in time or in space. This could come from an investigation of systems of diffusing particles or occurrences in time of some random phenomena. *Point processes* (PP) are models to describe such random objects: It describes the number of occurrence of events in a given time-frame or, alternatively, the number of points in regions of some space. For this work, we mainly focus on the latter application: density of points in regions of space; especially density of eigenvalues calculated from *random matrices*.

Random matrices can be defined in two ways: As a measurable function from a probability space into a set of matrices or, more concretely, as a matrix whose entries are random variables with some joint probability density function. In some cases there is an explicit joint density formula for the eigenvalues of such random matrix. We will be interested in these cases, which we call *Invariant Ensembles*. Moreover, we will be particularly interested in a subset of ensembles in which this density is expressed as the determinant of a *Kernel* - a matricial object that contains the statistical characterization of the system. This is called the *determinantal structure* and will lead us to the concept of *Determinantal Point Processes* (DPP).

What follows is our best attempt in introducing the concepts in *Random Matrix Theory* (RMT) that are pertinent to the scope of this work and of general interest, albeit inside the limitations of time, effort and experience. We wrote the proofs that were within the theoretical reach of the chapter and that further elucidated the concepts discussed. More technical or tangential results were only stated, with at times some indication on how one would go about the proof. Part of the definitions and technical structure about RMT are based on the well written Thesis by Rahman (2023). Furthermore, definitions and ideas for point processes and determinantal point processes follow very closely the well structured works of Serfozo (1990), Johansson (2006) and Girotti (2014).

2.1 Random Matrix

The idea of *random matrices* - matrices with random entries - is a core concept in this work and the main theme of this section. As any random object, these matrices must be drawn from a set with some given probability density function (p.d.f.). If a matrix *models a system*, we think of each realization of this random object as one state the system can assume. Borrowing the concept from physics, an *ensemble* is thought of as a virtual collection of all possible states that a system can be in, with more probable states appearing more often in the collection. To consider an ensemble of random matrices is to associate each state to a probability. More concretely, in the context of Random Matrix Theory (RMT) we have the following definition.

Definition 2.1. *A* ***random matrix ensemble*** $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ *is a set of matrices* $\mathcal{S}$ *of fixed size equipped with p.d.f.* $\mathcal{P} : \mathcal{S} \to [0, \infty)$. *If we say that a matrix* $\hat{X}$ *is drawn from the random ensemble* $\mathcal{E}$, *we are saying that it is a random variable assuming values in* $\mathcal{S}$ *with p.d.f.* $\mathcal{P}$.

The construction of many ensembles sets $\mathcal{S}$ in RMT is done by applying some constraint to the set of real, complex or quaternionic matrices. The constraints used are typically those requiring elements of $\mathcal{S}$ to exhibit features such as orthogonality, unitarity, (skew-)symmetry, Hermiticity, invariance under complex conjugations or a reasonable combination of them.

Example 2.1. The $m \times n$ real, complex, and quaternionic *Ginibre Ensembles* are the set of n by m matrices with real, complex and quaternionic entries with probability density function

$$\mathcal{P}^{(Gin)}(\hat{G}) = \frac{1}{\mathcal{Z}^{(Gin)}_{n,m,\beta}} \exp\left(-\sqrt{nm}\,\mathrm{Tr}(\hat{G}^*\hat{G})\right) = \frac{1}{\mathcal{Z}^{(Gin)}_{n,m,\beta}} \prod_{i=1}^{m}\prod_{j=1}^{n} \exp\left(-\sqrt{nm}|g_{i,j}|^2\right),$$

defined with respect to the Lebesgue measure

$$\mathrm{d}\hat{G} = \prod_{i=1}^{m}\prod_{j=1}^{n}\prod_{s=1}^{\beta} \mathrm{d}g^{(s)}_{i,j}.$$

The coefficient β is called the Dyson index and counts the generic number of real components of each matrix entry $g_{i,j}$, so that $\beta = 1, 2, 4$ refers to real, complex and quaternionic entries. These are the Ginibre orthogonal, unitary and symplectic ensembles.

Normally, when constructing an ensemble of random matrices, one is trying to model some type of system or behavior - be that for a physical or mathematical reason. Historically, many ensembles were constructed with hermitian matrices to model large complex systems such as heavy nuclei, interacting particle gases on the line and other correlated systems. In such cases the real eigenvalues represented quantities as eigenenergies and particle positions. Although you can construct ensembles in many ways, there are two approaches that we will take interest on in this work. They are referred here as the *universality* and the *invariance* methods. The first approach is mainly considered for its contextual and historical importance, however, most results presented in the following chapters are constructed with the second approach in mind.

The construction by universality consists of assuming very little of the system and focusing on universal results, as suggested by its name. Its origins refer back to the works of Wigner in heavy nuclei as described in Chapter 1. The intent was to capture statistical information on the behavior of complex systems without the need to encode the finer details of the interactions, only demanding symmetries or general features to be displayed in the matrix. Surprisingly, doing so still allows you to characterize aspects that turn out to be common across a large range of random objects. More precisely, consider the definition below.

Definition 2.2. *Let every $\hat{Y}$ be a matrix $\hat{Y} = \{Y_N\}_{i,j}$ such that $Y_{i,j} = Y^*_{j,i}$ satisfying the following properties: $\{Y_{i,j}\}_{1\leq i\leq j\leq n}$ are independent; $\{Y_{i,j}\}_{i\neq j}$ are identically distributed as also are the $\{Y_{i,i}\}$; $\mathbb{E}(Y_{i,j}) = 0$ and $\mathbb{E}(Y^k_{i,j}) < \infty$, i.e. $r_k = \max\left(\mathbb{E}(Y^k_{i,j}), \mathbb{E}(Y^k_{i,j})\right) < \infty$. Set $r_2 < \infty$ with $\mathbb{E}(Y^2_{1,2}) > 0$. Then, we say $\hat{Y}$ is a **Wigner matrix**. Furthermore, if $\hat{X}_n = n^{-\frac{1}{2}}\hat{Y}_n$, $\hat{X}$ is what we call a **normalized Wigner matrix**.*

Example 2.2. Let $\hat{G}$ be drawn from the $n \times n$ real, complex or quaternionic Ginibre Ensemble introduced in Example 2.1. Then, the random matrix $\hat{H} = \frac{1}{2}(\hat{G}^* + \hat{G})$ represents an element of the corresponding real, complex or quaternionic *Gaussian ensemble* - also known as *Hermite*. The probability density function for such ensembles are then given by

$$\mathcal{P}^{(G)}(\hat{H}) = \frac{1}{\mathcal{Z}^G_{n,\beta}} \exp\left\{-n \operatorname{Tr}\left(\hat{H}^2\right)\right\}.$$

Although thinking of the Gaussian ensembles as a symmetrization of the Ginibre ensemble can be useful, one can also construct these ensembles thinking of it as a Wigner matrix. Let $\hat{M}$ have its upper-triangular entries $\{m_{i,j}\}_{i<j}$ independently and identically distributed as standard (real, complex or quaternionic) Gaussian random variables. Let also its diagonal entries $\{M_{i,j}\}_{i=j}$ be independently identically distributed as standard real Gaussian random variables with half the variance. Finally, let the matrix be self-adjoint, i.e. $\{M_{i,j}\} = \{M^*_{j,i}\}$ for $i \neq j$ and the eigenvalues scaled by $1/\sqrt{n}$. Such matrix $\hat{M}$ has distribution

$$\mathcal{P}^{(G)}(\hat{M}) = \prod_{i=1}^{n} \frac{e^{-nm^2_{i,i}}}{\sqrt{\pi}} \prod_{i<j}^{n} \frac{e^{-2nm^2_{i,j}}}{\sqrt{\pi/2}} = 2^{n(n+1)/4}\pi^{n(n+1)/4} \exp\left\{-n \operatorname{Tr}\left(\hat{M}^2\right)\right\}.$$

Representatives from the Gaussian ensembles are an instance of Wigner matrices as entries are Gaussian random variables satisfying the requirements in Definition 2.2.

The advantages of this approach for constructing ensembles becomes clearer when considering that many convergence results hold for independently and identically distributed centered random variables. We will present, without proof, a theorem that sits at the center of this ideology: It roughly states that the typical eigenvalue configuration, a set of random points, converges to a distribution called *Wigner Semi-Circle* as we increase the size of the matrix. To state it, we first introduce an *empirical spectral measure* of the eigenvalues.

Let $\hat{M}$ be a Wigner matrix, by the spectral theorem $\hat{M}$ has n real eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then, define $\delta_{\hat{M}}$, the *empirical spectral measure*, as

$$\delta_{\hat{M}} := \frac{1}{n}\sum_{k=1}^{n} \delta_{\lambda_k}, \tag{2.1}$$

where $\delta_{\lambda_k}(A) \equiv \mathbb{1}_A(\lambda_k)$ is the Dirac delta function. Let $\mathcal{E}$ denote the class of Borel sets and $\mathcal{B}$ the class of bounded Borel sets, both on the real line. One can notice that this function $n\delta_{\hat{M}}$ defines a *random counting measure* in the usual sense, i.e. a mapping $f : \mathcal{E} \to \{0, 1, \cdots, \infty\}$ such that $f(\varnothing) = 0$, and $f(B) < \infty$ for any $B \in \mathcal{B}$ and that, for disjoint sets $B_1, B_2, \cdots \in \mathcal{B}$ it is such that $f\left(\bigcup_k B_k\right) = \sum_k f(B_k)$. For Wigner matrices, it can be shown that this discrete measure converges to a continuous measure when we take the limit on the size n of the matrices - when considering a large number of eigenvalues. The is the result proved by Wigner (1958).

Theorem 2.1. *(Wigner's Semicircle Law) The empirical spectral measure $\delta_{\hat{M}}$ of Wigner matrices converges weakly to the Semi-Circle distribution $\mu^{(SC)}$ defined by*

$$\mu^{(SC)}(\mathrm{d}x) := \begin{cases} \frac{1}{\pi R^2}\sqrt{R^2 - x^2}\,\mathrm{d}x, & |x| \leq R, \\ 0, & |x| > R. \end{cases}$$

where R depends only on the variance of entries of the matrix.

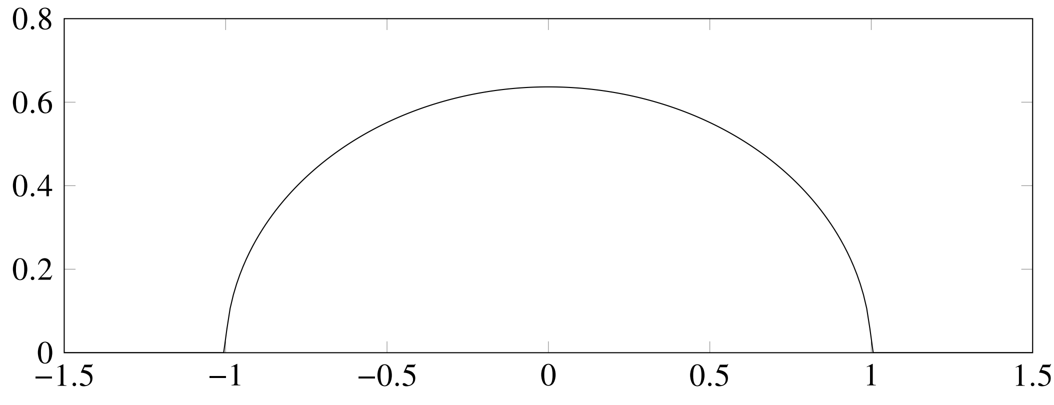

Figure 1 – Plot of the Wigner Semi-Circle distribution normalized such that $\mathrm{supp}\,\mu^{(SC)} = [-1, 1]$.

This is an instance of what is called *universality* in RMT, hence the approach's name. This will be better formalized when we discuss *determinantal point processes* in Section 2.2. For now, universality refers to the phenomena where different ensembles have the same large limit statistic, e.g. the distribution of eigenvalues. We know that Wigner matrices share the same limit distribution for eigenvalues; other classes of universality may share other statistics: Eigenvalue spacing and maximum eigenvalue distribution are some of them.

Example 2.3. (Numerics on the Semicircle Law) Take the Gaussian Orthogonal (GOE), Unitary (GUE) and Symplectic (GSE) Ensembles. By Theorem 2.1, we expect to see, as $n \longrightarrow \infty$, the empirical eigenvalue measure $\delta_{\hat{M}}$ approximating the Semi-Circle distribution μ^{SC}. We indicate such convergence in the simulations below.

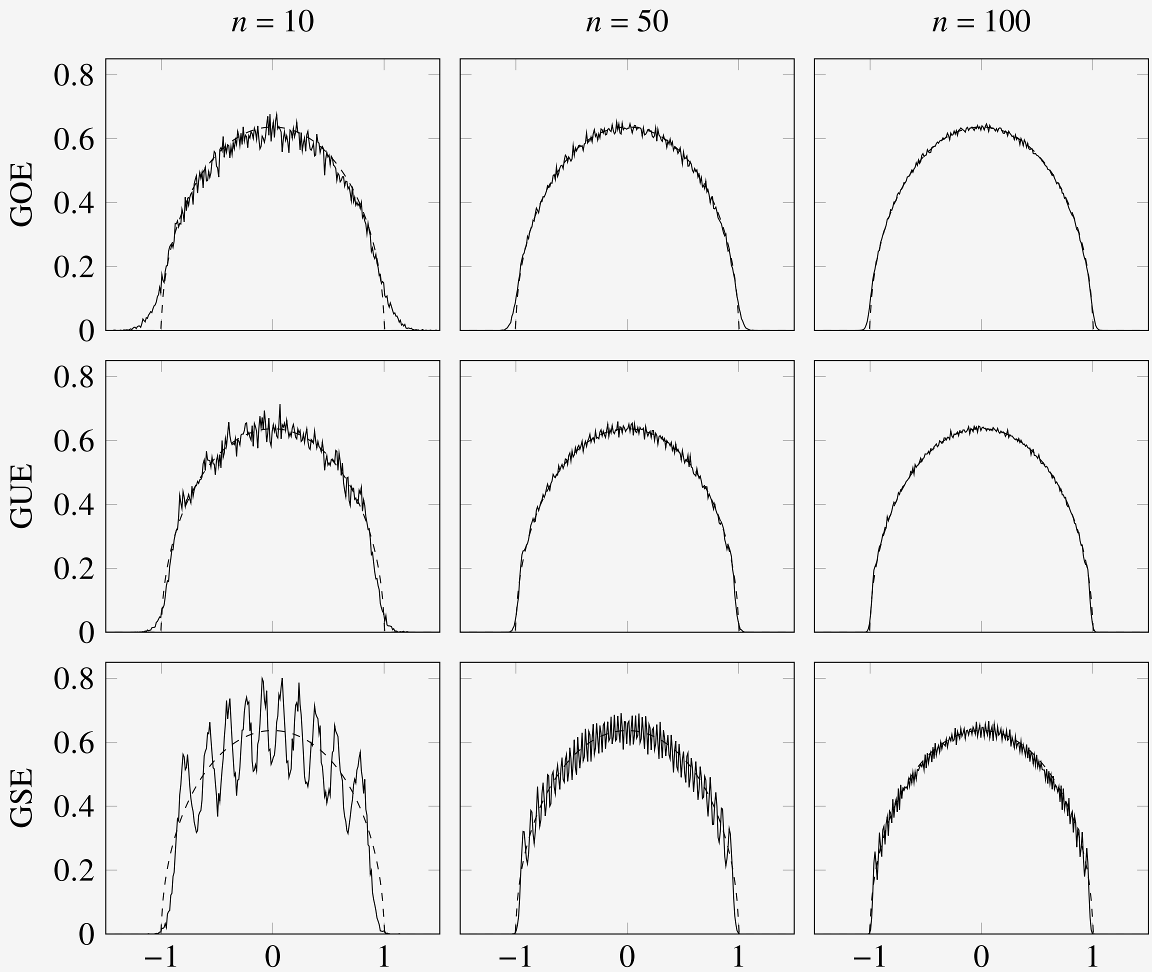

Figure 2 – The dashed line is the Semi-circle distribution, whereas in solid we display the average distribution of eigenvalues from $\hat{M}_n$. The columns correspond to $n = 10, 50, 100$ from left to right. The lines correspond to the GOE, GUE and GSE from top to bottom. We took 10000 matrix samples for each average distribution.

The second approach, of invariance, ask us to construct ensembles with invariant measure: We equip the set $\mathcal{S}$ of matrices with an $\mathcal{A}$-invariant measure $\mathcal{P}$, i.e. $\mathcal{P}(\hat{M}) = \mathcal{P}(\mathcal{A}(g, \hat{X}))$ for a group action $\mathcal{A}$. Both the *Ginibre Ensembles* introduced in Example 2.1 and the *Gaussian Ensembles* introduced in Example 2.2 are constructed considering orthogonal, unitary and symplectic conjugation invariance. Interestingly, when dealing with matrices with real eigenvalues, there is only one ensemble that is rationally invariant and has independent entries: The Gaussian ensembles. No other ensemble of Wigner matrices is also invariant (and vice-versa). Although the theory of real eigenvalues has its historical and theoretical place, our results concern complex matrices with complex eigenvalues. In the next section we discuss the details of this approach.

2.2 Invariant Ensembles

Invariant ensembles of random matrices are probability spaces invariant under some group action. However, *invariant matrix ensemble* usually refers to invariance under the conjugation mapping $\hat{X} \mapsto g\hat{X}g^*$ for g drawn from the appropriate compact group $\mathcal{G}$. *Conjugation invariance* is a very natural physical assumption and it carries its significance to mathematics; it simply means that all coordinate systems are equivalent. Think of these ensembles as having probability distribution depending on their eigenvalue configurations while their eigenvectors are independent and Haar (uniformly) distributed. According to Dyson (1962b), when a matrix ensemble underlying set of matrix is $\mathcal{S} \subseteq \mathbb{M}_{n,n}(\mathbb{R}), \mathbb{M}_{n,n}(\mathbb{C}), \mathbb{M}_{n,n}(\mathbb{H})$ - $n \times n$ matrices with real, complex and quaternionic entries respectively, the group $\mathcal{G}$ must be one of $O(n), U(n), Sp(2n)$ - the orthogonal, unitary and symplectic groups. Then, we say the ensemble is *Orthogonal*, *Unitary* or *Symplectic*.

Example 2.4. Consider the three ensembles from the aforementioned symmetries.
The *Orthogonal Ensembles* consist of $n \times n$ real symmetric matrices $\hat{M} = \hat{M}^* = \hat{M}^T$ together with a probability distribution that is invariant under orthogonal conjugation $\hat{M} \mapsto \hat{O}\hat{M}\hat{O}^T$, $\hat{O}$ orthogonal, i.e. $\hat{O}\hat{O}^T = \hat{O}^T\hat{O} = \hat{I}$.
The *Unitary Ensembles* consists of $n \times n$ Hermitian matrices $\hat{M} = \hat{M}^*$ together with a probability distribution that is invariant under unitary conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}^*$, $\hat{U}$ unitary, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$.
The *Symplectic Ensembles* consists of $2m \times 2m$ Hermitian self-dual matrices $\hat{M} = \hat{M}^* = \hat{J}\hat{M}^T\hat{J}^T$ where $\hat{J} = \operatorname{diag}(\sigma, \cdots, \sigma)$ with $\sigma = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ together with a probability distribution that is invariant under unitary-symplectic conjugation $\hat{M} \mapsto \hat{U}\hat{M}\hat{U}$, with $\hat{U}$ unitary and symplectic, i.e. $\hat{U}\hat{U}^* = \hat{U}^*\hat{U} = \hat{I}$ and $\hat{U}\hat{J}\hat{U}^T = \hat{J}$.

For this manuscript, we are mainly concerned with statements of ensembles constructed with *normal matrices* of complex entries, and therefore, with *unitary invariance*. Then, we make formal the following definition of *invariance*.

Definition 2.3. *(Invariance) Given a group $\mathcal{G}$ and a group action $\mathcal{A} : \mathcal{G} \times \mathcal{S} \to \mathcal{S}$, a random matrix ensemble $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ is said to be* ***$\mathcal{A}$-invariant*** *if the probability measure* $\mathrm{d}\mathcal{P}(\hat{X}) := \mathcal{P}(\hat{X})\mathrm{d}\hat{X}$ *satisfies*

$$\mathrm{d}\mathcal{P}(\mathcal{A}(g, \hat{X})) = \mathrm{d}\mathcal{P}(\hat{X})$$

for every $g \in \mathcal{G}$. In our particular case we say $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ to be ***unitarily invariant*** *if the probability measure* $\mathrm{d}\mathcal{P}(\hat{X})$ *satisfies*

$$\mathrm{d}\mathcal{P}(\hat{U}\hat{X}\hat{U}^*) = \mathrm{d}\mathcal{P}(\hat{X})$$

for every $\hat{U} \in U$, the unitary group of matrices.

To construct ensembles with unitary invariance we need to take $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ such that $d\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{X})d\hat{X}$ is invariant under unitary conjugation. This means we should establish that both the p.d.f. $\mathcal{P}(\hat{X})$ and the measure $d\hat{X}$ are invariant under the aforementioned action. Invariance of the function $\mathcal{P}(\hat{X})$ is straightforward if we take

$$\mathcal{P}(\hat{X}) = \phi\left(\mathrm{Tr}\, Q(\hat{X})\right) \tag{2.2}$$

for some function ϕ in terms of the trace of a polynomial Q of the matrix. Note that, because of the trace properties, if $\hat{A} = \hat{U}\hat{B}\hat{U}^*$, then $\mathcal{P}(\hat{A}) = \mathcal{P}(\hat{B})$ since

$$\phi\left(\mathrm{Tr}\, Q(\hat{A})\right) = \phi\left(\mathrm{Tr}\, Q(\hat{U}\hat{B}\hat{U}^*)\right) = \phi\left(\mathrm{Tr}\, Q(\hat{U}^*\hat{U}\hat{B})\right) = \phi\left(\mathrm{Tr}\, Q(\hat{B})\right).$$

Therefore, if a p.d.f. is such as in (2.2), then it is invariant by our definition. Moreover, since $\hat{X}$ will be a normal $n \times n$ matrix, we know by the spectral theorem that it is similar to a diagonal matrix by unitary conjugation. Therefore, for some $\Lambda = \mathrm{diag}\{\lambda_1, \lambda_2, \dots, \lambda_n\}$, ϕ takes the form

$$\phi\left(\mathrm{Tr}\, Q(\hat{X})\right) = \phi\left(\mathrm{Tr}\, Q(\hat{U}\hat{\Lambda}\hat{U}^*)\right) = \phi\left(\mathrm{Tr}\, Q(\hat{\Lambda})\right) = \phi\left(\sum_{i}^{n} Q(\lambda_i)\right).$$

The invariance of the measure $d\hat{X}$ is more involved. (HOUGH *et al.*, 2009) The main idea is that given $\hat{X} = \{x_{i,j}\}_{i,j}^{n}$ a matrix with complex entries and complex eigenvalues $\{z_1, z_2, \dots, z_n\}$, one can choose a set of auxiliary variables $\mathbf{u}$ ($2n(n-1)$ of them) so that the transformation $T(z, \mathbf{u}) = (x_{i,j})$ is essentially one-to-one and has Jacobian determinant

$$f(\mathbf{u}) \prod_{i<j} |z_i - z_j|^2$$

for some function f. This was discussed for non-hermitian complex matrices by Ginibre (1965), where he extended the classical argument used for Hermitian matrices to *Ginibre matrices* $\hat{G}$ using the conjugation transformation $\hat{G} = \hat{P}\hat{\Lambda}\hat{P}^{-1}$. Note that this is reasonable since the subspace of matrices with simple spectra (unique eigenvalues) is dense on the space of complex matrices.

> **Proposition 2.1.** *The set of complex matrices with simple spectra is open and dense in $\mathcal{M}_n$ (n-dimensional complex matrices) and has full Lebesgue measure on the euclidean space of matrices, that is, generally, complex matrices have simple eigenvalues.*

Proof. Let the matrix $\hat{M}$ have the decomposition

$$\hat{M} = \hat{P}\hat{\Lambda}\hat{P}^{-1},$$

with $\hat{\Lambda}$ diagonal with eigenvalues $\{\lambda_1, \lambda_2, \dots, \lambda_n\}$. To prove denseness of matrices with simple eigenvalues we note that there exists real variables $\epsilon_1, \epsilon_2, \dots, \epsilon_n$, with $\sum_{i=1}^{n} |\epsilon_i| \to 0$, such that

$$\hat{M}_\epsilon = \hat{P} \begin{bmatrix} \lambda_1 + \epsilon_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n + \epsilon_n \end{bmatrix} \hat{P}^{-1} = \hat{M}_\epsilon^*$$

has a simple spectrum. Because $\hat{M}_\epsilon \to \hat{M}$, matrices with simple spectra are dense in $\mathcal{M}_n$, the space of n-dimensional matrices. Furthermore, to show that the set of matrices with simple eigenvalues has full Lebesgue measure, we shall prove that its complement is the union of lower dimensional submanifolds. First, consider the map

$$\begin{aligned} f_i \colon \mathcal{M}_n &\to \mathbb{C} \\ \{m_{i,j}\} &\mapsto a_i\left(\{m_{i,j}\}\right) \end{aligned}$$

where a_i is the i-th coefficient for the characteristic polynomial of $\hat{M}$, function of the matrix' entries. Introduce also the discriminant of the characteristic polynomial $A(x) = \sum_{i=0}^{n} a_i x^i$, which is defined in terms of the *resultant* of A and its derivative A' as

$$D_x(A) = \frac{(-1)^{n(n-1)/2}}{a_n} \operatorname{Res}_x(A, A').$$

The resultant, by definition, is the determinant of a Sylvester matrix of A and A'. Therefore it is a multivariate polynomial in terms of the coefficients of A. Moreover, it is zero if and only if the polynomials have a common root. Therefore, $D_x(A)$ is zero if the matrix $\hat{M}$ itself has repeated eigenvalues. Therefore, composing both functions in the appropriate manner we can write

$$D_x \circ f(\hat{M}) = P\left(a_0\left(\{m_{i,j}\}\right), \ldots, a_{n-1}\left(\{m_{i,j}\}\right)\right),$$

where P is some multivariate polynomial function on the entries $m_{i,j}$ of the matrix. However, it is useful to think of it as a polynomial of one of the variables, such as $m_{1,1}$, and factor out $l_{i,j}$ factors of the type $(m_{i,j} - g_{i,j}^{(1)}(\{m_{l,k}\}_{k,l\neq i,j}))$. Note that $l_{i,j}$ is finite and may be zero for some of the entries. Using this, we write

$$D_x \circ f(\hat{M}) = \prod_{i,j} \prod_{k=1}^{l_{i,j}} (m_{i,j} - g_{i,j}^{(k)}(\{m_{l_1,l_2}\}_{l_1,l_2\neq i,j})) =: \prod_{i,j} \prod_{k=1}^{l_{i,j}} p_{i,j}^{(k)}.$$

Note also that, denoting the set of zeros for a function g as $Z(g)$, we must have by definition of the functions p that the set of zeros of $D_x \circ f$ must be the union of the zeros of each $p_{i,j}$ as

$$Z(D_x \circ f) = \bigcup_{i,j,k} Z\left(p_{i,j}^{(k)}\right).$$

Note that, more importantly, the set of matrices with simple eigenvalues is precisely $\mathcal{M}_n \setminus Z(D_x \circ f)$, by construction. Furthermore, each of the polynomials $p_{i,j}^{(k)}$ defines a submersion between the manifold of matrices and the complex plane since

$$\frac{\partial p^{(k)} i,j}{\partial m_{i,j}} \neq 0.$$

Therefore, by the local form of the submersion theorem $Z\left(p_{i,j}^{(k)}\right)$ is a manifold with dimension $n^2 - 2$ and must have null Lebesgue measure. The same holds for the union of all these functions and consequently for $Z(D_x \circ f)$. □

Especially interesting for us is the analogous statement regarding normal matrices, which is an adaptation of the argument above. Moreover, if we arbitrarily order eigenvalues and impose that the first eigenvector obeys some condition we make the conjugation transformation one-to-one, and therefore

$$\int_{\mathcal{N}_n(\mathbb{C})} \mathcal{P}(\hat{X} \mapsto \{\hat{\Lambda}, \hat{U}\})\mathrm{d}\hat{X} = \int_{\mathbb{C}^n} \int_{\frac{U(n)}{U(1)^n}} \mathcal{P}(\hat{X})\mathrm{d}(\Lambda, \hat{U}) = \int_{\mathbb{C}^n} \int_{\frac{U(n)}{U(1)^n}} \mathcal{P}(\hat{X}) \prod_{i<j} |z_i - z_j|^2 \mathrm{d}\hat{\Lambda}\mathrm{d}\hat{U}.$$

Then we can integrate over the eigenvector space to get the induced measure on eigenvalues. The last equality comes from a result which we call *Weyl's integration formula*.

Theorem 2.2. *(Weyl's integration formula - Unitary case) The measure* $\mathrm{d}\hat{X}$ *decomposes almost everywhere in* $\mathcal{N}(\mathbb{C})$ *as*

$$\mathrm{d}\hat{X} = \mathrm{d}\hat{U} \prod_{1 \le i < j \le n} |z_i - z_j|^2 \prod_{i=1}^{n} \mathrm{d}^2 z_i,$$

where $\{z_i\}_{i=1}n$ *is the spectrum of* $\hat{X}$*,* $\hat{M} = \hat{U}\hat{\Lambda}\hat{U}^*$ *for* $\hat{\Lambda} = \mathrm{diag}\{z_i\}_{i=1}^n$*,* $\hat{U} \in U(n)/U(1)^n$ *and the measure*

$$\mathrm{d}\hat{U} = 2^{\frac{n(n-1)}{4}} \prod_{i<j} \mathrm{d}^2 u_{i,j}, \quad \mathrm{d}^2 u_{i,j} = (\hat{U}^*)_{i,k} \mathrm{d}^2 \hat{U}_{k,j}$$

on $U(n)/U(1)^n$ *is induced by the measure on* $U(n)$ *and is a Haar measure.*

With Theorem 2.2, we can directly check that $\mathrm{d}\hat{X}$ is invariant by its decomposition. Naturally, $\prod_{1 \le i<j \le n} |z_i - z_j|^2 \prod_{i=1}^n \mathrm{d}^2 z_i$ is invariant under unitary conjugation as it is a function of the spectrum of $\hat{X}$. Moreover, $\mathrm{d}U$ is a Haar measure, i.e. has exactly the properties of invariance by left and right unitary transformations. While eigenvectors may be of interest, eigenvalue statistics have been noticeably more explored; justified by the uniform distribution on its eigenvectors. This gives a reason for the larger interest in eigenvalue statistics. Before advancing we discuss some objects that are fundamental when studying RMT; namely, we introduce the joint probability density function, the correlation function and the univariate density.

2.2.1 Orthogonal Polynomial Ensembles

The subclass of invariant matrix ensembles this section refers to are those constructed with normal matrices as described in Elbau and Felder (2005) . The p.d.f.s $\mathcal{P}(\hat{X})$ on this ensembles are such that the joint probability density function on eigenvalues can be written as

$$p_n(\lambda_1, \lambda_2, \ldots, \lambda_n) = \frac{1}{\mathcal{Z}_n^w} |\Delta_n(\lambda)|^2 \prod_{i=1}^{n} w(\lambda_i)$$

where $\{\lambda_i\}$ are the indistinguishable eigenvalues of $\hat{X}$, $w(\lambda)$ is some continuous weight function that is real valued and decays fast enough as $|\lambda| \to \infty$, $|\Delta_n(\lambda)|$ is the Vandermonde determinant and $\mathcal{Z}_n^w$ is a normalization function.The second condition on the weight function guarantees that normalization constant for the j.p.d.f. considered is well-defined. We are interested in the probability distribution of the eigenvalues $\lambda_1, \cdots, \lambda_n$ of $\hat{X}$. By the spectral theorem, for any normal matrix we have $\hat{U}$ an unitary matrix such that $\hat{U}^*\hat{X}\hat{U} = \hat{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

To arrive at the expression for the eigenvalue j.p.d.f. we integrate the eigenvector dependent terms so that we are left with only the part depending on the eigenvalues. This integral on $\mathcal{P}$ can be evaluated using Theorem 2.2 and the fact that $\mathcal{P}(\hat{X}) = \mathcal{P}(\hat{\Lambda})$ for $\hat{X} = \hat{U}\hat{\Lambda}\hat{U}^*$. Then, using Theorem 2.2 we show that the integral on the quotient space with respect to d$\mathcal{P}$ is given by

$$\begin{aligned} \mathrm{d}p_n(\{\lambda_i\}_{i=1}^n) = \int_{U(n)/U(1)^n} \mathcal{P}(\hat{X})\mathrm{d}\hat{X} &= \frac{1}{\mathcal{Z}_n^{\mathcal{P}(\Lambda)}} \mathcal{P}(\hat{\Lambda}) \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=1}^n \mathrm{d}^2\lambda_i \\ &= \frac{1}{\mathcal{Z}_n^w} \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=1}^n w(\lambda_i) \prod_{i=1}^n \mathrm{d}^2\lambda_i. \end{aligned}$$

where we set $\mathcal{P}(\Lambda) = \prod_{i=1}^n w(\lambda_i)$. Since we took $w(\{\lambda_i\}_{i=1}^n) = \phi\left(\sum_i^n Q(\lambda_i)\right)$,

$$p_n(\{\lambda_i\}_{i=1}^n) = \frac{1}{\mathcal{Z}_n^Q} \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=1}^n \phi\left(\sum_i^n Q(\lambda_i)\right).$$

and $\mathcal{Z}_n^w = \int_{\mathbb{C}^n} p_n(\{\lambda_i\}_{i=1}^n)\mathrm{d}^2\lambda_i$.

Example 2.5. (Gaussian Measure) Let us make it explicit the weight function defining the Gaussian Unitary Ensembles introduced in Example 2.2. If U_n is the unitary group, we know by the previous calculation that

$$\begin{aligned} \int_{U_n} \mathcal{P}(\hat{M})\mathrm{d}\hat{M} &= \int_{\mathbb{V}_n} \mathcal{P}(\hat{\Lambda}, \hat{U}) \prod_{i<j} |\lambda_j - \lambda_i|^2 \mathrm{d}\hat{U}\mathrm{d}\hat{\Lambda}, \\ &= \frac{1}{\operatorname{Vol}(U_n)^{-1}} \mathcal{P}(\hat{\Lambda}) \prod_{i<j} |\lambda_j - \lambda_i|^2 \prod_{i=1}^n \mathrm{d}^2\lambda_i \end{aligned}$$

where $\hat{M} = U\Lambda U^*$. We also have seen that

$$\mathcal{P}(\hat{M}) = \exp\left\{-n \operatorname{Tr}\left(\hat{H}^2\right)\right\} = \exp\left\{-n \operatorname{Tr}\left(\hat{\Lambda}^2\right)\right\} = \prod_{i=1}^n \mathrm{e}^{-n\lambda_i^2} = \mathcal{P}(\hat{\Lambda})$$

which says that $w(x) = \mathrm{e}^{-nx^2}$ and lead us to write the j.p.d.f. as in Example 2.7

Definition 2.4. *Fix $n \in \mathbb{N}$ and let $w(z) = \mathrm{e}^{-nQ(z)}$ where Q is a polynomial function such that*

$$\liminf_{|z|\to\infty} \frac{Q(z)}{2\ln(|z|)} > 1.$$

Then, the j.p.d.f.

$$p_n(\lambda_1, \lambda_2, \ldots, \lambda_n) = \frac{1}{\mathcal{Z}_n^Q} |\Delta_n(\lambda)|^2 \prod_{i=1}^{n} w(\lambda_i) \tag{2.3}$$

where $|\Delta_n(\lambda)|$ is the Vandermonde determinant, describes the n-sized ***orthogonal polynomial ensemble*** *(OPE).*

If we stare at (2.3) strongly and for long, the terms become less mysterious. One may notice that this expression is simply the normalized product of some weight function $w(\lambda)$ and the Jacobian of the transformation $\hat{M} \mapsto \hat{U}\hat{\Lambda}\hat{U}^*$ given by the squared Vandermonde determinant. More than that, there is an observation to be made here in terms of the behavior of the eigenvalues. Notice that the Jacobian takes arbitrarily small values if there are equally arbitrarily close eigenvalues in the set. At first instance this means that our set is most likely simple, i.e. without repeated eigenvalues. Moreover it also means that the eigenvalues show some repulsion with each-other. Why then one might not expect the eigenvalues to scatter in the line? This is the role of the weight function. More than characterizing the ensemble, in the formula it has the role of ensuring integrability, i.e. holding the eigenvalues in a limited set. It must, therefore, decay sufficiently fast. If this feels like some physical repelling gas system, fear not, this interpretation is something we make use of and will be expanded in Section 3.1.

Example 2.6. A random matrix ensemble is said to be a unitary *Classical Matrix Ensemble* if its eigenvalue j.p.d.f. is of the form

$$p_n(\lambda_1, \lambda_2, \ldots, \lambda_n) = \frac{1}{\mathcal{Z}_n^w} \prod_{i=1}^{n} w(\lambda_i) |\Delta_n(\lambda)|^2,$$

where $\Delta_n(\lambda)$ is the Vandermonde determinant and w, the weight function has one of the following forms

$$w(\lambda) = \begin{cases} \mathrm{e}^{-n\lambda^2} & \text{Gaussian,} \\ \mathrm{e}^{-n(\lambda - a\ln(\lambda))}\, \mathbb{1}_{\{\lambda>0\}} & \text{Laguerre,} \\ \mathrm{e}^{-na(\ln(1+b\lambda))^2}\, \mathbb{1}_{\{0<\lambda<1\}} & \text{Jacobi.} \end{cases}$$

where $\mathbb{1}$ is the indicator function and $a, b > -1$. All of the classical ensembles defined as is are invariant under unitary conjugation. The names Hermite, Laguerre and Jacobi also refer to a class of polynomials, the reason for this and the ensemble's name will become clearer in Subsection 2.3.1.

2.3 Eigenvalue Statistics

We are mainly interested in studying eigenvalue configurations. This is especially true because we are dealing with ensembles for which the p.d.f. is uniform in the eigenvectors. Interestingly, setting a joint probability density function (j.p.d.f.) on a configuration of points is the building block of a commonly know and studied structure called *point process*. Suppose there are n particles at random locations in the complex plane $\mathbb{C}$ denoted by the configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$. We define a quantity $\xi(A)$, similar to (2.1), that measures the amount of particles ($\#\{\chi_n \cap A\}$) for any sub-region $A \subset \mathbb{C}$,

$$\xi(A) = \sum_{k=1}^{n} \mathbb{1}_A(z_k).$$

As before, this defines a *random counting measure* in the usual sense. We say that the counting measure ξ is simple if $\xi(\{z\}) \leq 1$ for all $z \in \mathbb{C}$. Denote by $\mathcal{E}$ the class of Borel sets and $\mathcal{N}(\mathbb{C})$ the space of all counting measures on $\mathbb{C}$.

Definition 2.5. *(Point Processes on the complex plane) A* ***point process*** $\mathbb{P}$ *on* $\mathbb{C}$ *is a probability measure on the space of all counting measures* $\mathcal{N}(\mathbb{C})$*. We say* $\mathbb{P}$ *to be a n-point process if* $\mathbb{P}(\xi(\mathbb{C}) = n) = 1$*. We also say that the point process is simple if* $\mathbb{P}(\xi \text{ is simple}) = 1$.

We argue that the ensemble p.d.f. of invariant random matrix ensembles induce a joint probability density function on the space of configurations of eigenvalues, i.e. induce a simple point process (simple because of Proposition 2.1). In fact, note that the mapping $\mathcal{E} \ni A \mapsto \mathbb{E}[\#(\chi \cap A)]$, that assigns to the Borel set $A \in \mathbb{C}$ the expected value of number of points in itself, under the configuration χ, is a measure on $\mathbb{C}$.

Although not true for general point processes, in the models we are interested, this measure can be defined as the integral of a function we call 1*-point correlation function*. This function measures the probability of finding a particle at a given infinitesimal interval.

Definition 2.6. *The* **1*-point correlation function*** p_1 *represents the infinitesimal probability of a point of the process being at a location z, i.e.* $\mathbb{P}(\xi(\mathrm{d}^2 z) = 1) = \mathrm{p}_1(z)\mathrm{d}^2 z$*. It is such that the* ***mean measure****, or* ***first moment****, of a point process on* $\mathbb{C}$*, defined by*

$$\mathfrak{p}(A) = \mathbb{E}[\xi(A)] := \mathbb{E}[\#(\chi \cap A)], \quad A \in \mathcal{E}.$$

is given by its integral over A

$$\mathfrak{p}(A) = \int_A \mathrm{p}_1(z)\mathrm{d}^2 z$$

where $\mathrm{d}^2 z$ *is the area measure.*

In RMT, the usual name for the 1-point correlation function p_1, when normalized, is *univariate eigenvalue density*, for which we use the symbol $\mu = \mathrm{p}_1/n$ henceforth. Moreover, we may also be interested in the infinitesimal probability that some k points of the process are in a certain configuration $\chi_k = \{z_1, z_2, \dots, z_k\}$ for k in between 2 and n. For this we define more generally all the *k-th point correlation functions* and *k-th moments*.

Definition 2.7. *The* k***-point correlation function*** $\mathfrak{p}_k$ *represents the infinitesimal probability of* k *points of the process being at a* k*-configuration* χ_k*. It is defined such that the* k***-th moment*** *of a point process on* $\mathbb{C}$*, defined by*

$$\mathfrak{p}_k(A_1 \times A_2 \times \cdots \times A_k) := \mathbb{E}\left[\xi(A_1)\xi(A_2)\cdots\xi(A_k)\right], \quad A_1, A_2, \ldots, A_k \in \mathcal{E},$$

where $A_j \cap A_i = \varnothing$ *for* $i \neq j$*, is given by the integral*

$$\mathfrak{p}_k(A_1 \times A_2 \times \cdots \times A_k) = \mathbb{E}\left[\prod_{j=1}^{k}(\#\chi \cap A_j)\right] = \int_{A_1} \cdots \int_{A_k} \mathfrak{p}_k(z_1, \cdots, z_k)\mathrm{d}^2 z_1 \cdots \mathrm{d}^2 z_k.$$

where $\mathrm{d}^2 z_i$ *is the area measure.*

For simple point processes on $\mathbb{C}$ there is a direct way to understand correlation functions as mentioned, $\mathfrak{p}_k(z_1, z_2, \ldots, z_k)$ is exactly the probability of finding particles at positions $x_1, x_2, \ldots, x_k$, including permutations. This means that $\mathfrak{p}_1(z)/n!$ is the density of particles at z. Now, if you have some $p_n(z_1, z_2, \ldots, z_n)$ a j.p.d.f. on $\mathbb{C}^n$, invariant under permutation of coordinates, then we can build a n-point process on $\mathbb{C}$ with correlation functions

$$\mathfrak{p}_k(z_1, z_2, \ldots, z_k) = \frac{n!}{(n-k)!}\int_{\mathbb{C}^{n-k}} p_n(z_1, z_2, \ldots, z_n)\mathrm{d}^2 z_{k+1} \cdots \mathrm{d}^2 z_n. \tag{2.4}$$

Moreover, the j.p.d.f. on the eigenvalues is just $p_n = \frac{1}{n!}\mathfrak{p}_n$.

Example 2.7. An analogous definition of point process also holds for the real line, consider for example the GUE (see Example 2.2). We have $\mathcal{E} = (\mathcal{H}_n, \mathcal{P}^{(G)})$ with the space of all $n \times n$ complex Hermitian matrices equipped with $\mathcal{P}^{(G)}$, the Gaussian measure. We note that for $\hat{M} \in \mathcal{H}_n$,

$$\hat{M} \mapsto \sum_{j=1}^{n} \mathbb{1}(\lambda_j(\hat{M}))$$

maps $\mathcal{P}^{(G)}$ to a finite point process $\mathbb{P}$ on $\mathbb{R}$. If $\mathrm{d}\hat{M}$ is the Lebesgue measure on $\mathcal{H}_n$,

$$\mathrm{d}\mathcal{P}(\hat{M}) = \frac{1}{\mathcal{Z}_n^{(G)}}\mathrm{e}^{-n\operatorname{Tr}\hat{M}^2}\mathrm{d}\hat{M} = \frac{1}{\mathcal{Z}_n^{(G)}}\mathrm{e}^{-n\operatorname{Tr}\hat{\Lambda}^2}\mathrm{d}\hat{M} = \frac{1}{\mathcal{Z}_n^{(G)}}\mathrm{e}^{-n\sum_i^n \lambda_i^2}\mathrm{d}\hat{M}$$

where $\{\lambda_1(\hat{M}), \cdots, \lambda_n(\hat{M})\}$ denotes the set of eigenvalues of $\hat{M} \in \mathcal{H}_n$. From a construction analogous to Theorem 2.2 we conclude that the j.p.d.f. on the eigenvalues is

$$p_n(\lambda_1, \lambda_2, \ldots, \lambda_k) = \frac{1}{\mathcal{Z}_n^{(G)}}\prod_{1 \le i < j \le n}^{n}|\lambda_i - \lambda_j|^2 \prod_{j=1}^{n}\mathrm{e}^{-n\lambda_j^2}.$$

The real point process $\mathbb{P}$ defined by the Gaussian unitary p.d.f. has correlation

$$\mathfrak{p}_k(x_1, x_2, \ldots, x_k) = \frac{n!}{\mathcal{Z}_n^{(G)}(n-k)!}\int_{\mathbb{R}^{n-k}}\prod_{1 \le i < j \le n}^{n}|x_i - x_j|^2 \prod_{j=1}^{n}\mathrm{e}^{-nx_j^2}\prod_{i=k+1}^{n}\mathrm{d}x_i.$$

2.3.1 Determinantal Structure

We stated that random eigenvalues can be described as point processes. However, when dealing with invariant ensembles there is more structure: There is a repulsion effect between eigenvalues. Repulsive random processes are, in general, a challenge to study as computing marginals and conditional probabilities might not be straightforward. However, there is a class of processes that is an exception to this rule: *Determinantal point processes* - or Fermionic point processes, in the physics literature. These are random processes which are both repulsive and tractable. (TREMBLAY, 2025) This property comes from the fact that their correlation functions can be expressed with respect to a determinant. We now make this precise.

Definition 2.8. *(Determinantal point processes on the plane) Consider a point process $\mathbb{P}$ on $\mathbb{C}$, with joint probability density function measure p_n, all of whose correlation functions* p_k *exist. If there is a function $K : \mathbb{C}\times\mathbb{C} \to \mathbb{C}$ such that*

$$\mathrm{p}_k(z_1, z_2, \dots, z_k) = \det(K(z_i, z_j))_{i,j=1}^{k}$$

for all $z_1, z_2, \dots, z_k \in \mathbb{C},\ k \geq 1$, then we say that $\mathbb{P}$ is a ***determinantal point process*** *on $\mathbb{C}$. We call K the* ***correlation kernel*** *of the process.*

This definition can be extended for a general complete separable metric space but its full extent is outside of our scope. We focus on weight functions as in Definition 2.4 and remind both the genral expression for correlation functions (2.4) and the j.p.d.f. for the polynomial ensemble given in Definition 2.4. Then, we can write the k-th correlation function for such ensemble as

$$\mathrm{p}_k(\lambda_1, \lambda_2, \dots, \lambda_k) = \frac{n!}{\mathcal{Z}_n^Q (n-k)!} \int_{\mathbb{C}^{n-k}} \prod_{i=1}^{n} \mathrm{e}^{-nQ(\lambda_i)} \prod_{i<j} |\lambda_i - \lambda_j|^2 \prod_{i=k+1}^{n} \mathrm{d}^2\lambda_i.$$

To show that we can find a function $K : \mathbb{C}\times\mathbb{C} \to \mathbb{C}$ such that $\mathrm{p}_k(z_1, z_2, \dots, z_k) \propto \det(K(z_i, z_j))_{i,j=1}^{k}$, we define *orthogonal polynomials* and prove a proposition regarding the determinant $|\Delta_n(\lambda)|$.

Definition 2.9. *With Q a suitable function, define the following inner products on polynomial space:*

$$\langle f, g\rangle_n = \int_{\mathbb{C}} f(z)\overline{g(z)}\, \mathrm{e}^{-nQ(z)}\, \mathrm{d}^2 z.$$

Let $\{q_{j,n}\}_{j=0}^{\infty}$ be a family of monic polynomials such that each $q_j(\lambda)$ is of degree j. Impose also that exists finite positive constants $\{h_{j,n}\}_{j=0}^{\infty}$ s.t.

$$\langle q_{j,n}, q_{k,n}\rangle_n = h_{j,n} \mathbb{1}_{\{j=k\}}$$

for any $j, k > 0$. We say that the family $\{q_{j,n}(z)\}_{j=0}^{\infty}$ ***is orthogonal with respect to the weight function*** *$w(z) = \mathrm{e}^{-nQ(z)}$.*

Note 2.1. (Expected characteristic polynomial for Gaussian Unitary Ensemble (GUE)) To calculate eigenvalues of some given matrix, one need to determine the *characteristic polynomial*. However, dealing with a random matrix $\hat{M}$, one of the quantities one might want to take on is the *average* or *expected characteristic polynomial*

$$P_n(x) = \mathbb{E}[\det\left(x\hat{I}_n - \hat{M}\right)], \quad n \in \mathbb{N}.$$

The characteristic polynomial has roots that coincide with the eigenvalues of the matrix. Therefore, morally, roots of an averaged characteristic polynomial can be thought of as the typical set of eigenvalues for the ensemble of $\hat{M}$. Taking the asymptotic of n - consequently the limit for the degree of the expected characteristic polynomial - the distribution of roots is expected to coincide with the distribution of eigenvalues.
Despite of this not being true for ensembles with non-real roots, for random Hermitian matrices distributed accordingly to some *weight function* w, the average characteristic polynomial is simply the monic orthogonal polynomial $P_n^w(x)$ of degree n with respect to $w(x)$ - as shown in Edelman (1989). The asymptotic of the appropriate orthogonal polynomials reveals information on the distribution of eigenvalues. The idea of orthogonal polynomials will come back as a way to express eigenvalue correlation functions.
We will indicate in Example 2.8 that *Hermite Polynomials* are the orthogonal polynomials with respect to the weight function $w(x) = \mathrm{e}^{-nx^2}$, which is the defining weight function of the Gaussian Ensemble. That means that for $\hat{M}$ a hermitian matrix of dimension n with probability density function given by

$$\frac{1}{\mathcal{Z}_n^{(G)}} \mathrm{e}^{-n \operatorname{Tr} \hat{M}^2} \mathrm{d}M,$$

one can show numerically that the concentration of roots of the appropriately normalized Hermite polynomial and the eigenvalues of $\hat{M}$ coincide in distribution; the roots represent the typical set of eigenvalues.

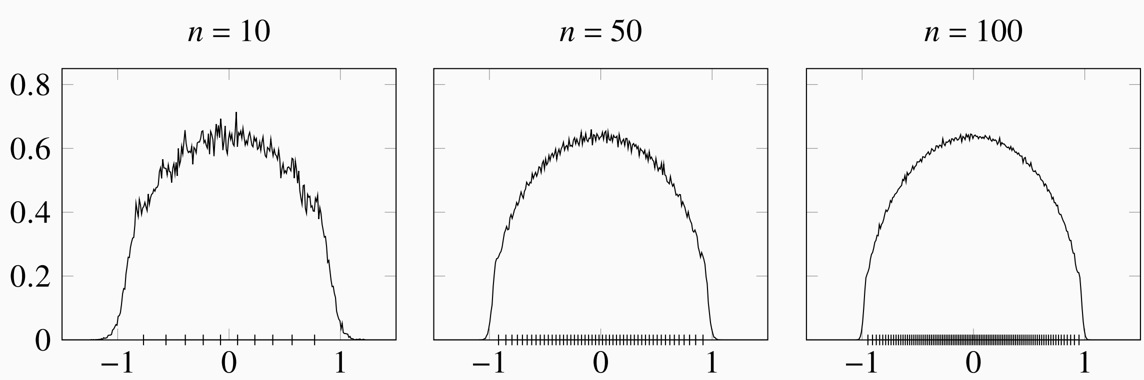

Figure 3 – For all graphs, the line is the average distribution of eigenvalues for the GUE. The columns correspond to the polynomial order (and matrix size) $n = 10, 50, 100$ from left to right. The dashes in the axis correspond to the roots of the n-th Hermite polynomial.

Now, we prove the following proposition such that we can use these orthogonal polynomials to construct the orthogonal polynomial Kernel in our correlation functions.

Proposition 2.2. *The product of the differences of the set* $\{\lambda_i\}_{i=1}^n$ *can be evaluated as a determinant known as the Vandermonde determinant. Moreover, it can be rewritten as the determinant of* $\left(q_{j-1}(\lambda_i)\right)_{j,i=1}^n$. *That is,*

$$\prod_{i<j}^{n} |\lambda_i - \lambda_j| = \det\left(q_{j-1}(\lambda_i)\right)_{n\times n}$$

where q_i *are monic orthogonal polynomials with respect to the real-valued function* w.

Proof. Consider a matrix $\hat{V}$ given by

$$\hat{V} = \begin{bmatrix} 1 & \alpha_1 & \alpha_1^2 & \dots & \alpha_1^{n-1} \\ 1 & \alpha_2 & \alpha_2^2 & \dots & \alpha_2^{n-1} \\ 1 & \alpha_3 & \alpha_3^2 & \dots & \alpha_3^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \alpha_n^2 & \dots & \alpha_n^{n-1} \end{bmatrix}.$$

We can use the determinant's multilinearity to subtract from each row r_i the product $\alpha_1 r_{i-1}$ starting from $i = n$ and ending at $i = 2$. Then subtract from every line the first line such that the first columns is exclusively zeros. Then, we factor out the term $\alpha_i - \alpha_1$ from each line, i.e.

$$\det\left(\hat{V}\right) = \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & \alpha_2-\alpha_1 & \alpha_2(\alpha_2-\alpha_1) & \dots & \alpha_2^{n-2}(\alpha_2-\alpha_1) \\ 0 & \alpha_3-\alpha_1 & \alpha_3(\alpha_3-\alpha_1) & \dots & \alpha_3^{n-2}(\alpha_3-\alpha_1) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \alpha_n-\alpha_1 & \alpha_n(\alpha_n-\alpha_1) & \dots & \alpha_n^{n-2}(\alpha_n-\alpha_1) \end{vmatrix} = \prod_{i=2}^{n} |\alpha_i - \alpha_1| \begin{vmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 0 & 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}.$$

Using *Laplace Theorem* for the determinant we know that there is only one co-factor that matters, then

$$\det\left(\hat{V}\right) = \prod_{i=2}^{n} |\alpha_i - \alpha_1| \begin{vmatrix} 1 & \alpha_2 & \dots & \alpha_2^{n-2} \\ 1 & \alpha_3 & \dots & \alpha_3^{n-2} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \alpha_n & \dots & \alpha_n^{n-2} \end{vmatrix}.$$

Note that the matrix is very similar to its initial form again. Repeating this procedure many times we get the expression we started with, that is,

$$\det\left(\hat{V}\right) = \prod_{i<j}^{n} |\alpha_j - \alpha_i|$$

Now, it is only a matter of showing that

$$\det\left((\alpha_i^j)_{i,j}^n\right) = \det\left((q_{j-1}(\alpha_i))_{i,j}^n\right)$$

but this is direct since it only takes fundamental operation, i.e. by multilinearity, the operations $\alpha_i^{j-1} \mapsto q_{j-1}(\alpha_i)$ do not change the determinant. Therefore for any polynomial base $\{q_j\}_0^{n-1} := \{\sum_{i=0}^{j-1} a_{i,j}\alpha_j^i\}$, we can, by column operations of the type $R_j \mapsto \sum_{i=0}^{j-1} a_{i,j}R_j$, rewrite

$$\det\left(\hat{V}\right) = \begin{vmatrix} 1 & a_{1,2}\alpha_1 & \sum_{i=0}^{2-1} a_{1,j}\alpha_1^i & \dots & \sum_{i=0}^{n-1} a_{1,j}\alpha_1^i \\ 1 & a_{2,2}\alpha_2 & \sum_{i=0}^{2-1} a_{2,j}\alpha_2^i & \dots & \sum_{i=0}^{n-1} a_{2,j}\alpha_2^i \\ 1 & a_{3,2}\alpha_3 & \sum_{i=0}^{2-1} a_{3,j}\alpha_3^i & \dots & \sum_{i=0}^{n-1} a_{3,j}\alpha_3^i \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_{n,2}\alpha_n & \sum_{i=0}^{2-1} a_{n,j}\alpha_n^i & \dots & \sum_{i=0}^{n-1} a_{n,j}\alpha_n^i \end{vmatrix} = \begin{vmatrix} 1 & q_1(\alpha_1) & q_1(\alpha_1) & \dots & q_{n-1}(\alpha_1) \\ 1 & q_2(\alpha_2) & q_1(\alpha_2) & \dots & q_{n-1}(\alpha_2) \\ 1 & q_1(\alpha_3) & q_2(\alpha_3) & \dots & q_{n-1}(\alpha_3) \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & q_1(\alpha_n) & q_2(\alpha_n) & \dots & q_{n-1}(\alpha_n) \end{vmatrix},$$

which give us exactly that $\det\left(\hat{V}\right) = \det\left(q_{j-1}(\lambda_i)\right)_{i,j}^n$. □

We still want to express the correlation function as a determinant of a Kernel function. We start by using the fact that $(\det(A))^2 = \det(\sum_{j=1}^n \overline{A_{ji}}A_{jk})$ and Proposition 2.2 to rewrite

$$\prod_{i<j}^n |\lambda_i - \lambda_j|^2 = \det\left(\sum_{l=1}^n \overline{q_{l-1,n}(\lambda_i)}q_{l-1,n}(\lambda_j)\right)_{n\times n}$$

where q_i are orthogonal polynomials with respect to a weight function w. Then, start by putting the weight function inside the determinant in the following manner: Take $w(\lambda_i) = \sqrt{w(\lambda_i)}\sqrt{w(\lambda_i)}$ and multiply the i-th column of the matrix by one of the terms. The other term we multiply by the i-th line of the matrix. We also introduce the norming constants inside the determinant by multiplying by one. We define K_n to be such that

$$\begin{aligned} \mathrm{p}_k(\{\lambda_i\}_{i=1}^k) &= \frac{n!\prod_{l=1}^n h_{n,l}}{\mathcal{Z}_n^Q (n-k)!}\int_{\mathbb{C}^{n-k}} \det\left(\sqrt{w(\lambda_i)w(\lambda_j)}\sum_{l=1}^n \frac{1}{h_{n,l}}\overline{q_{l-1,n}(\lambda_i)}q_{l-1,n}(\lambda_j)\right)_{i,j=1}^n \prod_{i=k+1}^n \mathrm{d}^2\lambda_i, \\ &= \frac{n!\prod_{l=1}^n h_{n,l}}{\mathcal{Z}_n^Q (n-k)!}\int_{\mathbb{C}^{n-k}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^n \prod_{i=k+1}^n \mathrm{d}^2\lambda_i. \end{aligned}$$

At this point it is still not clear why this would be a determinantal point process; we are still to show that the above integral results in a straightforward determinantal expression. To properly use this monic orthogonal polynomials structure, we will prove the fact that the orthogonal kernels $K_n(x,y)$ defined above are known to have a *reproducing property* which allow us to integrate the determinantal structure.

Theorem 2.3. *Let $\{q_{j,n}(x)\}_{j=0}^\infty, \{h_{j,n}\}_{j=0}^\infty$, be as in Definition 2.9 and let $n \in \mathbb{N}$. Consider the orthogonal kernel*

$$K_n(x,y) = \sqrt{w(x)w(y)}\sum_{k=1}^n \frac{1}{h_{k,n}}\overline{q_{k,n}(x)}q_{k,n}(y).$$

Then,

1. *The kernel has reproducing properties, i.e.*

$$\int_{\mathbb{C}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{k+1} \mathrm{d}^2\lambda_{k+1} = (n-k)\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}.$$

2. *The eigenvalue correlations functions of Definition 2.4 are given by*

$$\mathrm{p}_k(\lambda_1,\lambda_2,\ldots,\lambda_n) = \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}.$$

3. *The partition function of Definition 2.4 can be expressed as*

$$\mathcal{Z}_n^Q = n!\prod_{l=1}^{n} h_{n,l}.$$

Proof. We first note that the expression for the correlation function

$$\mathrm{p}_k(\{\lambda_i\}_{i=1}^k) = \frac{n!\prod_{l=1}^n h_{n,l}}{\mathcal{Z}_n^Q (n-k)!}\int_{\mathbb{C}^{n-k}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n} \prod_{i=k+1}^{n} \mathrm{d}^2\lambda_i$$

is such that the j.p.d.f. is

$$p_n(\{\lambda_i\}_{i=1}^n) = \frac{\prod_{l=1}^n h_{n,l}}{\mathcal{Z}_n^Q}\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}$$

and, if S_m is the m-th permutation group, we can calculate the $(n-1)$-th correlation function

$$\begin{aligned}
\mathrm{p}_{n-1}(\{\lambda_i\}_{i=1}^{n-1}) &= \frac{n!\prod_{l=1}^n h_{n,l}}{\mathcal{Z}_n^Q}\int_{\mathbb{C}} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n} \mathrm{d}^2\lambda_n,\\
&= \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1}\int_{\mathbb{C}} K_n(\lambda_n,\lambda_n)\mathrm{d}^2\lambda_n - \sum_{j=1}^{n-1}\int_{\mathbb{C}} K_n(\lambda_j,\lambda_n)\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1}\mathrm{d}^2\lambda_n\\
&= n\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} - \sum_{j=1}^{n-1}\sum_{\sigma\in S_{n-1}} \mathrm{sgn}\,(\sigma)\left(\prod_{i\neq j}^{n-1} K_n(\lambda_i,\lambda_{\sigma(i)})\right)\int_{\mathbb{C}} K_n(\lambda_j,\lambda_n)K_n(\lambda_n,\lambda_{\sigma(j)})\mathrm{d}^2\lambda_n,\\
&= n\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} - \sum_{j=1}^{n-1}\sum_{\sigma\in S_{n-1}} \mathrm{sgn}\,(\sigma)\left(\prod_{i\neq j}^{n-1} K_n(\lambda_i,\lambda_{\sigma(i)})\right)K_n(\lambda_j,\lambda_{\sigma(j)}),\\
&= n\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} - (n-1)\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1},\\
&= (n-(n-1))\det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1} = \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n-1}.
\end{aligned}$$

where we used, by its orthogonality properties, that

$$\int_{\mathbb{C}} K_n(\lambda_n,\lambda_n)\mathrm{d}^2\lambda_n = n, \text{ and } \quad \int_{\mathbb{C}} K_n(\lambda_j,\lambda_k)K_n(\lambda_k,\lambda_l)\mathrm{d}^2\lambda_k = K_n(\lambda_j,\lambda_l).$$

This proves (1). Now, using the above integration recursively for any number k of times, one can also check that (2) is true. Finally, using the above technique again,

$$\mathcal{Z}_n^Q = \prod_{l=1}^{n} h_{n,l}\int_{\mathbb{C}^n} \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{n}\prod_{i=1}^{n}\mathrm{d}^2\lambda_i = n!\prod_{l=1}^{n} h_{n,l}$$

□

At some point we turn our attention to the univariate eigenvalue density μ as it gives the distribution for a single variable. We note that $\mu = \mathrm{p}_1(\lambda)/n!$ and moreover

$$n!\mu(\lambda) = \det\left[K_n(\lambda_i,\lambda_j)\right]_{i,j=1}^{1} = K_n(\lambda,\lambda). \tag{2.5}$$

Therefore, if we understand the behavior of the correlation kernel, we can also state the behavior of the univariate eigenvalue density. Therefore, most results we study of a determinantal point process - such as the asymptotic of μ - follow from properties of the Kernel.

Example 2.8. (Eigenvalue for orthogonal polynomial ensembles) The result we proved for the correlation functions for normal matrix can be adapted for for hermitian matrices and results in a expression as follows

$$\begin{aligned}
\mathrm{p}_k(\lambda_1,\lambda_2,\dots,\lambda_k) &= \frac{n!}{\mathcal{Z}_n^Q(n-k)!}\int_{\mathbb{R}^{n-k}}\prod_{i=1}^{n}\mathrm{e}^{-nQ(\lambda_i)}\prod_{i<j}(\lambda_i-\lambda_j)^2\prod_{i=k+1}^{n}\mathrm{d}\lambda_i.\\
&= \frac{1}{(n-k)}\int_{\mathbb{R}^{n-k}}\det\left(\sqrt{w(\lambda_i)w(\lambda_j)}\sum_{l=1}^{n}\frac{1}{h_{j,n}}q_{l-1,n}(\lambda_i)q_{l-1,n}(\lambda_j)\right)_{i,j=1}^{n}\prod_{i=k+1}^{n}\mathrm{d}\lambda_i\\
&= \det\left[K_n^Q(\lambda_i,\lambda_j)\right]_{i,j=1}^{k}
\end{aligned}$$

where $w(\lambda_i)$ is the chosen weight function. This is the main connection for the theory of random matrices and DPPs - and one that we take major advantage of. On the real line, the orthogonal polynomials $\{q_j(\lambda)\}_{j=0}^{\infty}$ that compose the kernel are known to satisfy the three-term recurrence

$$\lambda q_{j,n}(\lambda) = b_{j-1,n}q_{j-1,n}(\lambda) + a_{k,n}q_{j,n}(\lambda) + b_{k,n}\phi_{k+1,n}(\lambda).$$

where

$$a_{k,n} = \int xq_{k,n}(x)^2\,\mathrm{e}^{-nQ(x)}\,\mathrm{d}x,\quad b_{k,n} = \int xq_{k,n}(x)q_{k+1,n}(x)\,\mathrm{e}^{-nQ(x)}\,\mathrm{d}x.$$

The polynomials which satisfy the above three-term recurrence for $w(x) = \mathrm{e}^{-nx^2}$ are the Hermite polynomials He_i. We denote the Kernel defined by the Hermite polynomials $K_n^{(H)}$. Moreover, as we are on the real line, we can use the *Christoffel-Darboux* formula

$$K_n^{(H)}(x,y) = \frac{\sqrt{w(x)w(y)}}{h_{n-1,n}}\frac{q_{n,n}(x)q_{n-1,n}(y) - q_{n-1,n}(x)q_{n,n}(y)}{x-y}$$

where $h_{k,n}$ is the norming constant and $h_{k,n}$ is the monic orthogonal polynomial orthogonal with respect to $w(x) = \mathrm{e}^{-nx^2}$. Furthermore, taking the limit $y \to x$ using L'Hopital's rule, for finite n we show that Christoffel-Darboux simplifies to

$$n!\mu^{(H)}(\lambda) = K_n^{(H)}(\lambda,\lambda) = \frac{w(\lambda)}{h_{n-1,n}}\left(q_{n-1}(\lambda)\partial_\lambda q_n(\lambda) - q_n(\lambda)\partial_\lambda q_{n-1}(\lambda)\right).$$

and if we have the asymptotic of the n-th and $(n-1)$-th Hermite polynomials, we get the asymptotic of $\mu^{(H)}$.

We now state a few result that will elucidate properties of determinantal point processes and allow us to examine different scales and regimes for the univariate eigenvalue density. Let $f \in C^1(\mathbb{C})$ be a bijection, with continuously differentiable inverse $f^{-1} = g \in C^1(\mathbb{C})$. Then, f maps configurations on $\mathbb{C}$ to configurations on $\mathbb{C}$. The push-forward of a point process $\mathbb{P}$ under such mapping is $f(\mathbb{P})$ and it is again a point process.

Proposition 2.3. *(GIROTTI, 2014) If K is the kernel of a DPP $\mathbb{P}$, then $f(\mathbb{P})$ is also a DPP with kernel*

$$\hat{K}(x,y) = \sqrt{g'(x)g'(y)}K(g(x),g(y))$$

where $f^{-1} = g$.

In particular, if we consider a linear function $f(x) = \alpha(x - x^*)$, for $\alpha > 0$ and $x^* \in \mathbb{C}$, then the transformed point process $f(\mathbb{P})$ is a scaled and translated version of the original point process $\mathbb{P}$. The new correlation kernel is

$$\hat{K}(x,y) = \frac{1}{\alpha}K\left(x^* + \frac{x}{\alpha}, x^* + \frac{y}{\alpha}\right).$$

Example 2.9. (Karlin-McGregor Theorem & Dyson Bridges (KUIJLAARS, 2014)) Consider n independent copies $X_1(t), \cdots, X_n(t)$ of a one-dimensional strong Markov Process with continuous sample paths conditioned so that $X_j(0) = a_j$, where $a_1 < a_2 < \cdots < a_n$ are certain given values. Let $p_t(x,y)$ be some transition probability function of the Markov process. Moreover, let $E_1, E_2, \ldots, E_n$ be Borel sets so that $\sup E_j < \inf E_{j+1}$ for $j = 1, \cdots, n-1$. Then

$$\int_{E_1} \cdots \int_{E_n} \det\left[p_t(a_i, x_j)\right]_{i,j=1}^{n} \mathrm{d}x_1 \ldots \mathrm{d}x_n$$

is equal to the probability that the paths have not intersected in the time interval $[0,t]$ and $X_j(t) \in E_j$ for $j = 1, \cdots, n$. This is the Karlin-McGregor theorem.

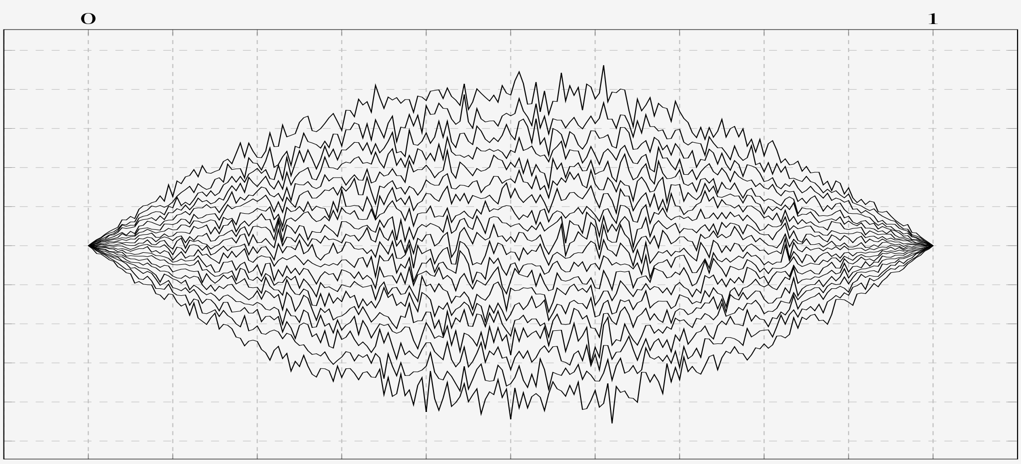

Figure 4 – 20 Non-intersecting Brownian paths for $0 \leq t \leq 1$ s.t. they start and end at $y = 0$.

Take $p_t(a,x) = 1/\sqrt{(2\pi t)}\exp\{-(x-a)^2/2t\}$ a Gaussian transition probability, and consider that we condition our path to not only start at positions $a_1, a_2, \ldots, a_n$ but finish at positions $b_1, b_2, \ldots, b_n$ at some time $T > t$. If we let $a_j \to a$ and $b_j \to b$, the new j.p.d.f. is non-trivially calculated for any t as

$$\begin{aligned}
P(x_1,\ldots,x_n) &= \frac{1}{\mathcal{Z}_n}\det\left[p_t(a_i,x_j)\right]_{i,j=1}^n \det\left[p_{T-t}(x_i,b_j)\right]_{i,j=1}^n \\
&= \frac{1}{\mathcal{Z}_n}\det\left[x_j^{i-1}\right]_{i,j=1}^n \prod_{j=1}^n \mathrm{e}^{-\frac{x_j^2}{2t}} \det\left[x_j^{i-1}\right]_{i,j=1}^n \prod_{j=1}^n \mathrm{e}^{-\frac{x_j^2}{2(T-t)}} \\
&= \frac{1}{\mathcal{Z}_n}\det\left[x_j^{i-1}\right]_{i,j=1}^n \det\left[x_j^{i-1}\right]_{i,j=1}^n \prod_{j=1}^n \mathrm{e}^{-\frac{Tx_j^2}{2t(T-t)}} \\
&= \frac{1}{\mathcal{Z}_n}\prod_{1\le i<j\le n}(x_j-x_i)^2 \prod_{j=1}^n \mathrm{e}^{-\frac{Tx_j^2}{2t(T-t)}}
\end{aligned}$$

Two things are worth noting on this point. The first is that this resembles the j.p.d.f. of the Gaussian Unitary Ensemble! The second thing is that it has a scaling in the weight function based on t. This means that for every t the distribution obeys the same law with a different scaling - it keeps being an determinant process as specified in Proposition 2.3. One can simulate such Brownian motions as below for $T = 1$ as in Figure 4. Notice also on Figure 5 below that the distribution of points is kept the same modulo the re-scaling of $x/((2t(1-t))\sqrt{N})$.

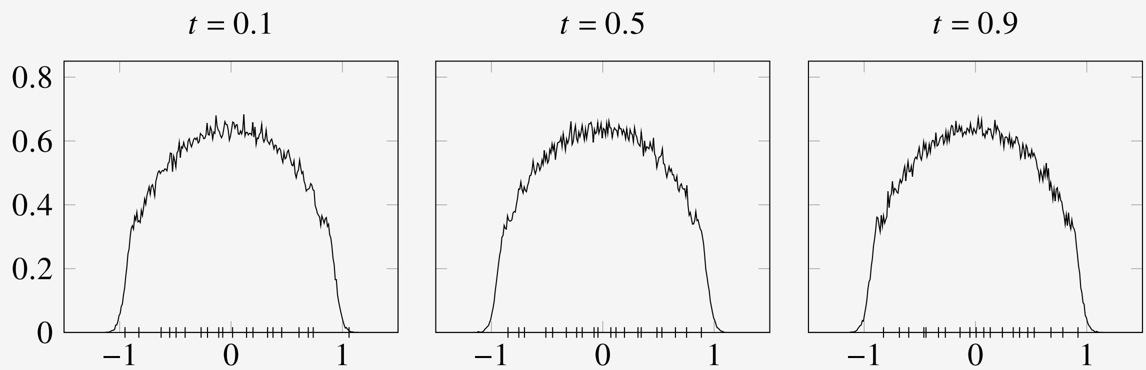

Figure 5 – Three conditions simulated form the 20 Dyson Bridges taken from $t = 0.1, 0.5, 0.9$. For each one the experimental density (line plot) and one sampled position (dashes in axis) were plotted. The three plots are re-scaled by $((2t(1-t))\sqrt{N})$.

We know we can reexamine the Kernels centralized on different points and under different scales. However, so far we only examined finite n Kernels. An essential aspect of RMT, however, is what happens on the thermodynamic limit - when we take $\lim_{n\to\infty} K_n(x,y)$. Therefore, we now consider sequences of finite DPP $\mathbb{P}_n$ with correlation kernel K_n and understand how the limit behaves.

Proposition 2.4. *(GIROTTI, 2014) Let $\mathbb{P}$ and $\mathbb{P}_n$ be determinantal point processes with kernels K and K_n respectively. Let K_n converge point-wise to K*

$$\lim_{n\to\infty} K_n(x,y) = K(x,y)$$

uniformly in x,y over compact subsets of $\mathbb{C}$. Then, the point processes $\mathbb{P}_n$ converges to $\mathbb{P}$ weakly.

The limit K is a kernel that corresponds to a determinantal point process with an infinite number of points. A very important case is that of the limit of an re-scaled and centered determinantal point process. Consider a sequence of kernels $K_n(x,y)$ and a fixed point c of interest. We first perform a centering and re-scaling of the form $x \mapsto Cn^{\alpha}(x-c)$ for some suitable value of $C,\gamma > 0$. We defined a determinantal point process with the re-scaled kernel

$$\tilde{K}_n(x,y) = \frac{1}{Cn^{\alpha}} K_n\left(c + \frac{x}{Cn^{\alpha}}, c + \frac{y}{Cn^{\alpha}}\right).$$

If $\lim_{n\to\infty} \tilde{K}_n(x,y) = K(x,y)$, K represents the kernel of the infinite DPP in this scaling. Interestingly, in many situations, the same limit K may appear. This phenomenon in known as **universality** in RMT.

Example 2.10. Consider again the Kernel described in Example 2.8 for the Gaussian Unitary Ensemble, $K_n^{(H)}(x,y)$. If

$$\lim_{n\to\infty} \frac{1}{n} K_n^{(H)}(x,x) = \mu_V(x), \quad x \in \mathbb{R},$$

then the asymptotic distribution of the points in the point process (eigenvalues of $\hat{M}_n$) is given by the measure with density μ_V:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{j=1}^{n} f(x_j) = \int f(x)\mu_V(x)\mathrm{d}x.$$

Moreover, suppose μ_V to be supported on one interval $[a,b]$, then for $x^* \in (a,b)$ with $\mu_V(x^*) > 0$, one has

$$\lim_{n\to\infty} \frac{1}{\mu_V(x^*)n} K_n^{(H)}\left(x^* + \frac{x}{\mu_V(x^*)n}, x^* + \frac{y}{\mu_V(x^*)n}\right) = \frac{\sin\pi(x-y)}{\pi(x-y)}.$$

Finally, if near the endpoint b

$$\mu_V(x) = \frac{c}{\pi}(b-x)^{\frac{1}{2}}(1+o(1)), \quad x \to b,$$

then

$$\lim_{n\to\infty} \frac{1}{(cn)^{\frac{2}{3}}} K_n^{(H)}\left(b + \frac{x}{(cn)^{\frac{2}{3}}}, b + \frac{y}{(cn)^{\frac{2}{3}}}\right) = \frac{\mathrm{Ai}(x)\,\mathrm{Ai}'(y) - \mathrm{Ai}'(x)\,\mathrm{Ai}(y)}{x-y},$$

where Ai is the Airy function. Both these limit Kernels are common to all Wigner Matrices.

CHAPTER

3

EQUILIBRIUM PROBLEMS

We have seen in Chapter 2 that the j.p.d.f. of eigenvalues of certain random matrices defines a determinantal point processes. We will see that not only that, but eigenvalues can have the same correlation functions as another point process: The so called *Coulomb gas*. These can be though of as a measure on a physical gas of repelling ions restricted to some conductor under a confining potential. With this interpretation one wonders what happens to these gases in the *thermodynamic limit*. Physically, this is the regime we are used to work with; in thermodynamics we often study macroscopic properties of gases such as *entropy* or *temperature*. When considering a large enough number of particles, fluctuations of these quantities become negligible. Moreover, under the appropriate conditions, to be discussed further in the chapter, these particles are known to converge to a limit equilibrium configuration. In our matrices, these equilibrium configurations will refer to the eigenvalue univariate density function and, as discussed in Chapter 2, will be expressed by the correlation kernel.

Furthermore, in this chapter we also expand on the usual formulation of minimization of a particle system's energy problems. We begin discussing *equilibrium measures* of eigenvalues - the unique measure that minimizes our functional problem. This minimization lead us to a type of problem can be solved by considering the *Euler-Lagrange*, or *Variational Equations*. We use these results to calculate the equilibrium measures for the ensemble of random matrices we work with in Chapter 5. Finally, we finish this chapter doing some informal calculations that intend to motivate the asymptotic expression we get for the partition function on Chapter 5. This also gives a basis estimate to which we compare the order of approximation obtained in Byun, Kang and Seo (2023) and explored in Chapter 5.

We based Section 3.2, on equilibrium measures, on the works of Saff and Totik (2024) and Deift (1999). The argument to derive an expression for the free energy in Section 3.3 is presented in Livan, Novaes and Vivo (2018).

3.1 Coulomb Gases

We start by describing a familiar distribution for physicists: The Boltzmann-Gibbs distribution. This is a probability distribution that represents the probability of a given state for a system of n particles as a function of the system state's *energy* ($\mathcal{H}$) and the *temperature* ($1/(\beta n^2)$). Physically, this is the distribution that maximizes the entropy: lower energy states have higher probability of being occupied than states with higher energy. This distribution is common on thermodynamics by this very reason. Let's introduce a system that has states partitioned accordingly to such distribution. Suppose a set of n particles at positions $x_1, x_2, \ldots, x_n \in \Sigma$, a *conductor* of dimension d, or in some configuration $\chi_n = (x_1, x_2, \ldots, x_n)$, embedded $\mathbb{R}^{dn}$.

Definition 3.1. *If* $\mathrm{d}p_n$ *is a probability measure on* $(\mathbb{R}^d)^n$, *we name it* ***Coulomb gas*** *if it has the form of a Boltzmann-Gibbs probability measure, namely if*

$$\mathrm{d}p_n(\chi_n) = \frac{\mathrm{e}^{-\beta n^2 \mathcal{H}(\chi_n)}}{\mathcal{Z}_{n,\beta}^{Q}} \prod_{i=1}^{n} \mathrm{d}^d x_i, \tag{3.1}$$

where

$$\mathcal{H}(\chi_n) = \frac{1}{2n}\sum_{i=1}^{n} Q(x_i) + \frac{1}{n^2}\sum_{i<j}^{n} \mathfrak{g}(x_i - x_j)$$

is usually called the ***Hamiltonian****, or more concretely, the* ***configurational energy*** *of the system. The function* $\mathfrak{g}\colon \Sigma \to (-\infty, \infty]$ *is the Coulomb Interacting Kernel, solution to the Poisson equation and given by* $\mathfrak{g}(x) = -\ln(|x|)$ *for* $d = 1, 2$. *The function* $Q\colon \mathbb{R}^d \cong \Sigma \to \mathbb{R}$ *is the external potential and* $\mathcal{Z}_{n,\beta}^{Q}$ *is the* ***partition function****, or normalization function, given by*

$$\mathcal{Z}_{n,\beta}^{Q} = \int_{\mathbb{R}^{dn}} \mathrm{e}^{-\beta n^2 \mathcal{H}(\chi_n)} \prod_{i=1}^{n} \mathrm{d}^d x_i.$$

The measure $\mathrm{d}p_n$ models an Coulomb-interacting gas of charged indistinguishable particles under some external field Q. We wish $\mathrm{d}p_n$ to be a probability measure, then one must ensure that Q is such that the normalizing constant $\mathcal{Z}_{n,\beta}^{Q}$ is finite for all n. This is all to say that the potential should be confining: As the number of particles increases, it must counter the repulsion effect and keep limited the region where particles move around. Recall the expression for the orthogonal polynomial ensembles in Definition 2.4. As we are dealing with point processes on the plane, we take the gas to be in a two dimensional conductor ($d = 2$) and $\beta = 2$. Rearranging the terms in (3.1) we get

$$\mathrm{d}p_n(z_1, z_2, \ldots, z_n) = \frac{1}{\mathcal{Z}_n^{Q}} \prod_{i<j}^{n} |z_i - z_j|^2 \prod_{j=i}^{n} \mathrm{e}^{-nQ(z_i)}\, \mathrm{d}^2 z_i.$$

The similarity to Definition 2.4 indicates that eigenvalues from invariant ensembles share some statistics with Coulomb gases. This opens the way for interpretations of the distribution of eigenvalues but also to computations using simulation methods for particle dynamics, see

Example 3.1. This also motives why most often the weight functions used for invariant ensembles will have the form, for Q polynomial,

$$w(x) = \mathrm{e}^{-nQ(x)}.$$

This directly relates to the weight function in Definition 2.4 and gives a way to trap particles in a limited set - giving smaller probabilities for configurations with more diffused particles.

Example 3.1. As explored in Chafaï and Ferré (2018), a few classical algorithms of simulation are very successful in replicating the measures for Coulomb gas systems. Equivalently, if one is interested in sampling eigenvalues for our invariant ensembles, one can sample from its configuration equilibrium $\mathrm{p}_Q(\lambda)$ with respect to some potential Q using simulation of the particles. Consider for example the non-trivial case explored in Balogh, Grava and Merzi (2016), where, taken $d = 2$ and $n = 2$ we will have $W(x) = g(x) = \log|x|$ and impose the potential

$$Q(z) = |z|^{2a} - \mathrm{Re}\, t z^a.$$

This potential has disjoint intervals for some combinations of the parameters t and a - it presents a transition in $t_c \approx \sqrt{(1/a)}$. Analytically, it was just recently solved. However we can easily sample from this measure using particle simulations, as shown below. Data was taken from Pimenta (2024).

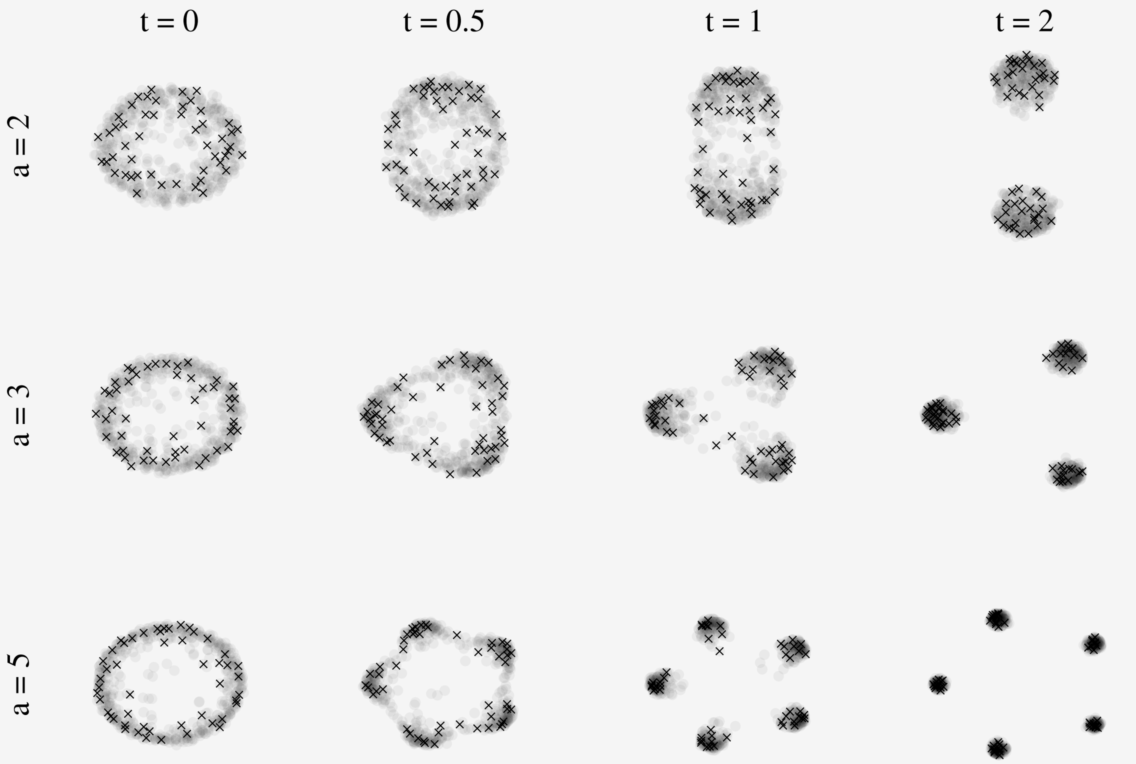

Figure 8 – Shown in the shadow is the mean configurations - in some an approximation of the limit distribution of points, and in the scatter is a random $n = 50$ sample from the distribution. We show configurations for combinations of the parameters $t = 0, 0.5, 1, 2$ from left ro right and $a = 2, 3, 5$ from top to bottom.

3.2 Equilibrium Measures

The description of the eigenvalues as a *Coulomb gas* discussed in Section 3.1 and as a *point process* in Section 2.1 raises some ways to think of this system of particles. Especially, when dealing with the large n limit, the physical interpretation of this system as a interacting gas lead us to the idea of minimizing the energy of the system. In such minimization the particles should find themselves in what we call an equilibrium configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$. In the large n limit, under some mild conditions to be discussed, the counting measure defined with respect to this configuration converges to a continuous measure with compact support. This limit object is what we call an *equilibrium measure*. To clarify the point, suppose we are dealing with the j.p.d.f.

$$p_n(\chi_n) = \frac{\mathrm{e}^{-n^2\mathcal{H}(\chi_n)}}{\mathcal{Z}_n^Q}, \quad \mathcal{H}(\chi_n) = \frac{1}{n}\sum_{i=1}^{n} Q(z_i) + \frac{1}{n^2}\sum_{i\neq j}^{n} \ln\frac{1}{|z_i - z_j|}. \tag{3.2}$$

For any n-configuration $\chi_n = \{z_1, z_2, \dots, z_n\}$, let μ_{χ_n} be the normalized counting measure on $\mathbb{C}$,

$$\mu_{\chi_n} = \frac{1}{n}\sum_{i=1}^{n}\delta_{z_i}, \quad \text{such that} \quad \int_{\mathbb{C}} \mathrm{d}\mu_{\chi_n} = 1.$$

Then

$$\frac{1}{n^2}\sum_{i\neq j} \ln|z_i - z_j|^{-1} + \frac{1}{n}\sum_{i=1}^{n} Q(z_i)$$

may be written in the form

$$\int\int_{t\neq s} \ln|t-s|^{-1}\mathrm{d}\mu_{\chi_n}(t)\mathrm{d}\mu_{\chi_n}(s) + \int Q(s)\mathrm{d}\mu_{\chi_n}(s).$$

This lead us, think of the convergence of $\mu_{\chi_n} \to \mu$, to the energy minimization problem

$$\begin{aligned} E^{(Q)} &= \inf_{\mu\in\mathcal{M}^1(\mathbb{C})}\left[\int\int_{t\neq s} \ln|t-s|^{-1}\mathrm{d}\mu(t)\mathrm{d}\mu(s) + \int Q(s)\mathrm{d}\mu(s)\right] \\ &= \inf_{\mu\in\mathcal{M}^1(\mathbb{C})}\left[I_Q(\mu)\right] \end{aligned}$$

where

$$\mathcal{M}^1(\mathbb{C}) = \left\{\mu \in \text{Borel measures on } \mathbb{C} \,\middle|\, \int \mathrm{d}\mu = 1\right\}.$$

The quantity $E^{(Q)}$ is the *equilibrium energy* of the system and $I_Q(\mu_{\chi_n})$ is the *energy integral*. For μ_Q, the equilibrium measure for the minimization problem above, to be compact, Q the confining potential, must essentially be finite in such set and increase sufficiently fast around infinity. However, to analyze this case, we first consider the simplified version without an external potential. Let $\Sigma \subseteq \mathbb{C}$ be a compact subset of the complex plane and $\mathcal{M}(\Sigma)$ be the collection of all positive Borel measure of compact support in Σ. Let $\mu \in \mathcal{M}(\Sigma)$ be such a measure in the complex plane $\mathbb{C}$.

Definition 3.2. *The associated* ***logarithmic potential*** *of a finite measure μ is defined by*

$$U^{\mu}(z) := \int \ln \frac{1}{|z-t|} \mathrm{d}\mu(t).$$

Also, $I(\mu)$, called the ***logarithmic energy*** *of $\mu \in \mathcal{M}(\Sigma)$ is defined by*

$$I(\mu) := \int \int_{z \neq t} \ln \frac{1}{|z-t|} \mathrm{d}\mu(z)\mathrm{d}\mu(t),$$

the ***energy*** *V of Σ is*

$$V := \inf\{I(\mu) | \mu \in \mathcal{M}(\Sigma)\}.$$

If V is finite, then the measure μ_{Σ}^{V} which minimizes the energy, is called the ***equilibrium measure****.*

This logarithmic potential can be easily interpreted if one reconsiders the Coulomb gas analogy discussed in Section 3.1. Note that the interaction between two particles of gas we considered is logarithmic. Then, taking the empirical counting measure μ_{χ_n},

$$U^{\mu_{\chi_n}}(z) = \int \ln \frac{1}{|z-t|} \mathrm{d}\mu_{\chi_n}(t) = \sum_{i=1}^{n} \ln \frac{1}{|z-\lambda_i|}.$$

This is precisely the potential created by the particles interactions on the point z. There are a few useful properties one can extract about this quantity. A similar argument can also be made to justify the logarithmic energy.

Proposition 3.1. *(SAFF; TOTIK, 2024) Let μ be a finite positive Borel measure of compact support on the complex plane $\mathbb{C}$. Then the logarithmic potential is super-harmonic on the whole plane, that is, it has the following properties.*

1. *$U^{\mu}(z) > -\infty$ for all z,*
2. *U^{μ} is lower semi-continuous,*
3. *$U^{\mu}(z) \geq \frac{1}{2\pi} \int_{-\pi}^{\pi} U^{\mu}(z + r\,\mathrm{e}^{\mathrm{i}\theta})\mathrm{d}\theta$ for all z and all $r \geq 0$.*

Proof. Firstly, property (1) follows directly by (2), since a lower semi-continuous function is everywhere $> -\infty$ and μ has compact support. Property (2) is proven noticing that

$$U^{\mu}(z) = \lim_{M\to\infty} \int \min\left(M, \ln \frac{1}{|z-t|}\right) \mathrm{d}\mu(t).$$

The functions on the right-hand side are continuous and increasing with M, and limit of an increasing sequence of functions must be lower semi-continuous. To prove (3), note that the kernel $\ln(1/|z-t|)$ is super-harmonic in $\mathbb{C}$ for each t. If some F is analytic in a domain D and $\rho > 0$, then $|F(z)|^{\rho}$ is sub-harmonic in D; furthermore, $\ln|F(z)|$ is sub-harmonic in D provided

F is not identically zero. Thus,

$$\frac{1}{2\pi}\int_{-\pi}^{\pi}\ln\frac{1}{|z-r\,\mathrm{e}^{\mathrm{i}\theta}-t|}\mathrm{d}\theta \le \ln\frac{1}{|z-t|}, \quad z,t\in\mathbb{C}.$$

Using the Fubini-Toneli theorem and the inequality above

$$\begin{aligned}\frac{1}{2\pi}\int_{-\pi}^{\pi}U^{\mu}(z+r\,\mathrm{e}^{\mathrm{i}\theta})\mathrm{d}\theta &= \int\frac{1}{2\pi}\int_{-\pi}^{\pi}\ln\frac{1}{|z-r\,\mathrm{e}^{\mathrm{i}\theta}-t|}\mathrm{d}\theta\mathrm{d}\mu(t),\\ &\le \int\ln\frac{1}{|z-t|}\mathrm{d}\mu(t) = U^{\mu}(z).\end{aligned}$$

□

For technical reasons we also state Lemma 3.1 regarding the logarithmic energy that will become useful in a later result but to which we do not provide a proof.

Lemma 3.1. *(SAFF; TOTIK, 2024) Let $\mu = \mu_1 - \mu_2$ be a signed Borel measure with compact support and total mass $\mu(\mathbb{C}) = 0$. Suppose further that each of the positive measures μ_1 and μ_2 has finite logarithmic energy. Then the logarithmic energy of μ*

$$I(\mu) = \int\int\ln\frac{1}{|z-t|}\mathrm{d}\mu(z)\mathrm{d}\mu(t),$$

is non-negative and it is zero if and only if $\mu \equiv 0$.

Suppose that there is a unique measure $\mu_{\Sigma}^{V} \in \mathcal{M}(\Sigma)$ for which the minimum energy $V = \inf\{I(\mu)|\mu \in \mathcal{M}(\Sigma)\}$ is finite and reachable. Such a measure μ_{Σ}^{V} is called the *equilibrium distribution* or *equilibrium measure* of the compact set Σ. These concepts have a natural electrostatic interpretation. Suppose Σ is a conductor and charged particles are placed on it. If the force between the particles is proportional to the reciprocal of their distance and Q is some external field to which they interact to, then μ_{Σ}^{V} is the configuration which minimizes the energy, or maximizes the entropy, of the system. This measure gives the distribution to which this particles are in equilibrium.

We should at this point reintroduce the external potential Q. Most often than not, an external force will change the minimum energy state and equilibrium configurations. Particles tend to accumulate at the minimum of such potential. The reason why the distribution will not be a single point in space is that the logarithmic energy, from the interactions, explodes when particles are close by. For this case, a weighted version of the problem was developed. The main difference now is that Σ is not required to be compact; we only impose that there is no positive mass around infinity. There is a natural way to enforce these conditions. Let $\Sigma \subset \mathbb{C}$ be a closed set and $w : \Sigma \to [0,\infty)$. This function w is called a *weight function* on Σ. Certain weight functions make it so that there is no positive mass around infinity; we call such function *admissible*.

Definition 3.3. *Define also the logarithmic capacity to be* $\operatorname{cap}(\Sigma) = e^{-V}$ *with* $V = \inf\{I(\mu) | \mu \in \mathcal{M}(\Sigma)\}$. *A weight function* $w = e^{-nQ(z)}$ *is said to be* ***admissible*** *is it satisfies the following three conditions*

1. *Q is upper semi-continuous;*
2. *$\Sigma_0 := \{z \in \Sigma | Q > -\infty\}$ has positive capacity;*
3. *if Σ is unbounded, then* $\liminf_{|z|\to\infty} \frac{Q(z)}{2\ln(|z|)} > 1$ *as* $|z| \to \infty$, $z \in \Sigma$.

Let $\mathcal{M}(\Sigma)$ be the collection of all positive unit Borel measures μ with support in Σ. Our definition for the weighted energy integral now reads

$$I_Q(\mu) = \int U^{\mu}(s)\mathrm{d}\mu(s) + \int Q(s)\mathrm{d}\mu(s),$$

for $\mu \in \mathcal{M}(\Sigma)$ and is recovered if both integrals exist and are finite. Note that this is to say that

$$\begin{aligned} I_Q(\mu) &= \int U^{\mu}(s)\mathrm{d}\mu(s) + \int Q(s)\mathrm{d}\mu(s), \\ &= \int\int_{s\neq t} \ln\frac{1}{|s-t|}\mathrm{d}\mu(s)\mathrm{d}\mu(t) + \frac{1}{2}\left[\int Q(s)\mathrm{d}\mu(s) + \int Q(t)\mathrm{d}\mu(t)\right], \\ &= \int\int_{s\neq t}\left(\ln\frac{1}{|s-t|} + \frac{Q(t)}{2} + \frac{Q(s)}{2}\right)\mathrm{d}\mu(s)\mathrm{d}\mu(t), \\ &= \int\int_{s\neq t}\kappa^Q(s,t)\mathrm{d}\mu(s)\mathrm{d}\mu(t) \end{aligned}$$

converges. Denote

$$\phi^Q(x) = Q(x) - \ln(x^2+1)$$

and note that

$$\ln|t-s|^{-1} \geq -\frac{1}{2}\ln(1+t^2) - \frac{1}{2}\ln(1+s^2).$$

Therefore

$$\kappa^Q(s,t) \geq \frac{1}{2}\phi^Q(t) + \frac{1}{2}\phi^Q(s) \geq c_Q > -\infty$$

since $\lim_{|x|\to\infty}\phi^Q(x) = \infty$ and $\phi^Q(x) \geq c_Q > -\infty$. Thus, $I_Q \geq c_Q > -\infty$ for every $\mu \in \mathcal{M}^1(\mathbb{C})$. Remember that we are dealing with weight functions of the form $w(z) = e^{-nQ(z)}$ for $z \in \mathbb{C}$ and we want Q to be such that the weighted logarithmic energy is finite - the weight is admissible. Then, by the definition, $Q : \Sigma \to (\infty, \infty]$ must be lower semi-continuous. To avoid degeneracy, we imposed that $Q < \infty$ in some set of positive capacity. Furthermore, we must have that $\liminf_{|z|\to\infty}\frac{Q(z)}{2\ln(|z|)} > 1$ such that the potential energy grows faster than the interaction logarithmic energy. Remember that this is similar to the condition imposed on the orthogonal polynomial ensembles in Definition 2.4. On the next section we move on to the matter of determining the *equilibrium measure* μ_Q for a given external field Q.

Theorem 3.1. *(SAFF; TOTIK, 2024) Let w be an admissible weight function on the closed set Σ and let*

$$V_w := \inf\{I_w(\mu) | \mu \in \mathcal{M}(\Sigma)\}.$$

Then, V_w is finite and there exists a unique element $\mu_w \in$ such that $I_w(\mu_w) = V_w$.

On the next section we move on to the matter of determining the *equilibrium measure* μ_Q for a given external field Q.

Example 3.2. Take the Coulomb gas measure described in Section 3.1, and restrict the conductor Σ to be the real line under the external potential

$$Q(x) = tx^{2m}.$$

By finding configurations that minimizes the Hamiltonian, one also finds the more likely configurations. These configurations, when $n \to \infty$ converges to a measure μ_Q that minimizes the logarithmic energy. It can be shown that this measure is given by

$$\operatorname{supp} \mu_Q(x) = [-a, a], \quad \mu_Q(x) = \frac{mt}{\pi}\sqrt{a^2 - x^2}\, h(x),$$

where

$$a := \left(mt \prod_{l=1}^{m} \frac{2l-1}{2l} \right) \quad \text{and} \quad h(x) := x^{2m-2} + \sum_{j=1}^{m-1} x^{2m-2-2j} a^{2j} \prod_{l=1}^{j} \frac{2l-1}{2l}.$$

See Deift (1999) for details. Note that this simplifies to the Gaussian case if $Q(x) = x^2/2$ ($t, m = 1$). In such case we already shown that the equilibrium measure is defined by the Semicircle distribution

$$\operatorname{supp} \mu_Q(x) = [-\sqrt{2}, \sqrt{2}], \quad \mu_Q(x) = \frac{1}{\pi}\sqrt{2 - x^2}.$$

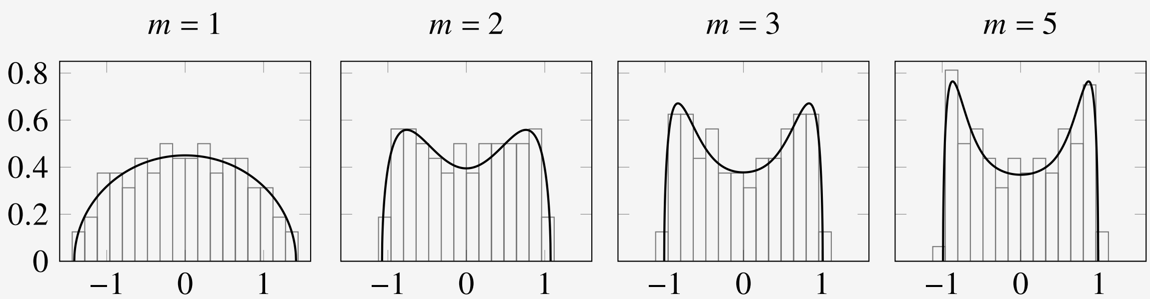

Figure 9 – Four conditions for the Monic potential, the equilibrium measure is expressed by the line plot. For all cases we used $t = 1$ and set $m = 1, 2, 3, 5$ from left to right. The bin-plot is the result of the simulation of the equivalent Coulomb Gas taken from Pimenta (2024).

3.2.1 Euler-Lagrange

Associated with the variational problem introduced in Section 3.2 defined by

$$
\begin{aligned}
E^w &= \inf_{\mu \in \mathcal{M}^1} I_Q(\mu), \\
&= \inf_{\mu \in \mathcal{M}^1} \left[\int \int \ln|t-s|^{-1} \mathrm{d}\mu(t)\mathrm{d}\mu(s) + \int Q(s)\mathrm{d}\mu(s) \right], \\
&= I_Q(\mu_Q),
\end{aligned} \tag{3.3}
$$

one defines the so-called *Euler-Lagrange* type variational equations. They are important as they provide a tool to solve these equations for the equilibrium measure μ_Q.

Theorem 3.2. *(Variational Equations) The equilibrium measure μ_Q for the given variational problem in* (3.3)*, where $\Sigma = \mathbb{C}$ satisfies the following conditions. There exists a real constant l such that*

1.
$$\int \left[2 \int \ln|x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) \right] \mathrm{d}\tilde{\mu}(x) \geq l$$
for all $\tilde{\mu} \in \mathcal{M}^1(\mathbb{C})$ of compact support with $I_Q(\tilde{\mu}) < \infty$.

2.
$$2 \int \ln|x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) = l$$
for μ_Q-almost everywhere.

Conversely, if $\mu \in \mathcal{M}^1(\mathbb{C})$ has compact support, satisfies conditions (1) *and* (2) *above, and $I_Q(\mu) < \infty$, then $\mu = \mu_Q$.*

Proof. The argument of the proof is based on (DEIFT, 1999).

($\Longrightarrow$) We recall that μ_Q has compact support and $I_Q(\mu_Q) = E^w < \infty$. Suppose $\mu^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I_Q(\mu^*) < \infty$. Then, defining a convex combination

$$\mu_t = t\mu^* + (1-t)\mu_Q = \mu_Q + t(\mu^* - \mu_Q) \in \mathcal{M}^1(\mathbb{C}), \quad 0 \leq t \leq 1,$$

we have

$$
\begin{aligned}
I_Q(\mu_t) = \int \int \ln|t-s|^{-1} \mathrm{d}\mu_Q(t)\mathrm{d}\mu_Q(s) + \int Q(s)\mathrm{d}\mu_Q(s) \\
+ 2t \int \int \ln|t-s|^{-1} \mathrm{d}\mu_Q(t)\mathrm{d}(\mu^* - \mu_Q)(s) \\
+ t \int Q(s)\mathrm{d}(\mu^* - \mu_Q)(s) \\
+ t^2 \int \int \ln|t-s|^{-1} \mathrm{d}(\mu^* - \mu_Q)\mathrm{d}(\mu^* - \mu_Q)(s).
\end{aligned}
$$

Each term on the right is finite since μ_Q, μ^* have compact support and $I_Q(\mu_Q), I_Q(\mu^*) < \infty$. As $I_Q(\mu_t) \geq I_Q(\mu_Q)$ for $0 \leq t \leq 1$ by assumption of minimality and Lemma 3.1, is is necessary that

$$\int \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) \right] \mathrm{d}(\mu^* - \mu_Q)(x) \geq 0$$

and therefore

$$\int \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) \right] \mathrm{d}\mu^*(x) \geq l \tag{3.4}$$

where

$$\int \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) \right] \mathrm{d}\mu_Q(x) := l$$

proving the first assertion in (1). Now, let

$$B = \left\{ x \,\middle|\, 2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) < l \right\}.$$

Then, B must be a bounded set of $\mathbb{C}$. Suppose $\mu^*(B) > 0$ and set

$$\mu_B^* = \frac{\chi_B}{\mu^*(B)} \mu^*$$

where χ_B is the characteristic function of the set B. Then $\mu_B^* \in \mathcal{M}^1(\mathbb{C})$ has compact support and $I_Q(\mu^*) < \infty$. inserting $\mathrm{d}\mu_B^*$ in (3.4), we find

$$l \leq \int \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) \right] \mathrm{d}\mu_B^*(x) < l$$

a contradiction. Therefore,

$$\mu^*(B) = \mu^* \left(\left\{ x \,\middle|\, 2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) < l \right\} \right) = 0$$

for all measures $\mu^* \in \mathcal{M}^1(\mathbb{C})$ with support compact and $I_Q(\mu^*) < \infty$. in particular, it is true for $\mu^* = \mu_Q$. Since,

$$\begin{aligned} 0 &= \int \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) - l \right] \mathrm{d}\mu_Q(x) \\ &= \int_{\mathbb{C}-B} \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu_Q(y) + Q(x) - l \right] \mathrm{d}\mu_Q(x) \end{aligned}$$

and (2) follows.

($\Longleftarrow$) If, now, $\mu \in \mathcal{M}^1$ satisfies (1) and (2), $I_Q(\mu) < \infty$, and suppose μ has compact support. Then write $\mu_Q = \mu + (\mu_Q - \mu)$. We have, as before

$$\begin{aligned} I_Q(\mu) &\geq I_Q(\mu_Q) \\ &= I_Q(\mu) + \int \left[2 \int \ln |x-y|^{-1} \mathrm{d}\mu(y) + Q(x) - l \right] \mathrm{d}(\mu_Q - \mu)(x) \\ &\qquad + \int \int \ln |t-s|^{-1} \mathrm{d}(\mu_Q - \mu)(t) \mathrm{d}(\mu_Q - \mu)(s). \end{aligned}$$

Using (2),

$$\int \left[2\int \ln|x-y|^{-1} \mathrm{d}\mu(y) + Q(x) - l \right] \mathrm{d}\mu_Q(x) - l \geq l - l = 0$$

but also the third term is strictly positive by Lemma 3.1 unless it is identically null. If, however, $\mu_Q - \mu \neq 0$, then we showed that

$$I_Q(\mu) \geq I_Q(\mu_Q) > I_Q(\mu),$$

a contradiction. Thus, if μ satisfies $(1),(2)$, then it must be μ_Q. □

If, in addition we assume that

$$\mathrm{d}\mu_Q(z) = \mu(z)\mathrm{d}^2 z \tag{3.5}$$

for some continuous function μ of compact support then the previous theorem takes the following simpler form.

Theorem 3.3. *(Strong Variational Equations) Suppose that* $\mathrm{d}\mu_Q = \mu(z)\mathrm{d}^2 z$ *for some continuous function* μ *of compact support. Then the conditions of Theorem 3.2 is replaced by*

1.

$$2\int \ln|z-t|^{-1} \mathrm{d}\mu_Q(t) + Q(z) \geq l$$

for all $z \in \mathbb{C}$.

2.

$$2\int \ln|z-t|^{-1} \mathrm{d}\mu_Q(t) + Q(z) = l$$

on $\{z : \mu(z) > 0\}$.

We mentioned in (3.5) the possibility of knowing the density μ of the equilibrium measure μ_Q with respect to the Lebesgue measure. We will now determine μ. The support the equilibrium measure is an important quantity in determining the equilibrium measure corresponding to a weight function w. Suppose for example this support to be bounded with a smooth border and w to be a continuous function. Then, by the previous theorem, inside of the support U^{μ_w} coincides with $-Q$ plus some constant. Thus, we get a way of determining the the equilibrium logarithmic potential. From this potential, there is a way to recover the equilibrium measure. Let $\mathrm{d}^2 z$ denote the two-dimensional area measure on $\mathbb{C}$ and Δ denote the usual Laplacian operator

$$\Delta F = \frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2}.$$

Then, we have the theorem below which makes a statement about the density function.

Theorem 3.4. *(SAFF; TOTIK, 2024) If in a region R the potential U^{μ} has continuous second partial derivatives, then μ is absolutely continuous in R with respect to two-dimensional area measure $\mathrm{d}^2 z$, and on R we have the formula*

$$\mathrm{d}\mu = -\frac{1}{4\pi}\Delta U^{\mu}\mathrm{d}^2 z$$

3.2.2 Radially Symmetric Potentials

In this section we will calculate the equilibrium measure for a large class of matrices in the *random normal ensemble of matrices*. Especially, we are going to focus on ensembles such that the external potential Q is radially symmetric. This symmetry will be essential in the calculations that follows in Chapter 5. The measure calculated in this section is used to perform asymptotic analysis on the partition function for these normal ensembles. We begin by defining such ensemble.

Definition 3.4. *The* ***random normal matrix*** *ensemble $(\mathcal{S}, \mathcal{P})$ is constructed by considering the set of n-sized normal matrices with complex entries $\mathcal{S} = \mathcal{N}_n(\mathbb{C})$ equipped with a p.d.f.*

$$\mathcal{P}(\hat{M}) = \mathrm{e}^{-n\,\mathrm{Tr}\,Q(\hat{M})}$$

where Q is some polynomial function and $\hat{M} \in \mathcal{N}_n(\mathbb{C})$.

We first note that because we are dealing with random matrices we rewrite this p.d.f. as the following expression

$$\mathcal{P}(\hat{M}) = \mathrm{e}^{-n\,\mathrm{Tr}\,Q(\hat{M})} = \mathrm{e}^{-n\sum_{i=1}^{n} Q(\lambda_i)}.$$

Together with Theorem 2.2, we say that, on the complex plane we are considering the measure on the set of eigenvalues given by

$$\mathrm{d}p_n(z_1, \cdots, z_n) = \frac{1}{\mathcal{Z}_n^Q}\prod_{j<k}^{n}|z_j - z_k|^2 \prod_{j=1}^{n}\mathrm{e}^{-nQ(z_j)}\,\mathrm{d}^2 z_j.$$

We now define the potential Q to further specify the working model.

Definition 3.5. *We deal with $\mathcal{E} = (\mathcal{S}, \mathcal{P})$ a* ***normal ensemble*** *in which Q, which defines the weight function $w(x) = \mathrm{e}^{-nQ(x)}$ is some complex potential that satisfies*

1. *The potential Q is radially symmetric, i.e.*

$$Q(z) = q(|z|)$$

for some $q : \mathbb{R} \to \mathbb{R}$;

2. *The potential Q grows sufficiently fast, i.e.*

$$\liminf_{|z|\to\infty} \frac{Q(z)}{2\ln(|z|)} > 1;$$

3. *Furthermore, assume Q to be smooth, sub-harmonic in $\mathbb{C}$, and strictly sub-harmonic in a neighborhood of the support of the equilibrium measure μ_Q.*

These assumptions are essential for what follows. To require radial symmetric is to say that the integral of a function of Q with respect to the area measure can be exchanged by an angle component and a line component, as all directions are homogeneous. The growth requirement comes from the need for the confinement of particles in the system - note that the interaction energy has logarithmic behavior. The conditions on sub-harmonicity can be though of as convexity and will ensure uniqueness of the minima and connectedness of the support of the equilibrium measure. We will be dealing mainly with the real potential q as we make use of the symmetry of Q. Therefore, we also state the following proposition.

Proposition 3.2. *The real potential $q\colon \mathbb{R} \to \mathbb{R}$ defined as in Definition 3.5 is such that $rq'(r)$ is **increasing** in the complex plane and **strictly increasing** in the support of the equilibrium measure μ_Q.*

Proof. Because Q is sub-harmonic on the plane, i.e. $\Delta Q(z) \geq 0$ for all $z \in \mathbb{C}$, then

$$\Delta Q(z) = \frac{1}{r}(rq'(r))' \geq 0 \implies rq'(r) \text{ is increasing,}$$

and strictly increasing in the support of the equilibrium measure μ_Q by the same relation with a strict inequality. □

A very important feature of the partition function of normal ensembles is that it can be expressed as a summation of norming constants of the correspondent orthogonal polynomial. This was proved on Theorem 2.3. We have stated a few properties of the model surrounding the potential $Q(z) = q(|z|)$. Knowing this to be an orthogonal polynomial ensemble, there is still a question of what are exactly the orthogonal polynomials with respect to the weight function defined. Consider $\{P_m(z) = \sum_{j=0}^{m} a_j z^j\}_m^{(w)}$ a family of orthogonal polynomials with respect to w where P_m has degree m.

Lemma 3.2. *The set $\{z^j\}_j^{(w)}$ is the proper orthogonal family of polynomials in relation to the weight function $w(z) = \mathrm{e}^{-nQ(z)}$ defined as in as in Definition 3.5.*

Proof. We know by the assumption of orthogonality that

$$\int P_k(x)P_m(x)w(x)\mathrm{d}x = \langle P_k, P_m\rangle_w = \begin{cases} 0 & \text{if } m = k, \\ 1 & \text{if } m \neq k. \end{cases}$$

Let us take some arbitrary base of the polynomial space $\{z^j\}_j^{(w)}$. We apply the Gram Schmidt scheme to get the proper orthogonality with respect to the weight w. If $P_m = x^m$, then the orthogonal element of the basis $\tilde{P}_m$ should be

$$\tilde{P}_m = P_m - \sum_{j=0}^{m-1} \frac{\langle P_m, P_j \rangle_w}{\langle P_j, P_j, \rangle_w} P_j,$$

but notice that

$$\begin{aligned}
\langle P_m, P_j \rangle_w &= \int_{\mathbb{C}} z^m \bar{z}^j \, \mathrm{e}^{-nQ(z)} \, \mathrm{d}^2 z, \\
&= \int_0^{\infty} \int_0^{2\pi} r^{m-j} \, \mathrm{e}^{\mathrm{i}\theta(m-j)} \, \mathrm{e}^{-nq(r)} \, r \mathrm{d}r \mathrm{d}\theta, \\
&= \int_0^{2\pi} \mathrm{e}^{\mathrm{i}\theta(m-j)} \, \mathrm{d}\theta \int_0^{\infty} r^{m-j+1} \, \mathrm{e}^{-nq(r)} \, \mathrm{d}r, \\
&= 0, \quad \text{since } m \neq j.
\end{aligned}$$

Therefore, we have that $\{z^j\}_j^{(w)}$ is the proper orthogonal family. □

We also want to determine the support of the equilibrium measure for the eigenvalues of the planar normal ensembles. For that, we make an ansatz and use Theorem 3.3 to confirm it.

Theorem 3.5. *Let w be as in Definition 3.5. If r_0 is the largest solution of $rq'(r) = 0$ and r_1 is the smallest solution of $rq'(r) = 2$ then the support of μ_Q is given by the ring $\mathcal{S} = \{z | r_0 \leq |z| \leq r_1\}$ and $\mathrm{d}\mu_Q$ is given by*

$$\mathrm{d}\mu_Q(z) = \frac{1}{4\pi} (rq'(r))' \mathbb{1}_{\mathcal{S}} \mathrm{d}r \mathrm{d}\phi, \quad z = r \, \mathrm{e}^{\mathrm{i}\phi}.$$

The set $\mathcal{S}$ can be either an annulus (if $r_0 > 0$) or a disk (if $r_0 = 0$).

Proof. We will start on the *ansatz* that this support has two profiles; it can either be a disk centered at the origin with radius r_1 denoted by $D(r_1)$ or it can be an annulus centered at the origin with internal radius r_0 and external radius r_1 denoted by $A(r_0, r_1)$. Let $\mu \in \mathcal{M}^1(\mathbb{C})$ be a measure such that $\mathrm{d}\mu = \frac{1}{4\pi}(rq'(r))' \mathbb{1}_{\mathcal{S}} \mathrm{d}\phi \mathrm{d}r$. We want that

$$\int_{\mathbb{C}} \mathrm{d}\mu = 1,$$

that is,

$$r_1 q'(r_1) - r_0 q'(r_0) = 2.$$

Naturally, with r_0, r_1 as in the theorem, we have that this relation is satisfied. Now, by Theorem 3.3, if μ satisfies that

$$\begin{aligned}
2U^{\mu}(z) &\geq l - Q(z), \quad z \in \mathbb{C} \\
2U^{\mu}(z) &= l - Q(z), \quad z \in \{z \mid r_0 \leq |z| \leq r_1\}.
\end{aligned}$$

with $I_Q(\mu) < \infty$, then $\mu_Q = \mu$. We first prove the formula

$$\int_{\pi}^{-\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} \mathrm{d}\phi = \begin{cases} 2\pi \ln \frac{1}{r}, & |z| \le r, \\ 2\pi \ln \frac{1}{|z|}, & |z| > r. \end{cases}$$

We start by considering the case $|z| > r$. We write

$$\ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\theta}|} = \ln \frac{1}{r} + \ln \frac{1}{\left|\frac{z}{r} - \mathrm{e}^{\mathrm{i}\theta}\right|} = \ln \frac{1}{r} + \Re \ln \frac{1}{\omega - s}$$

where we set $s = \mathrm{e}^{\mathrm{i}\theta}$ and $\omega = z/r$. Then, $\mathrm{d}\theta = \mathrm{d}s/(\mathrm{i}s)$ and by Cauchy's theorem

$$\int_{\pi}^{-\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} \mathrm{d}\phi = \ln \frac{1}{r} + \Re \int_{\pi}^{-\pi} \ln \frac{1}{\omega - s} \frac{1}{\mathrm{i}s} \mathrm{d}s = -2\pi \ln(|z|).$$

In a very similar fashion, for $|z| \le r$,

$$\ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\theta}|} = \ln \frac{1}{|z|} + \ln \frac{1}{\left|\mathrm{e}^{-\mathrm{i}\theta} - \frac{r}{z}\right|} = \ln \frac{1}{|z|} + \Re \ln \frac{1}{s - \omega}$$

where we set $s = \mathrm{e}^{-\mathrm{i}\theta}$ and $\omega = r/z$. Then, $\mathrm{d}\theta = -\mathrm{d}s/(\mathrm{i}s)$ and by Cauchy's theorem again

$$\int_{\pi}^{-\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} \mathrm{d}\phi = \ln \frac{1}{|z|} - \int_{\pi}^{-\pi} \ln \frac{1}{s - \omega} \frac{1}{\mathrm{i}s} \mathrm{d}s = -2\pi \ln(r).$$

Now, with this, we calculate that

$$\begin{aligned} U^{\mu}(z) &= \frac{1}{4\pi} \int_{\mathbb{R}} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} (rq'(r))' \mathbb{1}_{\mathcal{S}} \mathrm{d}r \mathrm{d}\phi, \\ &= \frac{1}{4\pi} \int_{r_0}^{r_1} \int_{-\pi}^{\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} (rq'(r))' \mathrm{d}r \mathrm{d}\phi, \\ &= \frac{1}{4\pi} \int_{r_0}^{r_1} (rq'(r))' \int_{-\pi}^{\pi} \ln \frac{1}{|z - r\,\mathrm{e}^{\mathrm{i}\phi}|} \mathrm{d}\phi \mathrm{d}r. \end{aligned}$$

And, then, for $|z| < r_0$

$$\begin{aligned} U^{\mu}(z) = \frac{1}{2} \int_{r_0}^{r_1} (rq'(r))' \ln\left(\frac{1}{r}\right) \mathrm{d}r &= -\frac{1}{2}(\ln(r_1) r_1 q'(r_1) - \ln(r_0) r_0 q'(r_0) - q(r_1) + q(r_0)), \\ &= -\frac{1}{2}(2\ln(r_1) - q(r_1)) - \frac{1}{2} q(r_0), \end{aligned}$$

for $|z| > r_1$

$$U^{\mu}(z) = \frac{1}{2} \int_{r_0}^{r_1} (rq'(r))' \ln\left(\frac{1}{z}\right) \mathrm{d}r = \ln \frac{1}{|z|},$$

and, finally, for $r_0 \le |z| \le r_1$

$$\begin{aligned} U^{\mu}(z) &= \frac{1}{2} \int_{r_0}^{|z|} (rq'(r))' \ln\left(\frac{1}{r}\right) \mathrm{d}r + \frac{1}{2} \int_{|z|}^{r_1} (rq'(r))' \ln\left(\frac{1}{z}\right) \mathrm{d}r, \\ &= -\frac{1}{2}(2\ln(r_1) - q(r_1)) - \frac{1}{2} q(|z|). \end{aligned}$$

Define $l = -(2\ln(r_1) - q(r_1))$, then

$$2U^{\mu}(z) = \begin{cases} l - q(r_0), & |z| < r_0, \\ 2\ln\frac{1}{|z|}, & |z| > r_1, \\ l - q(|z|), & r_0 \leq |z| \leq r_1. \end{cases}$$

We, as in Theorem 3.3, need to prove that for $z \in \mathbb{C}$, we have $2U^{\mu}(z) \geq l - q(|z|)$ while on $r_0 \leq |z| \leq r_1$ we have $2U^{\mu}(z) = l - q(|z|)$.

Firstly, for $0 < |z| < r_0$ we need that $q(r_0) - q(|z|) \geq 0$ such that $2U^{\mu}(z) = l - q(r_0)$ is less than or equal to $l - q(|z|)$. This is true because of the growing condition on q. Moreover, we need for $|z| > r_1$, that $-2\ln|z| + q(|z|) \geq -2\ln(r_1) + q(r_1)$. The same argument on the growing of q and ln holds to state that this is also true. Therefore, $\mu_Q = \mu$ and we have found our equilibrium measure. □

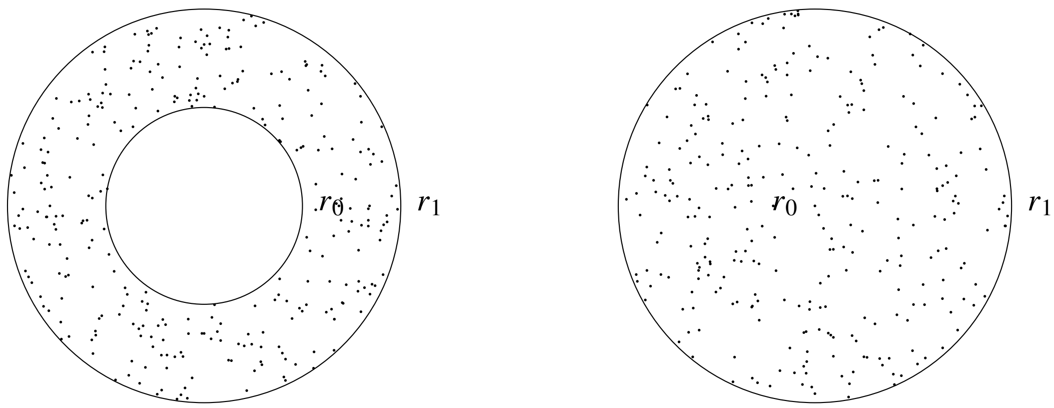

Figure 10 – The figure depicts the two topological cases for the support of the equilibrium measure - the droplet. On the left $r_0 > 0$ defining the annulus and on the right $r_0 = 0$ defining the disk.

Definition 3.6. *We call* ***Droplet*** *the support of μ_Q, which in our case is the set of all complex values z such that $r_0 \leq |z| \leq r_1$.*

3.3 Solving for the Free Energy

Remember the expression for the Gibbs-Boltzmann measure of Coulomb gases

$$\mathrm{d}p_n(\chi_n) = \frac{1}{\mathcal{Z}_n^Q} \mathrm{e}^{-\beta n^2 \mathcal{H}(\chi_n)} \prod_{i=1}^{n} \mathrm{d}^d x_i.$$

As discussed, this expression refers to a thermodynamic fluid of particles with positions $\chi_n = x_1, x_2, \ldots, x_n$ on a conductor Σ, in equilibrium at some inverse temperature β an submitted to a confining potential Q and a repulsive logarithmic term of interaction. Because of the term βn^2, taking the asymptotic $n \to \infty$ means at the same time zero temperature and the thermodynamic limit. Physically, understanding the equilibrium positions would be equivalent to minimizing the

Free Energy $F = -(1/\beta)\ln \mathcal{Z}_n^Q$. Let's describe the idea behind such minimization. This section *is not rigorous*, its only purpose is to connect the idea of equilibrium problems with the asymptotics we will perform in Chapter 5 and elucidate where the expression for the free energy comes from. This argument, in more detail, can be found in Livan, Novaes and Vivo (2018).

Lets first take the normalized counting measure, already introduced in Section 2.1,

$$\mathfrak{n}(x) = \frac{1}{n}\sum_{i=1}^{n}\delta(x - x_i).$$

Note that using this function we write the unity as

$$1 = \int \mathrm{d}[\mathfrak{n}(x)]\delta\left[\mathfrak{n}(x) - \frac{1}{n}\sum_{i=1}^{n}\delta(x-x_i)\right],$$

we rewrite our partition function from (3.2) as

$$\mathcal{Z}_n^Q = C_{n,\beta}\int \mathrm{d}[\mathfrak{n}(x)]\int_{\mathbb{R}^{dn}}\prod_{j=1}^{n}\mathrm{d}x_j\,\mathrm{e}^{-\beta n^2\mathcal{H}(\chi_n)}\,\delta\left[\mathfrak{n}(x) - \frac{1}{n}\sum_{i=1}^{n}\delta(x-x_i)\right].$$

We do not want to deal with summations, in this context, integrals can be much easier to deal with. Then, we also want to convert the expression for the Hamiltonian. The confining term is easier, we write

$$\frac{1}{n}\sum_{i=1}^{n}Q(x) = \frac{1}{n}n\int_{\mathbb{R}^d}n(x)Q(x)\mathrm{d}x.$$

For the interaction term, one must first note that

$$\frac{1}{2n^2}\sum_{i\neq j}\ln|x_i - x_j| = \frac{1}{2n^2}\left[\sum_{i,j}\ln|x_i - x_j| - \sum_i \Delta x_i\right]$$

where we call Δx_i is the cutoff distance and refers to the term $i = j$ in the discrete case and two neighboring charges approximating in the continuous case. This should ensure that the functional will still produce a finite result. Now we write

$$\frac{1}{2n^2}\sum_{i\neq j}\ln|x_i - x_j| = \frac{1}{2n^2}n^2\int\int_{\mathbb{R}^{2d}}\mathrm{d}x\mathrm{d}y\,\mathfrak{n}(x)\mathfrak{n}(y)\ln|x-y| - \frac{1}{2n^2}n\int_{\mathbb{R}^d}\mathrm{d}x\,\mathfrak{n}(x)\ln\Delta(x).$$

and then

$$\begin{aligned}\mathcal{Z}_n^Q &= C_{n,\beta}\int \mathrm{d}[\mathfrak{n}(x)]\,\mathrm{e}^{-\beta n^2\mathcal{H}(\mathfrak{n}(x))}\int_{\mathbb{R}^{dn}}\prod_{j=1}^{n}\mathrm{d}x_j\delta\left[\mathfrak{n}(x) - \frac{1}{n}\sum_{i=1}^{n}\delta(x-x_i)\right]\\ &=: C_{n,\beta}\int \mathrm{d}[\mathfrak{n}(x)]\,\mathrm{e}^{-\beta n^2\mathcal{H}(\mathfrak{n}(x))}\,I_n[\mathfrak{n}(x)],\end{aligned}$$

where

$$\mathcal{H}(\mathfrak{n}(x)) = \int_{\mathbb{R}^d}n(x)Q(x)\mathrm{d}x + \frac{1}{2}\int\int_{\mathbb{R}^{2d}}\mathrm{d}x\mathrm{d}y\,\mathfrak{n}(x)\mathfrak{n}(y)\ln|x-y| - \frac{1}{2n}\int_{\mathbb{R}^d}\mathrm{d}x\,\mathfrak{n}(x)\ln\Delta(x).$$

Physically it is possible to think of $I_n[\mathfrak{n}(x)]$ as a count of how many microstates χ_n are compatible with the macrostate $\mathfrak{n}(x)$. The logarithm of this quantity is proportional to the entropy of such system. Rewrite this function as

$$\begin{aligned} I_n[\mathfrak{n}(x)] &= \int \mathrm{d}[\mathfrak{n}(x)] \int_{\mathbb{R}^{dn}} \prod_{j=1}^{n} \mathrm{d}x_j \exp\left[\mathrm{i}n \int \mathrm{d}x \mathfrak{n}(x)\hat{\mathfrak{n}}(x) - \mathrm{i} \int \mathrm{d}x \hat{\mathfrak{n}}(x) \sum_{i=1}^{n} \delta(x - x_i) \right] \\ &= \int \mathrm{d}[\mathfrak{n}(x)] \exp\left[\mathrm{i}n \int_{\mathbb{R}^{dn}} \mathrm{d}x \mathfrak{n}(x)\hat{\mathfrak{n}}(x) \right] \left[\int_{\mathbb{R}^d} \mathrm{d}y \, \mathrm{e}^{\mathrm{i}\int_{\mathbb{R}^d} \mathrm{d}x \hat{\mathfrak{n}}(x)\delta(x-x_i)} \right]^n \\ &= \int \mathrm{d}[\hat{\mathfrak{n}}(x)] \, \mathrm{e}^{nS[\hat{\mathfrak{n}}(x)|\mathfrak{n}(x)]}, \end{aligned} \tag{3.6}$$

where

$$S[\hat{\mathfrak{n}}(x)|\mathfrak{n}(x)] = \mathrm{i} \int_{\mathbb{R}^{dn}} \mathrm{d}x \mathfrak{n}(x)\hat{\mathfrak{n}}(x) + \ln \int_{\mathbb{R}^d} \mathrm{d}y \, \mathrm{e}^{-\mathrm{i}\hat{\mathfrak{n}}(x)}.$$

Notice here that (3.6) is of a similar form from the one required by the Laplace asymptotic technique discussed in Chapter 4. We estimate it, if S has a maxima with strictly negative second derivative, by the evaluation at its maximum point. We find

$$\begin{aligned} 0 = \frac{\partial S}{\partial \hat{\mathfrak{n}}(x)} &= \mathrm{i}\mathfrak{n}(x) - \mathrm{i} \frac{\mathrm{e}^{-\mathrm{i}\hat{\mathfrak{n}}(x)}}{\int_{\mathbb{R}^d} \mathrm{d}y \, \mathrm{e}^{-\mathrm{i}\hat{\mathfrak{n}}(y)}} \\ \implies -\mathrm{i}\hat{\mathfrak{n}}(x) &= -\ln(\mathfrak{n}(x)) - \ln \int_{\mathbb{R}^d} \mathrm{d}y \, \mathrm{e}^{-\mathrm{i}\hat{\mathfrak{n}}(y)}, \end{aligned}$$

for the critical counting function. Then, substituting,

$$I_n[\mathfrak{n}(x)] \approx \exp\left[-n \int_{\mathbb{R}^{nd}} \mathrm{d}x \mathfrak{n}(x) \ln(\mathfrak{n}(x)) \right],$$

which has the form of the Shannon entropy of the density $\mathfrak{n}(x)$. There is a classical argument that states that $\Delta(x) \approx c/(n\mathfrak{n}(x))$, i.e., the higher the density of particles around a point x, the smaller the average distance between them. Thus,

$$\begin{aligned} \int_{\mathbb{R}^d} \mathrm{d}x \mathfrak{n}(x) \ln \Delta(x) &\approx \int_{\mathbb{R}^d} \mathrm{d}x \mathfrak{n}(x) \ln\left(\frac{c}{n\mathfrak{n}(x)} \right) \\ &= \int_{\mathbb{R}^d} \mathrm{d}x \mathfrak{n}(x) \left(\ln c - \ln n + \ln \mathfrak{n}(x) \right) \\ &= \int_{\mathbb{R}^d} \mathrm{d}x \mathfrak{n}(x) \ln c - \int_{\mathbb{R}^d} \mathrm{d}x \mathfrak{n}(x) \ln n + \int_{\mathbb{R}^d} \mathrm{d}x \mathfrak{n}(x) \ln \mathfrak{n}(x). \end{aligned}$$

Return now to the partition function and use the relations found to write

$$\begin{aligned} \mathcal{Z}_n^Q &\approx C_{n,\beta} \int \mathrm{d}[\mathfrak{n}(x)] \exp\left[-\beta n^2 \mathcal{H}(\mathfrak{n}(x)) \right] \exp\left[-n \int_{\mathbb{R}^{nd}} \mathrm{d}x \mathfrak{n}(x) \ln(\mathfrak{n}(x)) \right], \\ &\approx C_{n,\beta} \int \mathrm{d}[\mathfrak{n}(x)] \exp\left[-\beta n^2 \mathfrak{F}_0[\mathfrak{n}(x)] + \frac{\beta n}{2} \mathfrak{F}_1[\mathfrak{n}(x)] + \frac{\beta}{2} n \ln(n) - \frac{\beta}{2} n \ln(c) \right] \exp\left[-n \mathfrak{F}_1[\mathfrak{n}(x)] \right], \\ &\approx C_{n,\beta} \int \mathrm{d}[\mathfrak{n}(x)] \exp\left[-\beta n^2 \mathfrak{F}_0[\mathfrak{n}(x)] + \frac{\beta}{2} n \ln(n) + n \left(\frac{\beta}{2} - 1 \right) \mathfrak{F}_1[\mathfrak{n}(x)] - \frac{\beta}{2} n \ln(c) \right], \end{aligned}$$

where

$$\mathfrak{F}_0[\mathfrak{n}(x)] := \int_{\mathbb{R}^d} \mathfrak{n}(x)Q(x)\mathrm{d}x + \frac{1}{2}\int\int_{\mathbb{R}^{2d}} \mathrm{d}x\mathrm{d}y\mathfrak{n}(x)\mathfrak{n}(y)\ln|x-y|,$$
$$\mathfrak{F}_1[\mathfrak{n}(x)] := \int_{\mathbb{R}^d} \mathrm{d}x\mathfrak{n}(x)\ln\mathfrak{n}(x).$$

Note that, now, one again perform a saddle point estimation for the asymptotic in a critical point $\mathfrak{n}^*(x)$ to get

$$\mathcal{Z}_n^Q \approx \exp\left[-\beta n^2\mathfrak{F}_0[\mathfrak{n}^*(x)] + \frac{\beta}{2}n\ln(n) + n\left(\frac{\beta}{2}-1\right)\mathfrak{F}_1[\mathfrak{n}^*(x)] - \frac{\beta}{2}n\ln(c)\right]$$

and finally for the Free Energy

$$\ln(\mathcal{Z}_n^Q) = -\beta n^2\mathfrak{F}_0[\mathfrak{n}^*(x)] + \frac{\beta}{2}n\ln(n) + n\left(\frac{\beta}{2}-1\right)\mathfrak{F}_1[\mathfrak{n}^*(x)] - \frac{\beta}{2}n\ln(c) + O(n).$$

Obtaining a expression for the Free Energy when $\beta = 2$, rigorously, is the main goal for Chapter 5.

CHAPTER 4

ASYMPTOTICS

There is a sense in the previous chapters that we want to study distributions of eigenvalues when our matrices are large enough - be that for feasibility of computations or approximation of infinite dimension operators. We also have mentioned that our main interest here is the invariant ensembles, specially the orthogonal polynomial ensemble. Because of the expression of the univariate eigenvalue density as the determinant of a orthogonal polynomial kernel function, we change our focus to the study of these polynomials. Studying the distribution of eigenvalue for large matrices is equivalent, in this sense, to studying the regime of large n for the kernels and consequently, for the orthogonal polynomials.

For some of those polynomials, a integral representation is available, allowing us to use asymptotic methods on integrals to solve for the large n behavior. These methods revolve around the idea that we want to integrate a function that accumulates its value in some discrete set of points in the domain as we take some of its parameters to infinity. This is usually an exponential with negative exponent depending on some parameter n that we identify with the size of our initial matrices. Then, as n grows, the approximation of this integral as proportional to the value of the function on these point of maxima decreases in error. This method roughly describes the *Steepest Descent* technique, which will be described in this chapter. The sections that precedes this descriptions are the building blocks for the more sophisticated technique; we go through *Watson's lemma* and the *Laplace technique*.

Before going through asymptotic techniques in integrals, we briefly describe a technical result we will make use of in Chapter 5. This is a quadrature technique or, more specifically, the *Euler-Maclaurin* formula. This formula will later allow us to approximate very complicated summations as integrals that we can compute analytically. Techniques and structuring here are based on the book by Olver *et al.* (2010) and Alpert (1999) for quadrature and on the lecture notes by Silva (2021) for asymptotics on integrals.

4.1 Quadrature Technique

Before doing asymptotics on integrals we discuss a technique that will become important when approximating large summations: quadratures, i.e., approximating integrals by summations and vice-versa. One can see that asymptotics on integrals can then also perform a role on discrete summations. We first start briefly introducing a useful tool in the development of the formula we wish to use: the Bernoulli numbers B_n, polynomials $B_n(x)$ and periodic polynomials $\tilde{B}_n(x)$. The Bernoulli numbers B_n are a sequence of rational numbers that frequently appears in the Taylor series expansion of trigonometric-type functions. These number can also be seen as special values obtained by the Bernoulli polynomials $B_n(x)$. This is because these polynomials can be written explicitly as

$$B_k(x) = \sum_{n=0}^{k} \binom{k}{n} B_n x^{k-n},$$

then, we must have that $B_k(0) = B_k$. More interestingly for us is the fact that these polynomials satisfy the differential equation

$$B_n'(x) = nB_{n-1}(x), \quad n = 1, 2, \cdots \tag{4.1}$$

Finally, the periodic Bernoulli polynomial is simply the Bernoulli polynomials evaluated at the fractional part of the argument, i.e., $\tilde{B}_n(x) = B_n(x - \lfloor x \rfloor)$. Now, let B_n and $B_n(x)$ denote the n-th Bernoulli number and polynomial, respectively, and $\tilde{B}_n(x)$ the n-th Bernoulli periodic function $B_n(x - \lfloor x \rfloor)$. Assume that a, m and n are integers such that $n > a$, $m > 0$, and $f^{(2m)}(x)$ is absolutely integrable over $[a, n]$. Then,

Theorem 4.1. *(Euler-Maclaurin Formula, (OLVER et al., 2010)) Following the context given above,*

$$\sum_{j=a}^{n} f(j) = \int_a^n f(x)\mathrm{d}x + \frac{1}{2}(f(a) + f(n)) + \sum_{s=1}^{m-1} \frac{B_{2s}}{(2s)!}\left(f^{(2s-1)}(n) - f^{(2s-1)}(a)\right) + \int_a^n \frac{B_{2m} - \tilde{B}_{2m}(x)}{(2m)!} f^{(2m)}(x)\mathrm{d}x.$$

This identity is called the *Euler-Maclaurin* formula. It can be extremely useful when estimating the asymptotic of the summation on the function f when its integral can be more easily evaluate. In essence, this is a quadrature technique; a numerical integration: the approximation of an integral $\int f(x)\mathrm{d}x$ of some function f by a discrete summation $\sum w_i f(x_i)$ over a set of points x_i with some weight function w_i. For example, in the classical Riemann sums, one approximates the integral $\int f(x)\mathrm{d}x$ over some interval $[a, b]$ with the limit of the summation $\sum f(x_i^*)\Delta x_i$, where $x_i^* \in [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. Different methods can be generated by choosing x_i^* differently.

The trapezoidal rule, for example, is a Riemann sum where x_i^* is chosen to be the mean point $x_i^* = (x_{i-1} + x_i)/2$. In this way, we would estimate the area using trapezoids, as the name suggests. Then, we write

$$\begin{aligned}\int_a^n f(x)\mathrm{d}x &\approx \sum_{j=0}^{n} f(x_j^*) = \sum_{j=0}^{n} \frac{f(x_i + x_{i-1})}{2},\\ &= \sum_{j=1}^{n-1} f(x_j^*) + \frac{1}{2}\left(f(n) + f(0)\right) = \sum_{j=0}^{n-1} f(x_j^*) + \frac{1}{2}\left(f(n) - f(n)\right).\end{aligned}$$

Of course, this has some error term. Suppose that for some (x_i, x_{i-1}) we have the same k-derivatives at both extreme points for every $k \geq 1$, i.e., the function is constant. Then, there is no error in the approximation. If however, the two extreme points differ only on the k-th derivative, there is some curvature defined by this difference where the error comes from. Therefore, there is indication of some error function given in respect to higher order derivatives.

Proof. Following the computations in Alpert (1999), take a function $f \in C^p(\mathbb{R})$ for $p \geq 1$. To derive the Euler-Maclaurin for the interval $[a,b]$, let $h = (b-a)/n$ and $c = a + jh$, then we express

$$\int_a^b f(x)\mathrm{d}x = \sum_{j=0}^{n-1} \int_{a+jh}^{a+(j+1)h} f(x)\mathrm{d}x$$

where we use (4.1) and integration by parts to rewrite each integral as

$$\begin{aligned}\int_c^{c+h} f(x)\mathrm{d}x &= h\int_0^1 B_0(1-x) f(c+xh)\mathrm{d}x\\ &= -h\frac{B_1(1-x)}{1!} f(c+xh)|_0^1 + h^2 \int_0^1 \frac{B_1(1-x)}{1!} f'(c+xh)\mathrm{d}x,\\ &\vdots\\ &= -\sum_{r=0}^{p-1} \frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!} f^{(r)}(c+xh)|_0^1 + h^{p+1}\int_0^1 \frac{h^{r+1}B_p(1-x)}{(p)!} f^{(p)}(c+xh)\mathrm{d}x,\\ &= h\frac{f(c)+f(c+h)}{2} - \sum_{r=0}^{p-1} \frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!}\left[f^{(r)}(c+h) - f^{(r)}(c)\right]\\ &\qquad + h^{p+1}\int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)\mathrm{d}x.\end{aligned}$$

where one can start to notice, for instance, the trapezoidal rule creeping out on the first term. Notice that it was necessary to use that $-B_1(0) = B_1(1) = 1/2$ and $B_n(0) = B_n(1) = B_n$ for $n \neq 1$. Returning to the summation, we rearrange terms as

$$\begin{aligned}h\left[\frac{f(a)}{2} + f(a+h) + \cdots + f(b-h) + \frac{f(b)}{2}\right] &= \frac{1}{2}\left(f(b) - f(a)\right) + \sum_{j=0}^{n-1} f(a+jh),\\ = \int_a^b f(x)\mathrm{d}x + \sum_{r=0}^{p-1} \frac{h^{r+1}B_{r+1}(1-x)}{(r+1)!}\left[f^{(r)}(b) - f^{(r)}(a)\right] &- h^{p+1}\int_0^1 \frac{B_p(1-x)}{(p)!} f^{(p)}(c+xh)\mathrm{d}x.\end{aligned}$$

□

4.2 Asymptotic on Integrals

The techniques we will expand on here refers to a specific type of complex function $\phi : \mathbb{C} \to \mathbb{R}$ in the regime where $\lambda \to \infty$. Those functions are of the form

$$\phi(\lambda) := \int_\Gamma g(s)\, \mathrm{e}^{-\lambda f(s)}\, \mathrm{d}s, \tag{4.2}$$

where Γ is a given contour in the plane and f, g are suitable functions. Suitable here means that we impose g and f to be continuous (or analytic); g is integrable; f have global minima with zero derivative and strictly positive second derivative at this point of minima. There is also a more technical regularity condition on f to ensure the approximations hold to still be made clear.

Example 4.1. (Airy Equation) Function such as in (4.2) are not uncommon. Consider the function that satisfies the equation $\psi''(x) = x\psi(x)$. Taking the Fourrier transform we have

$$-\omega^2\Phi(x) = \mathrm{i}\Phi'(x)$$

for which we solve with $\Phi(x) = \mathrm{e}^{-\mathrm{i}\omega^3/3}$ and then take the inverse Fourrier Transform to get

$$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \mathrm{e}^{\mathrm{i}/3\omega^3 + \mathrm{i}x\omega}\, \mathrm{d}\omega, \quad x \in \mathbb{R}.$$

The imaginary part of the argument is an odd function and therefore zero when integrated. The real part is expressed

$$\phi(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \cos\left(\frac{\omega^3}{3} + x\omega\right) \mathrm{d}\omega.$$

This is the Airy function. For this integral, the techniques applied in this chapter are of great use to understand its behavior as $\omega \to \infty$.

Consider $\phi : \mathbb{R} \to \mathbb{R}$, f with a single global minima and $\Gamma \equiv \mathbb{R}$ - for simplicity; the argument's idea should hold for other choices given the appropriate considerations. If ϕ is as in (4.2), and satisfying the conditions given for such function, then take x^* the single minimal critical point for f. We first note that if that is true, then, for any $|\epsilon| > 0$, the following limit should hold:

$$\lim_{\lambda\to\infty} \frac{\mathrm{e}^{-\lambda(f(x^*))}}{\mathrm{e}^{-\lambda(f(x^*-\epsilon))}} = \lim_{\lambda\to\infty} \mathrm{e}^{-\lambda(f(x^*)-f(x^*-\epsilon))} = \lim_{\lambda\to\infty} \mathrm{e}^{\lambda|c|} \to +\infty.$$

That is, the main contribution for the integral comes from a single point. The main idea then is to separate the integral in two regions: A first interval $(x^* - c/\sqrt{\lambda}, x^* + c/\sqrt{\lambda}), c > 0$, including this critical point and another interval that in theory should contribute only in error for the integral asymptotic. We start by approximating f around its global maximum by Taylor's theorem

$$f(x) = f(x^*) - \frac{1}{2}|f''(x^*)|(x - x^*)^2 + \epsilon(x - x^*)$$

and then perform a change of variables $\frac{x}{\sqrt{\lambda}} = s - x^*$, so that

$$\int_{x^*-\frac{c}{\sqrt{\lambda}}}^{x^*+\frac{c}{\sqrt{\lambda}}} g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}(s) = \frac{\mathrm{e}^{-\lambda f(x^*)}}{\sqrt{\lambda}} \int_{-c}^{c} g\left(x^* + \frac{x}{\sqrt{\lambda}}\right) \mathrm{e}^{\frac{1}{2}|f''(x^*)|x^2 - \epsilon\left(\frac{x^3}{\sqrt{\lambda}}\right)}\,\mathrm{d}x.$$

Now, using the technical assumption of regularity on the function f, one can argue that the error term ϵ goes to zero sufficiently fast, leaving a Gaussian Integral. Moreover, one needs to argue that the integral outside this small neighborhood of x^* is exponentially smaller then $\mathrm{e}^{-\lambda f(x^*)}$. This should follow by an argument where the asymptotic of the integral

$$\int_{x^*+\frac{c}{\sqrt{\lambda}}}^{\infty} |g(s)|\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}(s) \leq \mathrm{e}^{-\lambda f(x^*+c/\sqrt{\lambda})}\,||g||_{L^1}$$

is exponentially smaller than the previous region.

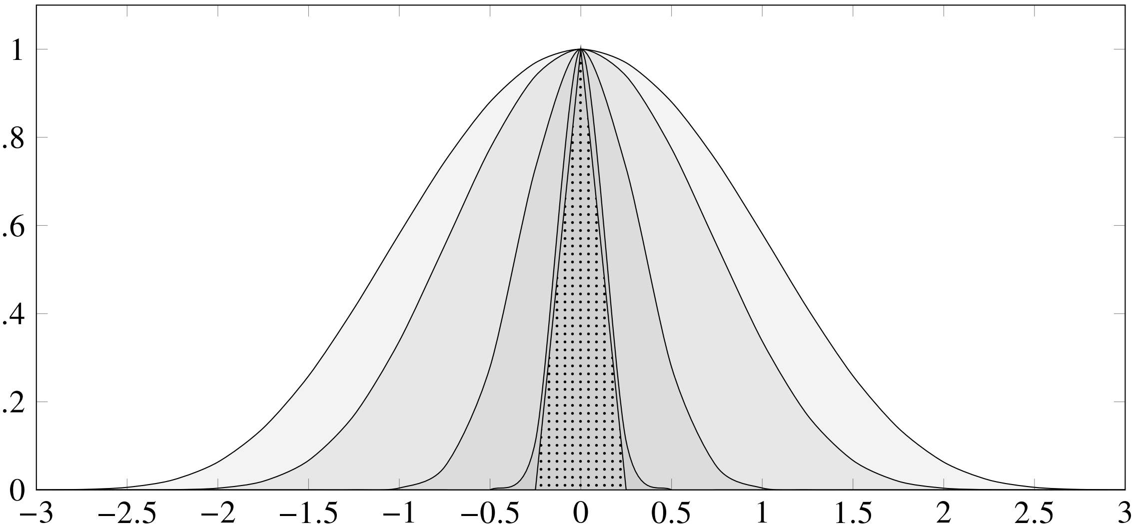

Figure 11 – In the Plot it is represented the plot for $\mathrm{e}^{-\lambda(\cosh(x)-1)}$ and an increasing $\lambda = 1, 2, 10, 70$ - darker represent larger values of the parameter. The dotted patter represents $\mathrm{e}^{-\lambda x^2}$ for large lambda.

With this, we only note a few details of interest. Firstly, more precise error estimates can be obtained if one uses higher order terms of the Taylor Series expansion of f. For maximizer x^* which have null second derivative (a double critical point), the Laplace Approximation can be used expanding to the third order terms of the Taylor Series $f(x) = f(x^*) = \frac{1}{6}f'''(x^*)(x-x^*)^3 + \epsilon(x-x^*)$ and making the change of variables $t/\sqrt[3]{n} = x - x^*$ instead. This argument and result could also be used on curves on the plane. This is motivated from the fact that every (smooth) curve is locally a straight line. This will lead us to the method of *Steepest Descent*. These next subsections are in the spirit of proving a bit more formally the expansion given by the *Laplace Method* and introducing the mentioned *Steepest Descent*.

4.2.1 Watson's Lemma

We start this topic on the real line by looking at the functions $F : \mathbb{R} \to \mathbb{R}$ of the form

$$F(\lambda) = \int_0^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t, \tag{4.3}$$

which one may think of as the Laplace transform of a function f supported on $[0,T]$. Note that F fits well with the models we described previously, and so one should expect the main contribution to come from the minima of the exponent function. Let us formalize the conditions for such expectation and the consequent result.

Theorem 4.2. *(Watson's Lemma) Take $f \in L^1(0,T)$ and suppose that for some $\delta > 0$ it is of the form*

$$f(t) = t^{\alpha} f_0(t), \quad 0 < t \le \delta,$$

where $\alpha \in (-1,\infty)$ and f_0 is continuously differentiable on $[0,\delta]$ with $f_0(0) \neq 0$. Then the function F in (4.3) satisfies

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1+O(\lambda^{-1}))$$

when $\lambda \to \infty$. Also, Γ is the usual Gamma function.

We must first mention that a weak estimation of the integral is given by

$$\int_0^T f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \le ||f||_{L^1(0,T)}$$

where we only use the fact that $f \in L^1(0,T)$. However, we have more information on the behavior of the function in the section $0 < t < \delta$. Because of that fact, it is expected that in this set we could refine the approximation.

Proof. We start by decomposing the function in two parts

$$F(\lambda) = \int_0^{\delta} f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + \int_{\delta}^{T} f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t =: (1) + (2).$$

For (2), we use the same weak estimate as commented before,

$$\int_{\delta}^{T} f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t \le \mathrm{e}^{-\lambda\delta}\int_{\delta}^{T} |f(t)|\mathrm{d}t = O(\mathrm{e}^{-\lambda\delta}\,||f||_{L^1(\delta,T)}).$$

We keep it this way as we have no further information on the f in this section of the real line. This leave us with

$$F(\lambda) = \int_0^{\delta} f(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + O(\mathrm{e}^{-\lambda\delta}\,||f||_{L^1(\delta,T)}).$$

Now, we remind of the condition of continuous differentiability and the hypothesis on $f(t)$ to get $f(t) = t^{\alpha}(f_0(0)+r(t))$ for some differentiable error function $r(t)$ satisfying $|r(t)| \le ||f_0'||_{L^\infty(0,\delta)}t$ with $||f_0'||_{L^\infty(0,\delta)} = \inf\{c \ge 0 \mid \mu|_{(0,\delta)}(\{|f_0'| > c\}) = 0\}$ as usual. Then, write (1) as

$$(1) = \int_0^{\delta} t^{\alpha}(f_0(0)+r(t))\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = f_0(0)\int_0^{\delta} t^{\alpha}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t + \int_0^{\delta} t^{\alpha} r(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t.$$

For these two integrals we simplify the problem by rewriting

$$\int_0^{\delta} t^{\alpha}\,\mathrm{e}^{-\lambda t} = \int_0^{\infty} t^{\alpha}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t - \int_{\delta}^{\infty} t^{\alpha}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - \int_{\delta}^{\infty} t^{\alpha}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t,$$

$$\begin{aligned}\left|\int_0^{\delta} t^{\alpha} r(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right| &\le ||f_0'||_{L^\infty(0,\delta)} \int_0^{\delta} t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t,\\ &= ||f_0'||_{L^\infty(0,\delta)} \left(\int_0^{\infty} t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t - \int_{\delta}^{\infty} t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right),\\ &= ||f_0'||_{L^\infty(0,\delta)} \left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - \int_{\delta}^{\infty} t^{\alpha+1}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right).\end{aligned}$$

We are now only missing one ingredient to get the result, the order of

$$G(\lambda,\beta) := \int_{\delta}^{\infty} t^{\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t, \quad \lambda > 1, \quad \beta > -1.$$

Using Cauchy-Schwartz we estimate

$$\begin{aligned}|G(\lambda,\beta)|^2 &\le \left(\int_{\delta}^{\infty} \mathrm{e}^{-\lambda t}\,\mathrm{d}t\right)\left(\int_{\delta}^{\infty} t^{2\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right),\\ &= \left(\frac{e^{-\lambda\delta}}{\lambda}\right)\left(\int_{\delta}^{\infty} t^{2\beta}\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right),\\ &\overset{(u=\lambda t)}{=} \left(\frac{e^{-\lambda\delta}}{\lambda}\right)\left(\int_{\delta}^{\infty} \mathrm{e}^{-u}\,u^{2\beta}\frac{1}{\lambda^{2\beta+1}}\mathrm{d}u\right),\\ &= \frac{\mathrm{e}^{-\lambda\delta}}{\lambda^{2\beta+2}} M_{\beta}\end{aligned}$$

where M_β is some constant independent of λ. That leave us with

$$G(\lambda,\beta) = O\left(\frac{\mathrm{e}^{-\lambda\delta/2}}{\lambda^{\beta+1}}\right) = O\left(\mathrm{e}^{-\epsilon\lambda}\right)$$

where $\epsilon > 0$ depends on β and δ. Finally we rewrite

$$\int_0^{\delta} t^{\alpha}\,\mathrm{e}^{-\lambda t} = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} - G(\lambda,\alpha) = \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(\mathrm{e}^{-\epsilon\lambda}\right),$$

$$\begin{aligned}\left|\int_0^{\delta} t^{\alpha} r(t)\,\mathrm{e}^{-\lambda t}\,\mathrm{d}t\right| &\le ||f'||_{L^\infty(0,\delta)}\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} - G(\lambda,\alpha+1)\right)\\ &= ||f'||_{L^\infty(0,\delta)}\left(\frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} + O\left(\mathrm{e}^{-\epsilon\lambda}\right)\right) = O\left(\lambda^{-\alpha-2}\right),\end{aligned}$$

and

$$F(\lambda) = O\left(\lambda^{-\alpha-2}\right) + f_0(0)\left(\frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} + O\left(\mathrm{e}^{-\epsilon\lambda}\right)\right) + ||f||_{L^1(\delta,T)}\,\mathrm{e}^{-\delta\lambda}.$$

By the dominance of the exponential error, we get the result

$$F(\lambda) = \frac{f_0(0)\Gamma(\alpha+1)}{\lambda^{\alpha+1}}(1 + O(\lambda^{-1})).$$

□

In the estimation above, the Gamma function and the factor of λ^{-1} is the result of integrating the linear exponent of $e^{-\lambda t}$. If we, however, change this exponent to be of the general type $-\lambda t^d$ we still have a similar result on the behavior of the function.

Proposition 4.1. *For $T, S > 0$, define the function*

$$G(\lambda) := \int_{-S}^{T} g(t)\, e^{-\lambda t^2}\, dt$$

and assume $g(x)$ absolutely integrable in $[-S,T]$, of class C^1 in a fixed neighborhood of the origin. Then the function satisfies

$$G(\lambda) = \frac{g(0)\Gamma(1/2)}{\lambda^{1/2}} + O\left(\frac{1}{\lambda^{3/2}}\right)$$

when $\lambda \to \infty$ and where Γ is the Gamma function.

Proof. The proof is a reduction of the problem to Watson's Lemma. Assume, $T > S$ and write

$$G(\lambda) = \int_{-S}^{S} g(t)\, e^{-\lambda t^2}\, dt + \int_{S}^{T} g(t)\, e^{-\lambda t^2}\, dt.$$

The second integral we know to be of order

$$\int_{S}^{T} |g(t)|\, e^{-\lambda t^2}\, dt \le ||g||_\infty\, e^{-\lambda S^2},$$

exponential on λ, so we drop it. For the first integral we rewrite

$$\int_{-S}^{S} g(t)\, e^{-\lambda t^2}\, dt = \int_{0}^{S} (g(t) + g(-t))\, e^{-\lambda t^2}\, dt,$$

$$\overset{(u=t^2)}{=} \frac{1}{2}\int_{0}^{S^2} \left(\frac{g\left(\sqrt{u}\right) + g\left(-\sqrt{u}\right)}{\sqrt{u}}\right) e^{-\lambda u}\, du$$

where now we use Watson's Lemma to finish the proof. □

Example 4.2. Take the function called Gamma given by the expression

$$\Gamma(x) = \int_{0}^{\infty} e^{-t}\, t^{x-1} dt. \tag{4.4}$$

We are interested in what happens to the function when we take $x \to +\infty$. To see what happens, let us fit it into Equation 4.2. Rewrite (4.4) as

$$\int_{0}^{\infty} e^{-t + x\log(t) - \log(t)}\, dt.$$

This lead us to identify ϕ as $-\log(t)$, which would give us $\phi'(t) < 0$, that is, we would have its minimum at $t = +\infty$. However $e^{\log(t)} = t^1$ is not integrable near the origin. Something need to change about the identification. Let us introduce a variable to rescale the integral:

$xs = t$, s.t.,

$$\Gamma(x) = x^x \int_0^\infty \mathrm{e}^{-\lambda(s-\log(s))}\,\mathrm{e}^{-s}\,\mathrm{d}s,$$

where we have used $\lambda = x-1$. Thus we identify $\phi(s) = s - \log(s)$ and $g(s) = \mathrm{e}^{-s}$. In fact, now ϕ has a global minimum at $s = 1$ and $\phi''(1) = 1 \neq 0$. Hence

$$\begin{aligned}\Gamma(x) &= \frac{\sqrt{2\pi} x^x\,\mathrm{e}^{-x}}{(x-1)^{(-1/2)}}(1+O(x^{-1})),\\ &= \sqrt{2\pi} x^{x+1/2}\,\mathrm{e}^{-x}(1+O(x^{-1})),\end{aligned}$$

which is called the Stirling formula.

4.2.2 Laplace Method

A seemly extension of the type of function we were dealing with are functions of the type

$$\Phi(\lambda) := \int_a^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \tag{4.5}$$

for $a, b \in \mathbb{R} \cup \{-\infty, \infty\}$ and $a < b$ and a well behaved function g. For now, ϕ is a real-valued function. To begin the study let us make some assumptions. The first one is that a is finite, that means that we set $a = 0$ without loss of generality. The second assumption is that ϕ is a increasing function on $[0, b]$, so that $\phi(0) < \phi(t)$ for any t in the interval. With that we note that the coefficient $\mathrm{e}^{-\lambda\phi(t)}/\mathrm{e}^{-\lambda\phi(0)}$ decreases exponentially, that is, the integral is accumulated as the neighborhood of 0, i.e.

$$\Phi(\lambda) \approx \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t.$$

As we know $\phi(t)$ to be increasing, its first non zero derivative must be of even order, so that when we expanded the function by Taylor,

$$\phi(t) \approx \phi(0) + \phi_{(2k+1)}(0)t^{2k+1},$$

for some $k \geq 0$ and $|t| < \delta$ where $\phi_{(2k+1)} := \phi_{(2k+1)}/(2k+1)!$. As ϕ_{2k+1}, the $(2k+1)$th-derivative, must be positive we also have that, as $\lambda \longrightarrow \infty$,

$$\begin{aligned}\Phi(\lambda) &\approx \mathrm{e}^{-\lambda\phi(0)} \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi_{(2k+1)}(t)\cdot t^{2k+1}}\,\mathrm{d}t,\\ &\overset{(u=t(\lambda\phi_{(2k+1)}(0))^{1/(2k+1)})}{\approx} \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{(2k+1)}(0))^{\frac{1}{2k+1}}} \int_0^\infty \mathrm{e}^{-u^{2k+1}}\,\mathrm{d}u,\\ &\overset{(s=u^{2k+1})}{\approx} \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(2k+1)(\lambda\phi_{(2k+1)}(0))^{\frac{1}{2k+1}}} \int_0^\infty \mathrm{e}^{-s}\,\mathrm{d}s.\end{aligned}$$

where the integral can be evaluated as a Gamma function. More than anything we need to focus on an important assumption that was made: that ϕ has a global minima at one of the endpoints of integration. If that were not the case, for example if $a < 0 < b$ and $\phi(t)$ had a global minima at $t = 0$ we now have to consider the expansion

$$\phi(t) \approx \phi(0) + \phi_{2k}(0)t^{2k}$$

which lead us to the expression

$$\Phi(\lambda) \approx \mathrm{e}^{-\lambda\phi(0)} \frac{g(0)}{(\lambda\phi_{2k}(0))^{\frac{1}{2k}}} \int_{-\infty}^{\infty} \mathrm{e}^{-u^{2k}} \, \mathrm{d}u.$$

This differentiates of the previously result more importantly by the limits of integration. This is closely related to physical boundaries of the problem. Before enunciating the proper Laplace Method we state the Implicit Function Theorem, which will become important in a series of results that follows.

Theorem 4.3. *(Implicit Function Theorem) Assume that S is an open subset of $\mathbb{R}^2$ and that $F : S \to \mathbb{R}$ is a function of class C^1. Assume also that (a,b) is a point in S such that*

$$F(a,b) = 0, \quad \textit{and} \quad D_y F(a,b) \neq 0.$$

Then

1. *There exist $r_0, r_1 > 0$ such hat for every $x \in \mathbb{R}$ such that $|x-a| < r_0$, there exists a unique $y \in \mathbb{R}$ such that $|y-b| < r_1$*

$$F(x,y) = 0.$$

   That is, defines a function $y = f(x)$ for $x \in \mathbb{R}$ near a and y close to b.

2. *Moreover, the function $f : B(r_0,a) \to B(r_1,b) \subset \mathbb{R}$ from item 1 is of class C^1, and its derivatives may be determined by differentiating the identity*

$$F(x, f(x)) = 0$$

   and solving to find the partial derivatives of f.

Now we turn to our main interest.

Theorem 4.4. *(Laplace's Method, endpoint contributions) Let Φ be as in (4.5), with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = a$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^{\alpha} g_0(t)$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[a, a+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is twice continuously differentiable on $[a, a+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies *an estimate of the form*

$$\Phi(\lambda) = \frac{1}{\lambda^{\alpha+1}} \frac{g_0(a)\Gamma(\alpha+1)}{\phi'(a)^{\alpha+1}} \, \mathrm{e}^{-\lambda\phi(a)} \left(1 + O(\lambda^{-1})\right)$$

as $\lambda \to \infty$.

Proof. Start by assuming, without loss of generality, that $a = 0$ and dividing the integral in two intervals, one in the region where we have information about the behavior of g and another in the rest of the interval,

$$\Phi(\lambda) = \int_0^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_\delta^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = (1) + (2).$$

Knowing that ϕ has a global minima at $t = 0$ we assume, changing δ as needed, that $\phi(t) > \phi(\delta)$ for any $t > \delta$. That give us a simple bound for (2):

$$|(2)| \leq \int_\delta^b |g(t)|\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \leq \mathrm{e}^{-\lambda\phi(\delta)} \int_\delta^b |g(t)|\mathrm{d}t \leq \mathrm{e}^{-\lambda\phi(\delta)}\,||g||_{L^1[0,b]}.$$

We now get back to (1) and start by studying the difference given by $\phi(t) - \phi(0) = s$ in a similar way that we would think of a change of variable from t to s. Define the multivariate function

$$H(t,s) := \phi(t) - \phi(0) - s$$

such that $H(0,0) = 0$ and $\partial_t H(0,0) \neq 0$. This means we are in the condition to use Theorem 4.3 to say that there exists a unique bijective and continuously differentiable function $t(s) := t$ such that

$$\begin{cases} \phi(t(s)) - \phi(0) - s = 0, \\ t'(0) = \frac{\partial_s H(0,0)}{\partial_t H(0,0)} = \frac{1}{\phi'(0)}. \end{cases}$$

If we set $T = t^{-1}(\delta)$ we then have for (1)

$$\int_0^\delta g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t = \int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s.$$

Now, to apply the assumption on g we write $g(t(s)) = t^\alpha(s) g_0(t(s))$. As t is C^2 and vanishes at the origin, using its Taylor expansion, we have

$$t(s) = st'(0)(1 + r(s)) = \frac{s}{\phi'(0)}(1 + r(s)).$$

Joining both results,

$$\begin{aligned} \int_0^T g(t(s))\,\mathrm{e}^{-\lambda\phi(t(s))}\,t'(s)\mathrm{d}s &= \mathrm{e}^{\lambda\phi(0)} \int_0^T g(t(s))\,\mathrm{e}^{-\lambda(\phi(t(s))-\phi(0))}\,t'(s)\mathrm{d}s \\ &= \mathrm{e}^{\lambda\phi(0)} \int_0^T t^\alpha(s) g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s \\ &= \frac{\mathrm{e}^{\lambda\phi(0)}}{\phi'(0)^\alpha} \int_0^T s^\alpha (1 + r(s))^\alpha g_0(t(s))\,\mathrm{e}^{-\lambda s}\,t'(s)\mathrm{d}s \end{aligned}$$

where, if we set $f(s) := s^\alpha(1 + r(s))^\alpha g_0(t(s))$ we apply Watson's Lemma to finish the result. □

Theorem 4.5. *(Laplace's Method, interior contributions) Let Φ be as in (4.5), with a finite and $g \in L^1(a,b)$. Assume that ϕ has a unique global minimum at $t = c \in (a,b)$. In addition, suppose that for some $\delta > 0$ and some $\alpha > -1$, the function g takes the form*

$$g(t) = (t-a)^\alpha g_0(t)$$

for $|t-a| < \delta$ and g_0 continuously differentiable in $[c-\delta, c+\delta]$ and $g_0(a) \neq 0$, whereas ϕ is three times continuously differentiable on $[c-\delta, c+\delta]$ with $\phi'(a) > 0$. Then the function Φ satisfies an estimate of the form

$$\Phi(\lambda) = \mathrm{e}^{-\lambda\phi(c)}\left(\frac{\sqrt{2\pi}g(c)}{\sqrt{|\phi''(c)|}\lambda^{1/2}} + O(\lambda^{-3/2})\right)$$

as $\lambda \to \infty$.

Proof. Although the proof may be similar there is an important distinction in the solution. Suppose we wanted to try the same approach. We first assume, w.l.g., that c = 0. Then, we write

$$\begin{aligned}
\Phi(\lambda) &= \int_a^b g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \\
&= \int_a^{-\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t + \int_{\delta}^{b} g(t)\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \\
&=: (1) + (2) + (3).
\end{aligned}$$

For (1) and (3) the result is as it was before

$$\begin{cases}
|(1)| \le \int_\delta^a |g(-t)|\,\mathrm{e}^{-\lambda\phi(-t)}\,\mathrm{d}t \le \mathrm{e}^{\lambda\phi(-\delta)}\,||g||_{L^1[a,b]}, \\
|(3)| \le \int_\delta^b |g(t)|\,\mathrm{e}^{-\lambda\phi(t)}\,\mathrm{d}t \le \mathrm{e}^{\lambda\phi(\delta)}\,||g||_{L^1[a,b]}.
\end{cases}$$

For (2) we try to define a function

$$H(t,s) := \phi(t) - \phi(0) - s^2.$$

However we can not apply Theorem 4.3 as the partial derivative $\partial_t H = \phi'$ vanishes at $t = 0$. A nice fact about this indetermination is that it does not come from the fact that there are no solutions to $H(t,s) = 0$ but from the fact that there are two. To solve this we will use a technique called *blow up*. First, we introduce a variable $v := v(t)$ such that $vs = t$. Now, we redefine our function of interest to be

$$L(s,v) := \frac{\phi(sv) - \phi(0)}{s^2} - 1.$$

for $s \in (-\delta, \delta) - \{0\}$. We think this function is also singular but note that, using the Taylor Expansion of ϕ we write

$$\phi(t) = \frac{\phi''(0)}{2}t^2 + t^3 R(t)$$

for some error function R, which leave us with

$$L(s,v) = \frac{\phi''(0)}{2}v^2 + sv^3R(sv) - 1,$$

and we can indeed apply the Implicit Function Theorem. In fact, we know $L \in C^2$ and as we want to solve for $L(s,v) = 0$, we need that the v_0 we take to satisfy $L(0,v_0) = 0$, ie,

$$v_0 = \sqrt{\frac{2}{\phi''(0)}}.$$

Now, we also know that

$$\partial_v L(0,v_0) = \phi''(0)v_0 \neq 0.$$

So we get a bijective and continuously differentiable function $v = v(s)$ such that

$$\begin{cases} L(s,v(s)) = 0 \\ v(0) = v_0 \end{cases}$$

But, $L(s,v(s))$ means exactly that

$$\phi(t) - \phi(0) = s^2$$

with $t = t(s) = sv(s)$. Getting back to our equation, we have that

$$\mathrm{e}^{\lambda\phi(0)} \int_{-\delta}^{\delta} g(t)\,\mathrm{e}^{-\lambda(\phi(t)-\phi(0))}\,\mathrm{d}t = \mathrm{e}^{\lambda\phi(0)} \int_{\alpha}^{\beta} g(sv(s))(v(s)+sv'(s))\,\mathrm{e}^{-\lambda s^2}\,\mathrm{d}s$$

where we calculate $\alpha = -\sqrt{\phi(-\delta)-\phi(0)}$ and $\beta = \sqrt{\phi(\delta)-\phi(0)}$. One can now again finish the proof with Proposition 4.1. □

Example 4.3. (NICA; ORTMANN, 2025) For $x \in \mathbb{R}$ fixed, i.e. $x = O(1)$, we have the following asymptotic as $n \longrightarrow \infty$:

$$\mathrm{He}_n(x) = \sqrt{2}\left(\frac{n}{\mathrm{e}}\right)^{n/2} \mathrm{e}^{x^2/2}\cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1+o(1)).$$

To see that, consider the integral representation of the Hermite Polynomial and re-write

$$\begin{aligned}\mathrm{He}_n(x) &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}(x+\mathrm{i}t)^n\,\mathrm{e}^{-t^2/2}\,\mathrm{d}t = \frac{2}{\sqrt{2\pi}}\Re\left\{\int_0^{\infty}(x+\mathrm{i}t)^n\,\mathrm{e}^{-t^2/2}\,\mathrm{d}t\right\} \\ &= \frac{2}{\sqrt{2\pi}}\Re\left\{\int_{\Re(z)=x,\mathrm{Im}(z)>0} \mathrm{e}^{n\ln(z)+\frac{1}{2}(z-x)^2}\,\frac{\mathrm{d}z}{\mathrm{i}}\right\} \\ &= \frac{2}{\sqrt{2\pi}}\Re\left\{\int_{\Re(z)=0,\mathrm{Im}(z)>0} \mathrm{e}^{n\ln(z)+\frac{1}{2}(z-x)^2}\,\frac{\mathrm{d}z}{\mathrm{i}}\right\}\end{aligned}$$

where you start with a line integral in $\mathbb{C}$ and move the contour over to the positive imaginary axis - justified by the Cauchy's Theorem and the exponential decay of the integrand near infinity. Now, we focus on $n\ln(z) + \frac{1}{2}(z-x)^2$ which has maximum at $z^* = \mathrm{i}\sqrt{n} + o(\sqrt{n})$.

The idea is to change variables by a factor of $\sqrt{z}$ so that the location of the maximum contribution stays $O(1)$ as n gets large. Specifically, we do

$$z = \mathrm{i}\sqrt{n}s, \quad \frac{\mathrm{d}z}{\mathrm{i}} = \sqrt{n}\mathrm{d}s, \quad \ln(z) = \ln(\sqrt{n}) + \ln(s) + \mathrm{i}\frac{\pi}{2}$$

which give us

$$\frac{1}{2}(z-x)^2 = -\frac{1}{2}ns^2 + \frac{1}{2}x^2 - \mathrm{i}\sqrt{n}sx$$

so that we remain with

$$\begin{aligned}\mathrm{He}_n(x) &= \frac{2}{\sqrt{2\pi}}\Re\left\{\int_0^\infty \mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{n\ln(s)}\,\mathrm{e}^{n\mathrm{i}\frac{\pi}{2}}\,\mathrm{e}^{\frac{1}{2}x^2}\,\mathrm{e}^{-\mathrm{i}\sqrt{n}sx}\sqrt{n}\mathrm{d}s\right\}\\ &= \frac{2}{\sqrt{2\pi}}\sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2}\int_0^\infty \mathrm{e}^{n\left(\ln(s)-\frac{1}{2}s^2\right)}\cos\left(n\frac{\pi}{2} - \sqrt{n}sx\right)\mathrm{d}s,\end{aligned}$$

where we can apply Laplace's approximation discussed before. Noting that $\phi(s) = \ln(s) - \frac{1}{2}s^2$ has $\phi'(s) = \frac{1}{s} - s$ and $\phi''(s) = -\frac{1}{s^2} - 1$, with global maximum at $s^* = 1$ with $\phi(1) = -\frac{1}{2}$ and $\phi''(1) = -2$, yielding

$$\begin{aligned}\mathrm{He}_n(x) &= \frac{2}{\sqrt{2\pi}}\sqrt{n}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}x^2}\left(\mathrm{e}^{n\phi(1)}\sqrt{\frac{2\pi}{n|\phi''(1)|}}\right)\cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1+o(1)),\\ &= \sqrt{2}\,\mathrm{e}^{n\ln(\sqrt{n})}\,\mathrm{e}^{\frac{1}{2}(x^2-n)}\cos\left(n\frac{\pi}{2} - \sqrt{n}x\right)(1+o(1)),\\ &\sqrt{2}\left(\frac{n}{\mathrm{e}}\right)^{n/2}\mathrm{e}^{x^2/2}\cos\left(\frac{n\pi}{2} - x\sqrt{n}\right)(1+o(1)).\end{aligned}$$

Remember that in Note 2.1 we mentioned the possibility of studying the distributions of eigenvalues from the asymptotic of the orthogonal polynomial. Not only that but in Example 2.8 we said that the Hermite polynomials are the orthogonal polynomials with respect to the Gaussian weight. Therefore, this asymptotic is directly related to the asymptotic of the distribution of eigenvalues of the GUE.

4.2.3 Steepest Descent

Let us now finally explore integrals over the complex plane with complex valued functions. We retake the integral

$$\phi(\lambda) := \int_\Gamma g(s)\,\mathrm{e}^{-\lambda f(s)}\,\mathrm{d}s, \tag{4.6}$$

where Γ is a contour in the complex plane and f, g are given complex-valued, analytic functions. Suppose that Γ is parametrized by $\gamma : [a,b] \to \mathbb{C}$ and let us decompose the functions we have into their real and imaginary parts such as

$$f(z) = u(z) + \mathrm{i}v(z), \quad g(z) = p(z) + \mathrm{i}q(z), \quad \gamma(t) = \alpha(t) + \mathrm{i}\beta(t)$$

with all functions in the decomposition being real valued. Of course we now have

$$\mathrm{e}^{-\lambda f(z)} = \mathrm{e}^{-\lambda u(z)}\,\mathrm{e}^{-\mathrm{i}\lambda v(z)},$$

and when λ grows large, we have the exponentially growing/decaying term $\mathrm{e}^{-\lambda u(z)}$ and a oscillating term coming from $\mathrm{e}^{\mathrm{i}\lambda v(z)}$ which is hard to control. The idea to get around this fact is to use paths in which this term is constant. If we have such a case, we write

$$\Phi(\lambda) = \mathrm{e}^{-\mathrm{i}\lambda v_0} \int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t$$
$$+\mathrm{i}\,\mathrm{e}^{-\mathrm{i}\lambda v_0} \int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t \quad (4.7)$$

where each integral can now be solved by Laplace's method.

Theorem 4.6. *(Steepest Descent) Consider Φ as in (4.6) with a smooth contour Γ. Assume that the exponent $f = u + \mathrm{i}v$ is an analytic function on a neighborhood of Γ that satisfies*

$$v(z) \equiv v_0, \quad z \in \Gamma$$

In addition, suppose that u has a unique minimum at an interior point of Γ, say on $z_0 \in \Gamma$, with $f''(z_0) \neq 0$, and also that g is analytic on a neighborhood of Γ and satisfies

$$\int_\Gamma |g(z)||\mathrm{d}z| < \infty.$$

Then Φ has a leading asymptotic as $\lambda \to +\infty$ given by

$$\Phi(\lambda) = \mathrm{e}^{-\lambda f(z_0)} \left(\frac{\sqrt{2\pi}g(z_0)\,\mathrm{e}^{\mathrm{i}\theta(z_0)}}{\lambda^{1/2}\sqrt{|f''(z_0)|}} + O(\lambda^{-3/2}) \right), \quad \lambda \to +\infty,$$

where $\theta(z_0)$ is the angle of the oriented tangent of Γ at z_0.

Proof. Fix a parametrization $\gamma : [a,b] \to \mathbb{C}$ of Γ with $\gamma(c) = z_0, a < c < b$, and decompose Φ as before in (4.7). By the assumptions on the functions f, g we apply Laplace's method on both equations, and yields, as $\lambda \to +\infty$,

$$\int_a^b (p(\gamma(t))\alpha'(t) - q(\gamma(t))\beta'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t = \sqrt{2\pi}\,\mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\alpha'(c) - q(z_0)\beta'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \epsilon \right)$$

and

$$\int_a^b (p(\gamma(t))\beta'(t) - q(\gamma(t))\alpha'(t))\,\mathrm{e}^{-\lambda u(\gamma(t))}\,\mathrm{d}t = \sqrt{2\pi}\,\mathrm{e}^{-\lambda u(z_0)} \left(\frac{p(z_0)\beta'(c) - q(z_0)\alpha'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + \epsilon \right)$$

with $\epsilon = O\left(\lambda^{-3/2}\right)$. Using that $v_0 + u(z_0) = f(z_0)$ and also that,

$$(p(z_0)\alpha'(c) - q(z_0)\beta'(c)) + \mathrm{i}(p(z_0)\beta'(c) - q(z_0)\alpha'(c)) = g(z_0)\gamma'(c),$$

and, combining the estimates for both parts we get

$$\Phi(\lambda) = \sqrt{2\pi}\,\mathrm{e}^{-\lambda f(z_0)} \left(\frac{g(z_0)\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}\lambda^{1/2}} + O\left(\lambda^{3/2}\right) \right)$$

and it suffices now to get an expression for the second derivative in the denominator. Knowing the imaginary part is constant we write

$$\frac{\mathrm{d}}{\mathrm{d}t}u(\gamma(t)) = \frac{\mathrm{d}}{\mathrm{d}t}f(\gamma(t)) = \gamma'(t)f'(\gamma(t))$$

and

$$\frac{\mathrm{d}^2}{\mathrm{d}t}u(\gamma(t)) = \frac{\mathrm{d}^2}{\mathrm{d}t}f(\gamma(t)) = (\gamma'(t))^2 f''(\gamma(t)) + \gamma''(t)f'(\gamma(t)).$$

Hence,

$$\frac{\gamma'(c)}{\sqrt{|(u\circ\gamma)''(c)|}} = \frac{1}{\sqrt{|f''(z_0)|}}\frac{\gamma'(c)}{|\gamma'(c)|}.$$

□

We now need do discuss why this assumption of constant value of $\mathrm{I}m(f(z))$ is not an assumption as strong as one might assume at first. Suppose we want to integrate in a arbitrary contour Γ of the plane. Of course, our assumption that the functions were analytic permit us to deform the contour keeping the value of the integral. We are going to use this to find a contour $\tilde{\Gamma}$ such that sections of the contour are such that $\mathrm{I}m(f(z))$ is constant. We already know that being analytic means satisfying the *Cauchy-Riemann* equations, $u_x(z) = v_y(z), \text{and} u_y(z) = -v_x(z)$. An immediate result from these equations is that the gradients of u and v must be normal. Indeed, if $\langle\cdot,\cdot\rangle$ is the inner product on $\mathbb{R}^2$,

$$\langle\nabla u(z), \nabla v(z)\rangle = u_x(z)v_x(z) + u_y(z)v_y(z) = u_x(z)v_x(z) - v_x(z)u_x(z) = 0.$$

As we are following the curves where $\mathrm{I}m(f(z))$ is constant, we are also following the path of Steepest Descent of the real part of the integral. As we also are looking to use Laplace's method on the integrals, we may want to have our paths going through the critical points of the function. This fully motivates the method described.

Example 4.4. There are three asymptotics regimes for $\mathrm{He}_n(x)$ when $x \to \infty$ depending on the value of $\lim_{N\to\infty} x/2\sqrt{N}$ for $N = n+1$, each of which has qualitatively different behavior

- Small x values $-2\sqrt{N} < x < 2\sqrt{N}$: in this regime the Hermite Polynomials will be oscillating, and a have a similar qualitative behavior as in Example 4.3, with modification to phase and amplitude.

- Large x values $|x| > 2\sqrt{N}$: In this regime the polynomial does not have more oscillations and is monotone increasing. As x grows larger, this approximation behaves as $x^N \mathrm{e}^{-N^2/2x^2}$.

- Edge x values $x = \pm 2\sqrt{N} + O(N^{-\frac{1}{6}})$: in this regime the polynomials will be approximated by the Airy Function. This interpolates between the oscillatory small x regime and the monotone large x regimes.

We write the integral representation given by:

$$\begin{aligned}\mathrm{He}_n(x) &:= \frac{n!}{2\pi\mathrm{i}}\int_\Gamma z^{-n-1}\,\mathrm{e}^{xz-\frac{z^2}{2}}\,\mathrm{d}z\\ &= \frac{n!}{2\pi\mathrm{i}}\int_\Gamma \mathrm{e}^{xz-\frac{z^2}{2}-(n+1)\ln(z)}\,\mathrm{d}z\end{aligned}$$

where Γ is a contour that goes around the origin coming and going from minus infinity such as $\Gamma = \{\lambda - \mathrm{i}\delta : \lambda < 0\} \cup \{\delta\,\mathrm{e}^{\mathrm{i}\theta} : -\pi/2 < \theta < \pi/2\} \cup \{\lambda + \mathrm{i}\delta : \lambda < 0\}$. This is equivalent to using $\Gamma = \{\mathrm{e}^{\mathrm{i}\theta} : 0 \le \theta < 2\pi\}$. To determine He_n, set $N = n+1$ and consider the scaling $x \mapsto N^{1/2}\alpha$ and $z = N^{1/2}\beta$

$$\begin{aligned}\mathrm{He}_n(N^{\frac{1}{2}}\alpha) &= \frac{(N-1)!}{2\pi\mathrm{i}}\int_\Gamma \mathrm{e}^{(N^{1/2}\alpha)(N^{1/2}\beta)-\frac{1}{2}(N^{1/2}\beta)^2-N\ln(N^{1/2}\beta)}\,\mathrm{d}\beta\\ &= \frac{(N-1)!N^{\frac{1}{2}(1-N)}}{2\pi\mathrm{i}}\int_\Gamma \mathrm{e}^{-N(-\alpha\beta+\frac{1}{2}\beta^2+\ln(\beta))}\,\mathrm{d}\beta,\end{aligned}$$

so that we have

$$\mathrm{He}_n(\sqrt{N}\alpha) = C_n^0\int_\Gamma \mathrm{e}^{-N\phi_\alpha(\beta)}\,\mathrm{d}\beta,\quad \phi_\alpha(\beta) = \frac{1}{2}\beta^2 - \alpha\beta + \ln(\beta).$$

As we want to calculate the asymptotic by steepest descent, we need to determine the critical points of such a function and then the steepest descent paths passing through this points. We can easily determine the critical points as $\beta_c^\pm = \frac{\alpha\pm\sqrt{\alpha^2-4}}{2}$ - solutions to the quadratic formula given by solving the critical point of the function (the zeros of its derivatives). Now, the behavior of the critical points will depend on the interval of the real parameter α,

$$\beta_c^\pm = \frac{\alpha\pm\sqrt{\alpha^2-4}}{2} = \begin{cases}\mathrm{e}^{\pm\gamma}, & \alpha = 2\cosh(\gamma) > 2, \quad \gamma > 0;\\ \mathrm{e}^{\pm\mathrm{i}\gamma}, & \alpha = 2\cos(\gamma) \in [0,2), \quad \gamma \in (0,\pi/2];\\ 1, & \alpha = 2,\end{cases}$$

each of the cases refers to one of the regimes we described before.

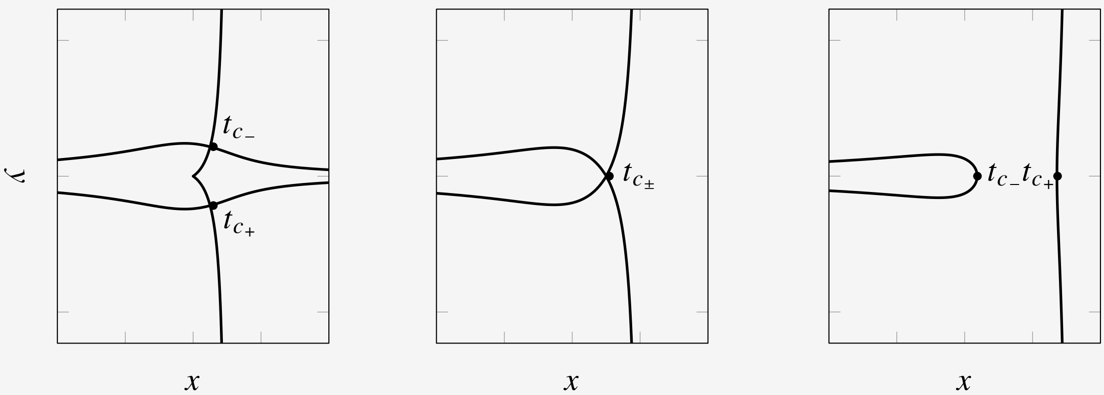

Figure 12 – The lines represent the solution to $\mathrm{I}m[f(z) - f(t_c)] = 0$ for the three possible cases of the Hermite Polynomial. the cases are from left to right $\alpha \in [0,2)$, $\alpha = 2$ and $\alpha > 2$. Only for the first case the critical points are complex. These are the curves we use to perform the asymptotic analysis on the integral. Note the case $\alpha = 2$; the coalescence of critical point gives origin to the Airy function.

CHAPTER

5

FREE ENERGY

We are going to discuss in this chapter the ideas developed in the paper Byun, Kang and Seo (2023), where asymptotics of partition functions were developed for a large class of random matrix ensembles taking advantage of the orthogonal polynomials present in these models. By the end, we hope to clarify with mathematical rigor the origin for the limit-behavior expression for the *Partition Function* $\mathcal{Z}_n^Q$ given by

$$\ln(\mathcal{Z}_n^Q) = -I_Q n^2 + \frac{1}{2} n \ln n + E_Q n + G \ln n + F^Q + o(1). \tag{5.1}$$

Starting from the p.d.f. of the ensemble, one can often express the j.p.d.f. of the set of eigenvalues that arise from realizations of random matrices. We have seen in Section 2.1 that these j.p.d.f. define point process - a measure on the space of configuration of eigenvalues. Partition functions perform the role of the normalization for the probability measure on j.p.d.f. of eigenvalues. In this instance, we deal with particles in the complex plane that come from *random normal matrix ensembles* as defined in Definition 2.4. We consider here confining potentials Q that are radially symmetric as specified in Definition 3.5. For a large class of such Q, the first term I_Q in (5.1) was known for a long time, given by the system's energy when distributed according to the equilibrium measure - a distribution that minimizes the system's energy. Note that this quantity was discussed in Section 3.2 when discussing the determination of the equilibrium measure. The coefficient $1/2$ of the order $n \ln n$, as well as the entropy term E_Q, were described in the work of Leblè and Serfaty (2017), also for rather large classes of Q. The remaining terms in this expansion appeared firstly at Byun, Kang and Seo (2023). The term G appears to depend only on the genus of the limiting spectrum - whether it is an annulus or a disk - which, surprisingly, seems to not have been observed in the previous literature. Although some recent work, such as in Hedenmalm and Wennman (2020), may indicate a direction to expand some of these results to larger classes of potentials, the symmetry of the external potential is, so far, still a necessary condition for the techniques presented henceforth.

5.1 Outline of the Technique

Given some measure $p_n(z_1, \cdots, z_n)$, one can think of its norming function. This is what is called here the partition function $\mathcal{Z}_n^Q$. Its logarithm represents the free energy of the system and is a quantity of interest in physics and mathematics. To get its asymptotics we will perform the following steps.

- *Partition functions can be expressed in terms of the norming constants*: Remind Theorem 2.3 and (2.5) that stated that normal ensembles' eigenvalues univariate density can be expressed as a Kernel; a function of the orthogonal polynomial with respect to the weight function w. An import result for normal ensembles gives the partition function as a function of the product of norming constants $h_j := h_{j,n}$. Then, as in 2.3, we can write the free energy as

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

- *Asymptotics of norming constant using the asymptotic Laplace technique*: If we set the weight function as $w(z) = \mathrm{e}^{-nQ(z)}$ for Q a radially symmetric potential, i.e. $Q(z) = q(|z|)$, the integral expression for the norming constants can be rewritten as a real line integral and approximated by the Laplace technique.

$$\begin{aligned} h_j &= \int_{\mathbb{C}} |z|^{2j}\, \mathrm{e}^{-nQ(z)}\, \mathrm{d}^2 z, \\ &= \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-n\left(q(r) - \frac{2j}{n}\ln(r)\right)}\, \mathrm{d}r, \\ &= \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-nV_{\tau(j)}(r)}\, \mathrm{d}r \approx \mathrm{e}^{-nV_{\tau(j)}(r_c(\tau(j)))}. \end{aligned}$$

- *Asymptotic of the summation on* $\ln h_j$ *using quadrature*: Finally, to approximate the summation on $\ln(h_j)$ we use the previously discussed idea of quadrature in Section 4.1 relating the limit summation with the corresponding integral

$$-\sum_{j=0}^{n-1} \ln h_j \approx -n^2 \int_{r_0}^{r_1} V(r) \frac{(rq'(r))'}{2} \mathrm{d}r = -n^2 I_Q[\mu_Q].$$

This technique can take us even further in order for the asymptotic expression. Above, we only used first order approximations, however in the following sections we show that doing the detailed computations, we can get all the terms down to the constant order for the asymptotic expansion. For the next section however we expand on this simplified version to give an outline for the more complex calculations. Modulo minor adaptations, the results we state for the random planar normal matrix ensemble are also valid in the context of the *planar symplectic ensemble* - defined with an additional complex conjugation symmetry. We however omit this case to focus on clarity for the technique.

5.1.1 Norming Constants

We must study the asymptotic behavior of the norming constant h_j described in Theorem 2.3. We expand on the idea in Section 5.1 where we stated that the integral could be approximated by its value in a critical point. This is, of course, based on the Laplace technique described in Subsection 4.2.2. Defining $z = r\,\mathrm{e}^{\mathrm{i}\theta}$ $(|z| = r)$, because of the symmetry on Q we write

$$\begin{aligned} h_j &= \int_{\mathbb{C}} |z|^{2j}\,\mathrm{e}^{-nQ(z)}\,\mathrm{d}^2 z \\ &= \int_{\Theta}\int_{\mathbb{R}_+} r^{2j}\,\mathrm{e}^{-nq(r)}\,\frac{r}{\pi}\mathrm{d}r\mathrm{d}\theta, \\ &= \frac{2\pi}{\pi}\int_{\mathbb{R}_+} r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r, \\ &= \int_{\mathbb{R}_+} 2r\,\mathrm{e}^{-n\left(q(r)-\frac{2j}{n}\ln(r)\right)}\,\mathrm{d}r. \end{aligned}$$

We redefine our potential using

$$V_{\tau(j)} = q(r) - 2\tau(j)\ln(r),$$

where $\tau(j) := j/n \in [0,1)$. Then, rewrite the integral as

$$h_j = \int_{\mathbb{R}_+} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r.$$

Note that now τ will replace the role of the index j marking the "degree" of the norming constant. Taking a look at the first derivative of the newly defined potential we find that the critical point r_c must satisfy

$$V'_\tau(r) = q'(r) - \frac{2\tau}{r} \qquad \Longrightarrow \qquad r_c q'(r_c) = 2\tau.$$

We found a critical point r_c of the potential, parameterized by the degree of the polynomial we are calculating (given by τ). We will denote $r_{\tau(j)}$ the r such that $rq'(r) = 2\tau(j)$, that is, for each τ, we denoted $r_{\tau(j)}$ its critical point. Of course, because we know the range of τ, it is worth defining the limit points when $\tau = 0$ and $\tau = 1$ as

$$\begin{cases} r_0, & \text{the largest solution of } rq'(r) = 0, \\ r_1, & \text{the smallest solution of } rq'(r) = 2. \end{cases}$$

Note that these two radius, r_0 and r_1 define what we have called the Droplet in Definition 3.6. They define the internal and external radius of the border for the equilibrium measure of the planar normal ensembles. Our main goal here is to evaluate the integral using the surroundings of the critical point. If this point is unique, then we only need to calculate the integral in a single small interval around $r_c(\tau(j)) := r_\tau(j)$. This is, of course, true: We mentioned that the properties on the potential directly implies that $rq'(r)$ is a strictly increasing function of $[r_0, r_1]$, then the solution must be unique.

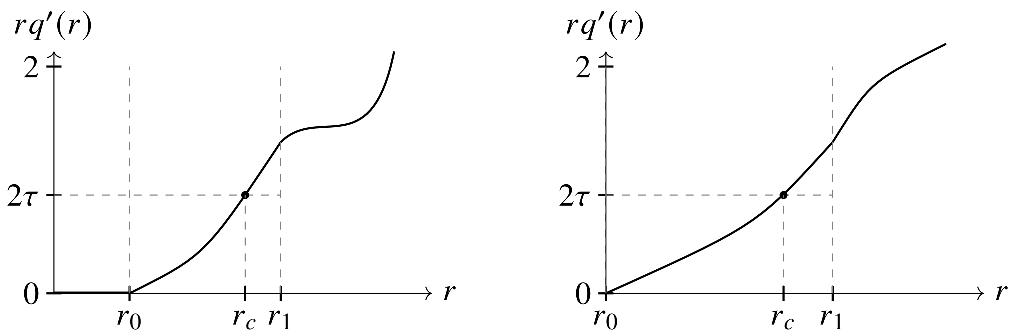

Figure 13 – Graphs for the two cases of the function $rq'(r)$. On the left, we show the case for $r_0 > 0$, corresponding to the annulus droplet. On the right, we show the case for $r_0 = 0$, corresponding to the disk droplet.

Now, as outside the droplet the growth is not strict, the definitions of r_0 and r_1 had to be as mentioned before: Respectively the largest and smallest solution to the proper instance of $rq'(r) = 2\tau$. An important distinction must be made between the cases when $r_0 = 0$ and $r_0 > 0$, corresponding to Section 5.3 and Section 5.2, respectively. We note that $r_\tau = r(\tau)$ defined such that $r_\tau q'(r_\tau) = 2\tau$ is an increasing function of τ and that $r_0 \le r_\tau \le r_1 \; \forall \; \tau \in [0,1)$. Whichever the case, we show that the integral is approximated by its value on the critical point as

$$\begin{aligned} h_j &= \int_{\mathbb{R}_+} 2r\, \mathrm{e}^{-nV_{\tau(j)}(r)}\, \mathrm{d}r, \\ &\approx \mathrm{e}^{-nV_{\tau(j)}(r_c(\tau(j))}. \end{aligned} \tag{5.2}$$

5.1.2 Approximating the Summation

Now, we remember that the expression for the free energy, accordingly to Theorem 2.3 is given in terms of the norming constants by

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j.$$

Having the asymptotic expression (5.2) for the norming constant, we now want to estimate the summation on the logarithm of such expression. We write

$$\ln \mathcal{Z}_n^Q = \ln n! - n \sum_{j=0}^{n-1} V_{\tau(j)}(r_c(\tau(j))).$$

Performing this sum is not a clear procedure at first. One way to do it is to reconsider the quadrature idea described in Section 4.1. With this, we can rewrite our summation as an integral, which may be easier to solve. In our case, the summation can be approximated as

$$n \sum_{j=0}^{n-1} V_{\tau(j)}(r_c(\tau(j)) \approx n^2 \int_0^1 V_\tau(r_c(\tau))\mathrm{d}\tau$$

We remember the expression for the derivative of the potential $V'_\tau(r) = q'(r) - 2\tau/r$. Note that the critical point $r_{\tau(j)}$ satisfies $r_{\tau(j)}q'(r_{\tau(j)}) = 2\tau(j)$.

Lemma 5.1. *The differential of r as a function of τ at $r(\tau)$ is given by*

$$\frac{\mathrm{d}r(\tau)}{\mathrm{d}\tau} = \frac{2}{r(\tau)V_\tau^{(2)}(r(\tau))} = \left.\frac{2}{(rq'(r))'}\right|_{r=r(\tau)}.$$

where $V^{(2)}$ is the second derivative of V.

Proof. Note the differential

$$\frac{\mathrm{d}r(\tau)}{\mathrm{d}\tau}q'(r) + \frac{\mathrm{d}r(\tau)}{\mathrm{d}\tau}rq''(r) = 2$$

which leads to

$$\begin{aligned}\frac{\mathrm{d}r(\tau)}{\mathrm{d}\tau} &= \frac{2}{q'(r)+rq''(r)},\\ &= \frac{2}{(rq'(r))'}.\end{aligned}$$

Evaluating at $r(\tau)$ we have

$$\left.\frac{2}{(rq'(r))'}\right|_{r=r(\tau)} = \frac{2}{q'(r(\tau))+r(\tau)q''(r(\tau))} = \frac{2}{\frac{2\tau}{r(\tau)}+r(\tau)q''(r(\tau))} = \frac{2}{r(\tau)V_\tau^{(2)}(r(\tau))}.$$

□

With Lemma 5.1, we write the integral in terms of r as

$$n\sum_{j=0}^{n-1} V_{\tau(j)}(r_c(\tau(j))) \approx n^2\int_{r_0}^{r_1} V(r)\frac{(rq'(r))'}{2}\mathrm{d}r.$$

This lead us to the main term in the expression (5.1), with order n^2. Note that this does not depend on the cases for r_0 bigger or equal to 0. This difference only appears in lower order approximations. This can be seen if one considers a more complete approximation and write

$$\begin{aligned}n\sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) =& n^2\int_{r_0}^{r_1} V(r)\frac{(rq'(r))'}{2}\mathrm{d}r\\ &-\frac{n}{2}\left(V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(0)}(r_{\tau(0)})\right)\\ &+\frac{n}{12}\left[\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0}\right]\\ &+O(n^{-3}).\end{aligned}$$

Namely, the dependency on the genus of the droplet is expressed on the coefficient of $\ln n$ order and changes from the disk to the annulus case. This general first-term approach does not have such detail. We will, in the following section, derive the same asymptotic using very similar techniques. However, we will show that this approach is capable of capturing even constant terms if done more carefully.

5.2 Annulus

We start by considering the annulus regime, when the droplet has $r_0 > 0$. When we say that $r_0 > 0$, we are saying that the equilibrium measure for model introduced in Definition 3.5 is non-zero in the region of space equivalent to the set of points z such that $|z| \in [r_0, r_1]$. Our objective with this section is to deduce the exact expression, analogous to (5.1), of the asymptotic of the free energy $\ln \mathcal{Z}_n^Q$ when $r_0 > 0$. For that, we will state the result and then follow the technique outlined in Section 5.1. We now state the main theorem of the section regarding the asymptotic for the partition function.

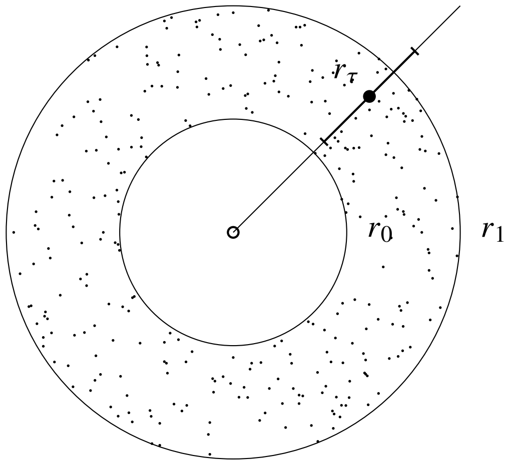

Figure 14 – The figure depicts an annulus referring to the equilibrium measure when $r_0 > 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for.

Theorem 5.1. *For the annulus case, where $r_0 > 0$, we have the following expression for the free energy asymptotically on n given by*

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{n}{2}\left(\ln(2\pi) - E_Q[\mu_Q] - 2\right) + \frac{n}{2}\ln(n) + \frac{\ln(2\pi)}{2} + \frac{1}{2}\ln(n) + F_V[A_{[r_0,r_1]}] + O(n^{-1})$$

where we set

$$F_V[A_{[r_0,r_1]}] := -\frac{2}{3}\ln\left(\frac{r_1}{r_0}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\,\mathrm{d}r,$$

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s) s V^{(2)}(s)\,\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right],$$

and

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)(sq'(s))'\mathrm{d}s.$$

We start by studying the asymptotic for the norming constants h_j. Then, we derive the formulas for the summation on its logarithm $\ln(h_j)$. These results should be enough to prove the previous theorem. The following subsections are referring to these two results.

5.2.1 Asymptotics of the Norming Constant

For this subsection, we will focus on getting the asymptotic for the integral

$$h_j = \int_{\mathbb{R}_+} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r$$

for each of the degrees j. With this, we can proceed to examine the summation over h_j needed top evaluate the free energy as it was shown in Theorem 2.3.

Theorem 5.2. *We have the following asymptotic of n for h_j, with $0 \le j \le n-1$,*

$$h_j = \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left[1+\frac{1}{n}\mathfrak{B}(r_{\tau(j)})+O(n^{-2})\right], \tag{5.3}$$

where we have set

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3}\frac{5}{24}.$$

The proof for such theorem passes trough Lemma 5.2 and Lemma 5.3. We begin as usual when applying asymptotic techniques in integrals; separating a central part where the integral value is concentrated and another part where we control the order of contribution. This is possible because the integrand is proportional to some exponential, with negative exponent, with a unique critical point, around which, the value of the expression concentrates, as discussed in Chapter 4. Define $\delta_n = \ln(n)/\sqrt{n}$ - this will be the region around the critical point which contribution we will consider - then, we have

$$h_j = \int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r + \int_{|r-r_j|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \tag{5.4}$$

As mentioned, the first integral must be somehow controlled and the second estimated as the main value for the result. We begin by showing that the first term has an error contribution.

Lemma 5.2. *We have the expression for the integral with $|r-r_j| > \delta_n$ as in (5.4) and*

$0 \le j \le n-1$, asymptotically on n, given by

$$\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r = \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\,\epsilon_n$$

where $\epsilon_n = O(\mathrm{e}^{-c(\ln(n))^2})$.

Proof. Let us first show that we can control the integral, for that, rewrite

$$\begin{aligned}
&\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\
&\qquad = \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-n\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \\
&\qquad = \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-n\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(M-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r, \\
&\qquad = \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(n-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r.
\end{aligned}$$

Now, we need some way to estimate the difference $(V_{\tau(j)}(r) - V_{\tau(j)}(r_{\tau(j)})$. If we are able to say that this difference is at least $O(n^{-1})$, we control the growth of the integral choosing M to be large enough such that

$$\int_{|r-r_j|>\delta_n} \left(\mathrm{e}^{-q(r)}\,r^{2\tau(j)}\right)^M r\mathrm{d}r$$

converges. Using the definition of the function V and the mean value theorem, we start by evaluating the difference $\Delta(j) = V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j)}(r_{\tau(j)})$ in the following manner

$$\begin{aligned}
\Delta(j) &= q(r_{\tau(j+1)}) - q(r_{\tau(j)}) - 2\left(\tau(j+1)\ln(r_{\tau(j+1)}) - \tau(j)\ln(r_{\tau(j)})\right), \\
&= q'(c_1)(r_{\tau(j+1)} - r_{\tau(j)}) - \frac{2}{n}\left(\ln(r_{\tau(j+1)}) + j\ln\left(\frac{r_{\tau(j+1)}}{r_{\tau(j)}}\right)\right), \\
&= q'(c_1)r'(c_2)(\tau(j+1)-\tau(j)) - \frac{2}{n}\left(\ln(r_{\tau(j+1)}) + jO(n^{-1})\right), \\
&= q'(c_1)r'(c_2)O(n^{-1}) - O(n^{-1}), \\
&= O(n^{-1}).
\end{aligned}$$

The last equality holds as both functions are continuous and c_1, c_2 are values that are limited; in a compact interval of the real line. Then, we add zero and use the estimation

$$V_{\tau(j+1)}(r_{\tau(j+1)}) - V_{\tau(j+1)}(r) + V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) = O(n^{-1})$$

to derive the result we are after. It is enough to estimate $V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)})$. For that, because $rq'(r)$ is increasing, for all $r > r_{\tau(j)} + \delta_n$ we say that

$$\begin{aligned}
V_{\tau(j+1)}(r) &\ge V_{\tau(j+1)}(r_{\tau(j)} + \delta_n), \\
&= V_{\tau(j+1)}(r_{\tau(j+1)}) + \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_n - r_{\tau(j+1)})^2 + O(\delta_n^3),
\end{aligned}$$

and therefore write

$$\begin{aligned}
V_{\tau(j+1)}(r) - V_{\tau(j)}(r_{\tau(j)}) &= \frac{1}{2}V''_{\tau(j+1)}(r_{\tau(j+1)})(r_{\tau(j)} + \delta_n - r_{\tau(j+1)})^2 + O(\delta_n^3),\\
&= \delta_n^2 c + O(n^{-1}) + O(\delta_n^3),\\
&\geq \delta_n^2 c.
\end{aligned}$$

Finally, using the above,

$$V_{\tau(j+1)}(r) - V_{\tau(j+1)}(r_{\tau(j+1)}) \geq c\delta_n^2.$$

Getting this inequality back to the integral expression end up with

$$\begin{aligned}
&\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r\\
&\quad= \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-(n-M)\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r,\\
&\quad\leq \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\,\mathrm{e}^{-c\delta_n^2(n-M)}\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r,
\end{aligned}$$

where one can check that

$$\begin{aligned}
&\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-M\left(V_{\tau(j)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r\\
&\quad= \int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-M\left((q(r)-2\tau(j)\ln(r))-(q(r_{\tau(j)})-2\tau(j)\ln(r_{\tau(j)}))\right)}\,\mathrm{d}r,\\
&\quad= 2\left(\mathrm{e}^{-q(r_{\tau(j)})}\,r_{\tau(j)}^{-2\tau(j)}\right)^M\int_{|r-r_j|>\delta_n}\left(\mathrm{e}^{-q(r)}\,r^{2\tau(j)}\right)^M r\,\mathrm{d}r.
\end{aligned}$$

By assumption, we chose M to be a big enough number such that the integral above converges. If that is true,

$$\int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r = \int_{|r-r_j|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \leq \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\,\epsilon_n,$$

with $\epsilon_n = O(\mathrm{e}^{-c(\ln(n))^2})$. □

We now move on to the value-contributing part of the integral. The other order term should be absorbed in this one if Theorem 5.2 is to be right.

> **Lemma 5.3.** *For the second part of the integral we have, for the integral with $|r - r_j| > \delta_n$ as in (5.4) and $0 \leq j \leq n-1$, asymptotically on n,*
>
> $$\int_{|r-r_j|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r = \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left[1 + \frac{1}{n}\mathfrak{B}(r_{\tau(j)}) + O(n^{-2})\right] \tag{5.5}$$
>
> *where $\mathfrak{B}(r_{\tau(j)})$ is as in Theorem 5.2.*

Proof. As we are dealing with a symmetric (in relation to the critical point) integral, we might as well set a centralized variable $x = r - r_\tau$. With this, the integral becomes

$$\begin{aligned}\int_{|r-r_j|<\delta_n} & 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\ &= \int_{-\delta_n}^{\delta_n} 2(x+r_{\tau(j)})\,\mathrm{e}^{-nV_{\tau(j)}(x+r_{\tau(j)})}\,\mathrm{d}x, \\ &= \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{-\delta_n}^{\delta_n} 2(x+r_{\tau(j)})\,\mathrm{e}^{-n\left[V_{\tau(j)}(x+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})\right]}\,\mathrm{d}x.\end{aligned}$$

Our interval grows with $\sqrt{n}/\ln(n)$. By re-scaling the reference variable using $y = \sqrt{n}x$ we get

$$\begin{aligned}\int_{|r-r_j|<\delta_n} & 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\ &= \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\int_{-\ln(n)}^{\ln(n)} 2\left(\frac{y}{\sqrt{n}}+r_{\tau(j)}\right)\mathrm{e}^{-n\left[V_{\tau(j)}(\frac{y}{\sqrt{n}}+r_{\tau(j)})-V_{\tau(j)}(r_{\tau(j)})\right]}\,\mathrm{d}y.\end{aligned}$$

Using Taylor expansion for $V_{\tau(j)}\left(\frac{y}{\sqrt{n}}+r_{\tau(j)}\right)$ near the critical point $r_{\tau(j)}$ to rewrite

$$\begin{aligned}&\int_{|r-r_j|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\ &= \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\int_{-\ln(n)}^{\ln(n)} 2\left(\frac{y}{\sqrt{n}}+r_{\tau(j)}\right)\mathrm{e}^{-n\left(\frac{y^2}{2n}V^{(2)}_{\tau(j)}(r_{\tau(j)})+\frac{y^3}{3!n^{\frac{3}{2}}}V^{(3)}_{\tau(j)}(r_{\tau(j)})+\frac{y^4}{4!n^2}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\left(\frac{y}{\sqrt{n}}\right)^5\right)\right)}\,\mathrm{d}y, \\ &= \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{n}}\int_{-\ln(n)}^{\ln(n)}\left(\frac{y}{r_{\tau(j)}\sqrt{n}}+1\right)\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})-\frac{y^3}{3!\sqrt{n}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!n}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)}\,\mathrm{d}y, \\ &= \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{(2r_{\tau(j)})^{-1}\sqrt{n}}\int_{-\ln(n)}^{\ln(n)}\left(\frac{y}{r_{\tau(j)}\sqrt{n}}+1\right)\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\left(\mathrm{e}^{-\frac{y^3}{3!\sqrt{n}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!n}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)}\right)\mathrm{d}y.\end{aligned}$$

Now we take the Taylor Series for

$$\begin{aligned}\mathrm{e}^{-\frac{y^3}{3!\sqrt{n}}V^{(3)}_{\tau(j)}(r_{\tau(j)})-\frac{y^4}{4!n}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)} &= 1-\left[\frac{y^3}{3!\sqrt{n}}V^{(3)}_{\tau(j)}(r_{\tau(j)})+\frac{y^4}{4!n}V^{(4)}_{\tau(j)}(r_{\tau(j)})+O\left(\frac{y^5}{n^{\frac{3}{2}}}\right)\right] \\ -\frac{1}{2}\left[\frac{y^6}{n}\left(\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6}\right)^2+\frac{y^8}{n^2}\left(\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24}\right)^2\right. &\left.+\frac{y^7}{n^3}\left(\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{6}\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{24}\right)+O\left(\frac{y^8}{n^2}\right)\right].\end{aligned}$$

To get the asymptotic expansion (since odd terms vanish) we reorganize the terms in the integral again to obtain

$$\begin{aligned}\int_{|r-r_j|<\delta_n} & 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\ &= 2r_{\tau(j)}\frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\left[1-\frac{y^4}{3!r_{\tau(j)}n}V^{(3)}_{\tau(j)}(r_{\tau(j)})\right. \\ &\qquad\left.-\frac{y^4}{4!n}V^{(4)}_{\tau(j)}(r_{\tau(j)})-\frac{y^6}{2\cdot(3!)^2n}V^{(3)}_{\tau(j)}(r_{\tau(j)})^2+O\left(\frac{y^6}{n^2}\right)\right]\mathrm{d}y, \quad (5.6)\end{aligned}$$

then, distributing the integral,

$$\int_{|r-r_j|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r$$

$$= 2r_{\tau(j)}\frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24n}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}n}\right)\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\,y^4\mathrm{d}y\right.$$
$$\left.-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72n}\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\,y^6\mathrm{d}y+\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\,\mathrm{d}y+O\left(n^{-2}\right)\right].$$

At this point, the result is directly evaluated using the Gaussian integrals in Proposition 5.1. So that we substitute the values to get

$$\int_{|r-r_j|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r$$

$$= \sqrt{2\pi}r_{\tau(j)}\frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24n}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}n}\right)\frac{3}{16}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{5}{2}}\right.$$
$$\left.-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72n}\frac{15}{64}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{7}{2}}+\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{1}{2}}+O\left(n^{-2}\right)\right],$$

and, rearranging the terms we get our result

$$\int_{|r-r_j|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r = \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left(\frac{8\pi r^2_{\tau(j)}}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left[1+\frac{1}{n}\mathfrak{B}(r_{\tau(j)})+O(n^{-2})\right] \tag{5.7}$$

where we defined $\mathfrak{B}(r_{\tau(j)})$ as

$$\mathfrak{B}(r_{\tau(j)}) = -\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{8}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})^3}\frac{5}{24}.$$

Notice that the error term of Lemma 5.2 is absorbed into the error order of (5.7). Therefore we have the asymptotic expression for h_j given by (5.3) and the proof for Theorem 5.2 follows directly. □

5.2.2 Asymptotics of the Summation

We return to Theorem 2.3 and turn to the calculation of each term of $\sum_{j=0}^{n-1}\ln h_j$ using the logarithm of the expansion calculated previously in Theorem 5.2

$$\sum_{j=0}^{n-1}\left[-nV_{\tau(j)}(r_{\tau(j)})+\frac{1}{2}\left(\ln(2\pi r^2_{\tau(j)})-\ln n-\ln\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)\right)+\frac{1}{n}\mathfrak{B}(r_{\tau(j)})+O(n^{-2})\right]. \tag{5.8}$$

That is, we need to prove the following theorem.

Theorem 5.3. *Asymptotically on n,*

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2}\ln n + n\left(\frac{\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) + F_V[A_{[r_0,r_1]}] + O(n^{-1})$$

where $F_V[A_{[r_0,r_1]}]$, $I_Q[\mu_Q]$, $E_Q[\mu_Q]$ are as in Theorem 5.1.

The proof comes from gathering the terms in (5.8) with its respective expressions derived in Lemma 5.4 and Lemma 5.5. We go term by term. Lemma 5.4 refers to the asymptotic of

$$\sum_{j=0}^{n-1} \left(-nV_{\tau(j)}(r_{\tau(j)})\right),$$

and Lemma 5.5 refers to the asymptotics of both terms

$$\sum_{j=0}^{n-1}\left(\ln(2\pi r_{\tau(j)}^2) - \ln\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)\right).$$

Then, the final expression appears by balancing the results.

Lemma 5.4. *Asymptotically on n,*

$$-n\sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6}\ln\left(\frac{r_0}{r_1}\right) + O(n^{-2})$$

where we have set $I_Q[\mu_Q]$ as in Theorem 5.1 and

$$-2U_{\mu_Q}(0) := \int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s = q(r_1) - q(r_0) + 2\ln(r_1).$$

Proof. Using the Euler-Maclaurin formula from Theorem 4.1,

$$\sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = \int_0^n V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\left(V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(0)}(r_{\tau(0)})\right) + \frac{1}{12}\left[\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0}\right] + O(n^{-3}). \quad (5.9)$$

The first term in (5.9) is solved doing a integral by parts. Setting $s = r_{\tau(t)}$,

$$\begin{aligned}\int_0^n V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= \int_0^n (q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)}))\mathrm{d}t,\\ &= \frac{1}{2}\int_{r_0}^{r_1}(q(s) - sq'(s)\ln(s))sV^{(2)}(s)n\mathrm{d}s,\\ &= n\left[\frac{1}{2}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \frac{1}{2}\int_{r_0}^{r_1} s^2q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s\right],\end{aligned}$$

and finally using that $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$ we apply the integration by parts on the second term to get

$$\begin{aligned}
\frac{1}{2}\int_{r_0}^{r_1} s^2 q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s &= \frac{1}{2}\int_{r_0}^{r_1} s^2 q'(s)\ln(s)\frac{1}{s}(sq'(s))'\mathrm{d}s,\\
&= \frac{1}{2}\int_{r_0}^{r_1} sq'(s)\ln(s)(sq'(s))'\mathrm{d}s,\\
&= \frac{1}{4}\int_{r_0}^{r_1} ((sq'(s))^2)'\ln(s)\mathrm{d}s,\\
&= \frac{1}{4}(sq'(s))^2\ln(s)\Big|_{r_0}^{r_1} - \frac{1}{4}\int_{r_0}^{r_1}(sq'(s))^2\frac{1}{s}\mathrm{d}s\\
&= \ln(r_1) - \frac{1}{4}\int_{r_0}^{r_1} sq'(s)^2\mathrm{d}s,\\
&= \ln(r_1) - \frac{1}{2}q(r_1) + \frac{1}{4}\int_{r_0}^{r_1} q(s)(sq'(s))'\mathrm{d}s.
\end{aligned}$$

Leaving us with, by definition of I_Q

$$\begin{aligned}
\int_0^n V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= n\left[\frac{1}{2}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) - \frac{1}{4}\int_{r_0}^{r_1} q(s)(sq'(s))'\mathrm{d}s\right],\\
&= n\left[\frac{1}{4}\int_{r_0}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1)\right],\\
&=: nI_Q[\mu_Q]. \qquad (5.10)
\end{aligned}$$

We also need an expression for $V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)})$. This is direct noting that

$$V_{\tau(j)}(r_{\tau(j)}) - V_{\tau(0)}(r_{\tau(0)}) = V_1(r_1) - V_0(r_0) = q(r_1) - q(r_0) + 2\ln(r_1) =: 2U_{\mu_Q}(0). \qquad (5.11)$$

And finally for the remaining term we use the Leibniz rule to obtain

$$\begin{aligned}
\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} &= \partial_t\left(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)})\right)_{t=n},\\
&= \partial_\tau\left(q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)})\right)_{\tau=1}\partial_t\left(\tau(t)\right)_{t=n},\\
&= \frac{1}{n}\left(\partial_\tau(q(r_\tau)) - 2\left(\ln(r_\tau) + \tau\partial_\tau\left(\ln(r_\tau)\right)_{\tau=1}\right)\right),\\
&= \frac{1}{n}\left[\partial_t(q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))_{\tau=1} - 2\ln(r_1)\right]
\end{aligned}$$

and

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{n}\left[\partial_t(q(r_{\tau(t)}) - 2\ln(r_{\tau(t)}))_{\tau=0} - 2\ln(r_0)\right].$$

Since we can calculate, from $r_1 q'(r_1) = 2$ and $r_0 q'(r_0) = 0$

$$\partial_\tau q(r_\tau)|_{\tau=0} = \left(\frac{2q'(r)}{(rq'(r))'}\right)_{r_0} = 0,$$

$$\partial_\tau q(r_\tau)|_{\tau=1} = \left(\frac{2q'(r)}{(rq'(r))'}\right)_{r_1} = \frac{4}{r_1(q'(r_1) + r_1 q''(r_1))},$$

$$\partial_\tau(2\ln(r_\tau))|_{\tau=1} = \left(\frac{4}{r(rq'(r))'}\right)_{r_1} = \frac{4}{r_1(q'(r_1)+r_1q''(r_1))},$$

we also express the difference as

$$\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{12n}\left[2(\ln(r_0)-\ln(r_1))\right] = \frac{1}{6n}\ln\left(\frac{r_0}{r_1}\right). \tag{5.12}$$

The proof for the summation $\sum_{j=0}^{n-1} V_{\tau(j)}(r_{\tau(j)})$ is complete if one gathers the terms in (5.10), (5.12) and (5.11).

□

Lemma 5.5. *Asymptotically on n,*

$$\sum_{j=0}^{n-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = nE_Q[\mu_Q] - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(n^{-1})$$

and

$$\sum_{j=0}^{n-1} \ln(2\pi r^2_{\tau(j)}) = n\ln(2\pi) + 2\left(-nU_{\mu_Q}(0) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + O(n^{-1})\right)$$

where $U_{\mu_Q}(0)$ as in Lemma 5.4 and $E_Q[\mu_Q]$ as in Theorem 5.1.

Proof. Lets start now with the second summation, to which we apply Euler-Maclaurin again

$$\sum_{j=0}^{n-1}\ln\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right) = \int_0^n \frac{1}{4}\ln V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) +$$
$$\frac{1}{12}\left[\partial_t \frac{1}{4}\ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t\frac{1}{4}\ln V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=0}\right] + O(n^{-3}).$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we have

$$\int_0^n \ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})\mathrm{d}t = \frac{n}{2}\int_{r_0}^{r_1}\ln\left(\frac{1}{4}V^{(2)}(s)\right)(sq'(s))'\mathrm{d}s.$$

And, noticing that

$$\partial_t\frac{1}{4}\ln V^{(2)}_{\tau(t)}(r_{\tau(t)}) = \partial_V\left(\ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\frac{1}{4}\partial_r\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{\left(\partial_r V^{(2)}\right)(r_{\tau(t)})}{r_{\tau(t)}\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)^2},$$

we have for the third term on the right hand side,

$$\partial_t\ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t\ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=0} = \frac{1}{2n}\left(\frac{\left(\partial_r V^{(2)}\right)(r_1)}{r_1\left(V_1^{(2)}(r_1)\right)^2} - \frac{\left(\partial_r V^{(2)}\right)(r_0)}{r_0\left(V_0^{(2)}(r_0)\right)^2}\right)$$
$$= O(n^{-1}).$$

Therefore,

$$\sum_{j=0}^{n-1} \ln \frac{1}{4} V^{(2)}_{\tau(j)}(r_{\tau(j)}) = \frac{n}{2}\int_{r_0}^{r_1} \ln\left(\frac{1}{4}V^{(2)}(s)\right)(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(n^{-1}),$$
$$:= nE_Q[\mu_Q] - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + O(n^{-1}).$$

Similarly, for the first summation, we apply Euler-Maclaurin again

$$\sum_{j=0}^{n-1} \ln(2\pi r^2_{\tau(j)}) = \ln(2\pi) + 2\left(\int_0^n \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right)\right.$$
$$\left. + \frac{1}{12}\left[\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0}\right] + O(n^{-3})\right).$$

We first tackle the integral on the right hand side, changing variables as $s = r_{\tau(t)}$ we have

$$\int_0^n \ln r_{\tau(t)}\mathrm{d}t = \frac{n}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s.$$

And, noticing that

$$\partial_t \ln r_{\tau(t)} = \partial_r \left(\ln(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right) = \frac{2}{V^{(2)}_{\tau(t)}(r_{\tau(t)})\left(r_{\tau(t)}\right)^2},$$

we have for the third term on the right hand side,

$$\partial_t \ln r_{\tau(t)}|_{\tau=1} - \partial_t \ln r_{\tau(t)}|_{\tau=0} = \frac{1}{2n}\left(\frac{2}{V^{(2)}(r_1)\,(r_1)^2} - \frac{2}{V^{(2)}(r_0)\,(r_0)^2}\right) = O(n^{-1}).$$

Then

$$\sum_{j=0}^{n-1} \ln(2\pi r^2_{\tau(j)}) = n\ln(2\pi) + 2\left(\frac{n}{2}\int_{r_0}^{r_1} \ln(s)(sq'(s))'\mathrm{d}s - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right) + O(n^{-1})\right).$$

□

For the following proof, and the analogous proof for the following section, we use the auxiliary result on Gaussian integrals.

Proposition 5.1. *The Gaussian integrals below are evaluated as follows.*

$$\int_{\mathbb{R}} \mathrm{e}^{-2ar^2}\mathrm{d}r = \sqrt{\frac{\pi}{2}}\frac{1}{a^{1/2}}, \quad \int_{\mathbb{R}} \mathrm{e}^{-2ar^2} r^4\mathrm{d}r = \sqrt{\frac{\pi}{2}}\frac{3}{16}\frac{1}{a^{5/2}} \quad \textit{and} \quad \int_{\mathbb{R}} \mathrm{e}^{-2ar^2} r^6\mathrm{d}r = \sqrt{\frac{\pi}{2}}\frac{15}{64}\frac{1}{a^{7/2}}.$$

Moreover,

$$\int_{\mathbb{R}} \mathrm{e}^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)y^2} = \sqrt{\frac{\pi}{2}}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{1}{2}},$$

$$\int_{\mathbb{R}} e^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)y^2} y^4 = \sqrt{\frac{\pi}{2}}\frac{3}{16}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{5}{2}},$$

$$\int_{\mathbb{R}} e^{-2\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right)y^2} y^6 = \sqrt{\frac{\pi}{2}}\frac{15}{64}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{7}{2}}.$$

We now move on to the proof of Theorem 5.3.

Proof. (Theorem 5.3) We only need to notice now that by Theorem 4.1

$$\begin{aligned}\frac{1}{n}\sum_{j=0}^{n-1}\mathfrak{B}(r_{\tau(j)}) &= \frac{1}{4}\int_0^{r_1}\mathfrak{B}(s)(sq'(s))'\mathrm{d}s + O(n^{-2}),\\ &= \frac{1}{4}\int_0^{r_1}\mathfrak{B}(s)sV^{(2)}(s)\mathrm{d}s + O(n^{-2}).\end{aligned}$$

Using the definition for $\mathfrak{B}(r_{\tau(j)})$ and integrating by parts,

$$\begin{aligned}&\frac{1}{n}\sum_{j=0}^{n-1}\mathfrak{B}(r_{\tau(j)}) =\\ &\frac{1}{4}\int_{r_0}^{r_1}\left[-\frac{V^{(4)}(r)}{V^{(2)}(r)^2}\frac{1}{8} - \frac{V^{(3)}(r)}{V^{(2)}(r)^2}\frac{1}{2r} - \frac{V^{(3)}(r)^2}{V^{(2)}(r)^3}\frac{5}{24}\right]rV^{(2)}(r)\mathrm{d}r + O(n^{-2}),\\ &= -\frac{1}{32}\int_{r_0}^{r_1}\frac{V^{(4)}(r)}{V^{(2)}(r)}r\mathrm{d}r - \frac{1}{8}\int_{r_0}^{r_1}\frac{V^{(3)}(r)}{V^{(2)}(r)}\mathrm{d}r - \frac{5}{96}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r + O(n^{-2}),\\ &= -\frac{1}{8}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\int_{r_0}^{r_1}\frac{V^{(4)}(r)}{V^{(2)}(r)}r\mathrm{d}r - \frac{5}{96}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r + O(n^{-2}),\end{aligned}$$

where

$$\int_{r_0}^{r_1}\frac{V^{(4)}(r)}{V^{(2)}(r)}r\mathrm{d}r = \frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)} - \ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) + \int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r.$$

Therefore

$$\begin{aligned}&\frac{1}{n}\sum_{j=0}^{n-1}\mathfrak{B}(r_{\tau(j)})\\ &= -\frac{3}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r + O(n^{-2}), \quad (5.13)\end{aligned}$$

and the theorem follows from

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) - \frac{1}{6}\ln\left(\frac{r_0}{r_1}\right) - \frac{n}{2}\ln(n) - \frac{1}{2}\left(nE_Q[\mu_Q] - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)\right)$$
$$+\frac{1}{2}\left(n\ln(2\pi) + 2\left(-nU_{\mu_Q}(0) - \frac{1}{2}\ln\left(\frac{r_1}{r_0}\right)\right)\right) - \frac{3}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)$$
$$-\frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(n^{-1})$$

that reduces to

$$\sum_{j=0}^{n-1} \ln h_j = -n^2 I_Q[\mu_Q] - \frac{n}{2}\ln(n) + n\left(\frac{\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) - \frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)$$
$$-\frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(n^{-1}).$$

□

Proof. (Theorem 5.1) From Theorem 5.2,

$$\sum_{j=0}^{n-1} \ln(h_j) = -n^2 I_Q[\mu_Q] - \frac{n}{2}\ln(n) + n\left(\frac{\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right) - \frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right)$$
$$-\frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r + O(n^{-1}).$$

We get the asymptotic expansion for the Free energy using that

$$\ln(n!) = n\ln(n) - n + \frac{1}{2}\ln(n) + \frac{1}{2}\ln(2\pi) + O(n^{-1})$$

that is,

$$\ln \mathcal{Z}_n^Q = \ln n! + \sum_{j=0}^{n-1} \ln h_j,$$
$$= n\ln(n) - n + \frac{1}{2}\ln(n) + \frac{1}{2}\ln(2\pi) + \left(-n^2 I_Q[\mu_Q] - \frac{n}{2}\ln(n) + n\left(\frac{\ln(2\pi)}{2} - \frac{1}{2}E_Q[\mu_Q]\right)\right.$$
$$\left.-\frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r\right) + O(n^{-1}),$$
$$= -n^2 I_Q[\mu_Q] + \frac{n}{2}\left(\ln(2\pi) - E_Q[\mu_Q] - 2\right) + \frac{n}{2}\ln(n) + \frac{1}{2}\ln(n) + \frac{\ln(2\pi)}{2}$$
$$-\frac{2}{3}\ln\left(\frac{r_0}{r_1}\right) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - \frac{1}{32}\left(\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_0 V_0^{(3)}(r_0)}{V_0^{(2)}(r_0)}\right) - \frac{1}{12}\int_{r_0}^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r O(n^{-1}),$$
$$=: -n^2 I_Q[\mu_Q] + \frac{n}{2}\left(\ln(2\pi) - E_Q[\mu_Q] - 2\right) + \frac{n}{2}\ln(n) + \frac{\ln(2\pi)}{2} + \frac{1}{2}\ln(n) + F_V[A_{[r_0,r_1]}] + O(n^{-1}),$$

□

5.3 Disk

We now consider the disk regime, when the droplet has $r_0 = 0$. Again, we are saying that the equilibrium measure for model introduced in Definition 3.5 is non-zero in the region of space equivalent to the set of points z such that $|z| \in [r_0, r_1]$. Our objective with this section is to deduce the exact expression, analogous to (5.1), of the asymptotic of the free energy $\ln \mathcal{Z}_n^Q$ now when $r_0 = 0$. For that, we will state the result and then follow the technique outlined in Section 5.1. Byun, Kang and Seo (2023) also shows that the expression we get is different from the expression one gets when taking the limit $r_0 \to 0$ using the expression from the case $r_0 > 0$. We now state the main theorem of the section regarding the asymptotic for the partition function.

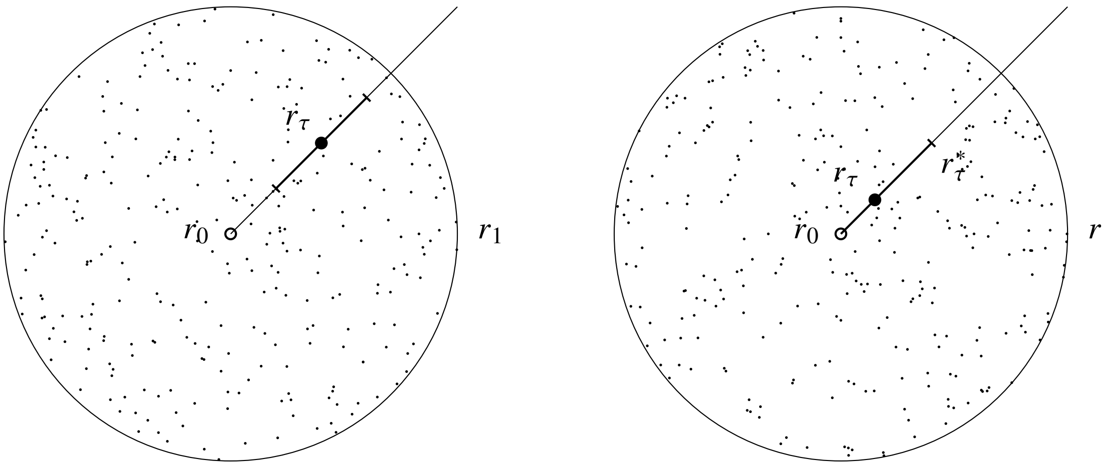

Figure 15 – The figure depicts an disk referring to the equilibrium measure when $r_0 = 0$. As the complex plane integral is reduced to a line integral, we draw the domain change and the point of interest r_τ where V has a critical point. Around this point we depict the region around the minima, which in the method that should account for the main order of the integral we solve for. The left image corresponds to the case when $r_{j/n} >> n^{-\epsilon}$ and the right image corresponds to $r_{j/n} << n^{-\epsilon}$. In this case $r*_\tau$ will define the contributing region to the integral.

Theorem 5.4. *For the annulus case, where $r_0 = 0$, we have the following, asymptotically on n, for the free energy*

$$\ln \mathcal{Z}_n^Q = -n^2 I_Q[\mu_Q] + \frac{1}{2} n \ln n + \frac{n}{2}\left(\ln(2\pi) - 2 - E_Q[\mu_Q]\right) - \frac{5}{12}\ln n \frac{\ln(2\pi)}{2} + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-1/12}(\ln n)^3).$$

where we have set

$$I_Q[\mu_Q] := \left[\frac{1}{4}\int_{r_0}^{r_1} q(s) s V^{(2)}(s) \mathrm{d}s - \ln(r_1) + \frac{1}{2} q(r_1)\right],$$

$$E_Q[\mu_Q] := \frac{1}{2}\int_{r_0}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)(s q'(s))' \mathrm{d}s$$

and

$$F_V[D_{r_1}] := \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)}\right) - \frac{1}{32}\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12}\int_0^{r_1} \frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r \mathrm{d}r.$$

Again, we start by studying the asymptotic for h_j and then for the summation $\sum \ln(h_j)$. This theorem then should follow from the lemmas concerning this two asymptotics.

5.3.1 Asymptotics of the Norming Constant

For this subsection, we will focus on getting the asymptotic for the integral

$$h_j = \int_{\mathbb{R}_+} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r$$

for each of the degrees j. With this, we can proceed to examine the summation over h_j needed top evaluate the free energy. For this case however there is one complication. One should distinguish the following two cases depending on a small constant $1/5 > \epsilon > 0$.

- Case 1: $r_{j/n} >> n^{-\epsilon}$. For the annular droplet case where $r_0 > 0$, this case covers all $j = 0, 1, \cdots, n-1$. On the other hand, for the disk droplet case where $r_0 = 0$, this case covers only $j = m_n, m_n + 1, \cdots, n-1$ for $m_n = n^\epsilon$ with some $\epsilon > 0$;
- Case 2: $r_{j/n} << n^{-\epsilon}$. This covers the remaining disk droplet case with $j = 0, 1, \cdots, m_n - 1$. Notably, the asymptotic expansion involves gamma functions in this case.

We separate this in two proofs, the first one will be regarding $j = m_n, m_n + 1, \cdots, n-1$. One might assume, and would be mainly correct, that nothing is that different in the disk and annulus regimes if the points are distant from r_0. The following results are direct adaptations of the previous arguments.

Theorem 5.5. *For $n-1 \geq j \geq m_n$, asymptotically on n,*

$$h_j = \frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}} \left(\frac{8\pi r_{\tau(j)}^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})} \right)^{\frac{1}{2}} \left(1 + \frac{1}{n}\mathfrak{B}(r_{\tau(j)}) + \epsilon_{n,1} \right)$$

where

$$\mathfrak{B}(r_{\tau(j)}) := -\frac{V_{\tau(j)}^{(4)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{8} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^2}\frac{1}{2r_{\tau(j)}} - \frac{V_{\tau(j)}^{(3)}(r_{\tau(j)})^2}{V_{\tau(j)}^{(2)}(r_{\tau(j)})^3}\frac{5}{24},$$

and

$$\epsilon_{n,1} := O(j^{-\frac{3}{2}}(\ln n)^\alpha)$$

for some $\alpha > 0$.

To actually expand on the details, we need some useful facts. We will make a lot of use of the behavior for r_τ, expressed in Proposition 5.2.

Proposition 5.2. *Due to strict sub-harmonicity of Q in a neighborhood of the droplet, r_τ, defined as before, satisfies*

$$r_\tau = \left(\frac{2\tau}{q''(0)}\right)^{\frac{1}{2}} + O(\tau) = \left(\frac{4\tau}{V_0^{(2)}(0)}\right)^{\frac{1}{2}} + O(\tau)$$

or more simply

$$r_\tau = O\left(\tau^{\frac{1}{2}}\right)$$

as $\tau \to 0$.

Proof. Begin defining a multivariate function

$$F(\tau, r) = rq'(r) - 2\tau$$

such that

$$\begin{cases} F(0,0) = rq'(r) - 2\tau = 0, \\ \partial_r F(\tau, r)\big|_{(0,0)} = q'(r) + q''(r)\cdot r\big|_{(0,0)} = 0 \end{cases}$$

since q has critical point at $r_\tau = 0$. This means that the assumptions in the Implicit Function Theorem 4.3 do not hold. We must use the *blow-up* technique. Consider a auxiliary variable $v := v(r)$ such that $v(r)\tau = r^2$ and look for solutions of the redefined

$$F(\sqrt{v(r)\tau}, \tau) := F'(v, \tau) = \frac{\sqrt{v\tau}\cdot q'(\sqrt{v\tau})}{2\tau} - 1$$

around $(v_0, 0)$, where v_0 is such that $F'(v_0, 0) = 0$. Note that, expanding $q'(r)$ in Taylor around $r = 0$ give us

$$q'(\sqrt{v\tau}) = q''(0)\cdot\sqrt{v\tau} + O(v\tau),$$

which, applied in the previous function,

$$F'(v, \tau) = \frac{q''(0)}{2}v - 1 + O(v^{3/2}\tau).$$

Now, we apply the Implicit Function Theorem on F' noting that the point v_0 must satisfy

$$v_0 = \frac{2}{q''(0)}$$

and that $q''(0) > 0$ by assumption of the global minimum. Then

$$\partial_v F'(v, \tau)\big|_{(v_0,0)} = \frac{q''(0)}{2} > 0.$$

Then, the theorem states that there is a bijective function $v(r\tau) \in C^2$ defined in a neighborhood to $\tau = 0$, for which

$$F'(v, \tau) = 0 \quad \text{and} \quad v(0) = v_0.$$

However, $F'(v,\tau)=0$ means precisely that

$$\sqrt{v\tau}\cdot q'(\sqrt{v\tau}) = 2\tau \quad \text{with} \quad r = r(\tau) = \sqrt{v(\tau)\tau}.$$

Leading us to believe that locally at v_0

$$r(\tau) = \left(\frac{2\tau}{q''(0)}\right)^{\frac{1}{2}} + O(\tau)$$

for some $\tau \longrightarrow 0$. Next to zero, all higher powers in the expansion are just order τ. $\square$

To actually go on and prove the stated theorem we start as usual dividing the integral into two sections

$$\begin{aligned} h_j &= \int_0^\infty 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r = \int_0^\infty 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r, \\ &= \int_{|r-r_{\tau(j)}|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r + \int_{|r-r_{\tau(j)}|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \end{aligned}$$

Let's start with a result on the outer integral, which we should be able to control the growth of.

Lemma 5.6. *Consider the case $m_n \le j \le n-1$. If we choose M to be such that*

$$\int_{|r-r_j|>\delta_n} \left(\mathrm{e}^{-q(r)}\,r^2\right)^M r\mathrm{d}r < \infty,$$

then, for the error-bound integral in which $|r - r_{\tau(j)}| > \delta_n$ we have the asymptotic for n given by

$$\int_{|r-r_{\tau(j)}|>\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r = \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\,O\left(\mathrm{e}^{-c_2(\ln(n))^2}\right).$$

for some positive constant c_2.

Proof. Note that for $d \ge 1$, we choose C_1 a constant uniformly for all τ and write

$$|V_\tau^{(d)}(r_\tau)| = |q^{(d)}(r_\tau) + (-1)^d 2\tau(d-1)!\; r_\tau^{-d}| \le C_1(1+\tau^{-\frac{d}{2}+1}).$$

Firstly, for $m_n \le j < n$, we have

$$r_{\tau(j+1/2)} - r_{\tau(j)} = O((jn)^{-\frac{1}{2}}).$$

Since the potential is increasing in (r_τ, ∞), the Taylor series expansion for V_τ is for all $r > r_{\tau(j)} + \delta_n$

$$\begin{aligned} V_{\tau(j+1/2)}(r) &\ge V_{\tau(j+1/2)}(r_{\tau(j)}+\delta_n), \qquad (5.14)\\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2}V^{(2)}_{\tau(j+1/2)}(r_{\tau(j+1/2)})(r_{\tau(j)}+\delta_n - r_{\tau(j+1/2)})^2 + O(\tau(j)^{-1/2}\delta_n^3). \end{aligned}$$

where $O(\tau(j)^{-1/2}\delta_n^3) = O(n^{-1}j^{-1/2}(\ln(n))^3)$. Similarly, since V_τ is decreasing in $(0,r_\tau)$, we have for all $r < r_{\tau(j)}-\delta_n$

$$\begin{aligned}V_{\tau(j+1/2)}(r) &\geq V_{\tau(j+1/2)}(r_{\tau(j)}-\delta_n),\\ &= V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) + \frac{1}{2}V^{(2)}_{\tau(j+1/2)}(r_{\tau(j+1/2)})(r_{\tau(j)}-\delta_n - r_{\tau(j+1/2)})^2 + O(\tau(j)^{-1/2}\delta_n^3).\end{aligned} \tag{5.15}$$

Using the Taylor series for V_τ again, we have

$$\begin{aligned}V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) &= V_{\tau(j)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) - \frac{1}{n}\ln r_{\tau(j+1/2)},\\ &= O((jn)^{-1}) - \frac{1}{n}\ln r_{\tau(j+1/2)}.\end{aligned}$$

Noting that

$$V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j)}(r_{\tau(j)}) = [V_{\tau(j+1/2)}(r_{\tau(j+1/2)}) - V_{\tau(j+1/2)}(r)] + [V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)})]$$

one rewrite using (5.15) and (5.14),

$$V_{\tau(j+1/2)}(r) - V_{\tau(j)}(r_{\tau(j)}) = -\frac{1}{n}\ln r_{\tau(j+1/2)} + O\left(\frac{\log(n)^2}{n}\right).$$

Thus, it follows from the last observations that

$$\begin{aligned}\mathrm{e}^{nV_{\tau(j)}(r_{\tau(j)})}\int_{|r-r_{\tau(j)}|>\delta_n} 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r &= \int_{|r-r_{\tau(j)}|>\delta_n} 2\,\mathrm{e}^{-n\left(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r,\\ &= \int_{|r-r_{\tau(j)}|>\delta_n} 2\,\mathrm{e}^{-(n-M)\left(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{e}^{-M\left(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r,\\ &\leq \mathrm{e}^{-c_1(\ln(n))^2}\,\mathrm{e}^{\frac{n-M}{n}\ln(r_{\tau(j+1/2)})}\int_{|r-r_{\tau(j)}|>\delta_n} 2\,\mathrm{e}^{-M\left(V_{\tau(j+1/2)}(r)-V_{\tau(j)}(r_{\tau(j)})\right)}\,\mathrm{d}r,\\ &= O(\mathrm{e}^{-c_2(\ln(n))^2}).\end{aligned}$$

for some constants $c_1, c_2 > 0$ and M such that the integral above converges. The details taken here are very similar to the ones in the annulus case. □

Lemma 5.7. *Consider the case $m_n \leq j \leq n-1$. For the value-contributing integral we have the asymptotic of n given by*

$$\int_{|r-r_{\tau(j)}|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r = \frac{\mathrm{e}^{-nV_{\tau(j)}}(r_{\tau(j)})}{\sqrt{n}}\left(\frac{8\pi r_{\tau(j)}^2}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{1}{2}}\left(1+\frac{1}{n}\mathfrak{B}(r_{\tau(j)})+\epsilon_{n,1}\right)$$

where $\mathfrak{B}(r_{\tau(j)})$ is as in Theorem 5.4 and

$$\epsilon_{n,1} := O(j^{-\frac{3}{2}}(\ln n)^\alpha)$$

for some $\alpha > 0$.

Proof. We now need to examine the integral near the critical point. That is,

$$
\begin{aligned}
&\int_{|r-r_{\tau(j)}|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\
&= \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{-\delta_n}^{\delta_n} 2(r_{\tau(j)}+t)\,\mathrm{e}^{-n\left(\frac{1}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})(t)^2+\frac{1}{3!}V^{(3)}_{\tau(j)}(r_{\tau(j)})(t)^3+\frac{1}{4!}V^{(4)}_{\tau(j)}(r_{\tau(j)})(t)^4+O(\tau(j)^{-\frac{3}{2}}t^5)\right)}\,\mathrm{d}t, \\
&= \mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}\int_{-\sqrt{n}\delta_n}^{\sqrt{n}\delta_n} 2\,\mathrm{e}^{-\frac{1}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})t^2}\left(r_{\tau(j)}+\frac{t}{\sqrt{n}}\right)\left(1-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})t^3}{6\sqrt{n}}\right. \\
&\qquad\left.-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})t^4}{24n}+\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2t^6}{72n}+\epsilon_{n,1}\right)\mathrm{d}t,
\end{aligned}
$$

where $\epsilon_{n,1}=O(j^{-\frac{3}{2}}(\ln n)^{\alpha})$ for some $\alpha>0$. From here, the expression is directly identified with (5.6) and the techniques for getting the result are one-to-one with the ones already done in that case. then, distributing the integral,

$$
\begin{aligned}
&\int_{|r-r_{\tau(j)}|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\
&= 2r_{\tau(j)}\frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24n}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}n}\right)\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\,y^4\mathrm{d}y\right. \\
&\qquad\left.-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72n}\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\,y^6\mathrm{d}y+\int_{-\ln(n)}^{\ln(n)}\mathrm{e}^{-\frac{y^2}{2}V^{(2)}_{\tau(j)}(r_{\tau(j)})}\,\mathrm{d}y+\epsilon_{n,1}\right].
\end{aligned}
$$

At this point, the result is directly evaluated using the Gaussian integrals in Proposition 5.1. So that we finally substitute the values to get

$$
\begin{aligned}
&\int_{|r-r_{\tau(j)}|<\delta_n} 2r\,\mathrm{e}^{-nV_{\tau(j)}(r)}\,\mathrm{d}r \\
&= \sqrt{2\pi}r_{\tau(j)}\frac{\mathrm{e}^{-nV_{\tau(j)}(r_{\tau(j)})}}{\sqrt{n}}\left[\left(-\frac{V^{(4)}_{\tau(j)}(r_{\tau(j)})}{24n}-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})}{6r_{\tau(j)}n}\right)\frac{3}{16}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{5}{2}}\right. \\
&\qquad\left.-\frac{V^{(3)}_{\tau(j)}(r_{\tau(j)})^2}{72n}\frac{15}{64}\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{7}{2}}+\left(\frac{4}{V^{(2)}_{\tau(j)}(r_{\tau(j)})}\right)^{\frac{1}{2}}+\epsilon_{n,1}\right],
\end{aligned}
$$

and, rearranging the terms we should get our result. □

Proof. (Theorem 5.5) This is now only a matter to notice that the expression derived in Lemma 5.6 is absolved in to the error term of the expression in Lemma 5.7 and the result follows. □

So far, nothing is considerably different from both cases as we are dealing with point distant from the origin. We must now deal with the more delicate case, for $j\leq m_n$. The behavior for h_j, in this regimen, is expressed by the following theorem.

Theorem 5.6. *For $0 \le j < m_n$, we will have the following asymptotic for the norming constant*

$$h_j = \mathrm{e}^{-nq(0)}\left[n^{-(j+1)}\Gamma(j+1)\left(\frac{1}{2}q''(0)\right)^{-(j+1)}\left(1+O(n^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln n)^3)\right)+\epsilon_n(j)\right]$$

where Γ is the usual Gamma function and

$$\epsilon_n(j) := O\left(\mathrm{e}^{-c_2(j+1)(\ln n)^2}\left(\frac{j+1}{n}(\ln(n))^2\right)^{(j+1)\frac{n-M}{n}}\right)$$

with $c_2 > 0$ a constant.

Proof. For $j < m_n$, we need to define some $r_\tau^* := r_\tau \ln(n)$ from which we separate the integral in a important part close to the origin and another part which has a lesser order. Consider then the decomposition

$$h_j = \int_0^\infty 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r = \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r + \int_{r^*_{\tau(j+1)}}^\infty 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r.$$

Due to strict sub-harmonicity of Q in a neighborhood of the droplet, r_τ satisfies $r_\tau = O(\tau^{\frac{1}{2}})$ as $\tau \to 0$, this is given by Proposition 5.2. Since V_τ has a global minimum at $r = r_\tau$ and increases for larger values, for all $r > r^*_{\tau(j+1)}$ we have

$$\begin{aligned}
V_{\tau(j+1/2)}(r) \ge V_{\tau(j+1)}(r)(r^*_{\tau(j+1)}) &= q(r^*_{\tau(j+1)}) - \frac{2j+1}{n}\ln(r^*_{\tau(j+1)}),\\
&= q(0) + \frac{1}{2}q''(0)(r^*_{\tau(j+1)})^2 - \frac{2j+1}{n}\ln(r^*_{\tau(j+1)}) + O(r^*_{\tau(j+1)})^3.
\end{aligned}$$

Note that $q'(0) = 0$ since $\Delta Q(z) \in (0,\infty)$ near the origin and $\Delta Q(z) \propto q'(0)/r + q''(0)$. Take

$$\begin{aligned}
\mathrm{e}^{-nq(0)}\int_{r^*_{\tau(j+1)}}^\infty 2r^{2j+1}\,\mathrm{e}^{-nq(r)+nq(0)}\,\mathrm{d}r &= \mathrm{e}^{-nq(0)}\int_{r^*_{\tau(j+1)}}^\infty 2\,\mathrm{e}^{(2j+1)\ln(r)-nq(r)+nq(0)}\,\mathrm{d}r,\\
&= \mathrm{e}^{-nq(0)}\int_{r^*_{\tau(j+1)}}^\infty 2\,\mathrm{e}^{-n\left(-\frac{2}{n}\left(j+\frac{1}{2}\right)\ln(r)-q(r)+q(0)\right)}\,\mathrm{d}r,\\
&= \int_{r^*_{\tau(j+1)}}^\infty 2\,\mathrm{e}^{-n(V_{\tau(j+1/2)}(r)-q(0))}\,\mathrm{d}r
\end{aligned}$$

and, using the inequality explicited above

$$\begin{aligned}
\mathrm{e}^{-nV_{\tau(j+1)}(r)+nq(0)} &= \mathrm{e}^{-nV_{\tau(j+1)}(r)+nq(0)+MV_{\tau(j+1)}(r)-MV_{\tau(j+1)}(r)},\\
&= \mathrm{e}^{-(n-M)V_{\tau(j+1)}(r)+nq(0)-MV_{\tau(j+1)}(r)},\\
&\le \mathrm{e}^{-(n-M)V_{\tau(j+1)}(r^*)+nq(0)-MV_{\tau(j+1)}(r)},\\
&= \mathrm{e}^{-(n-M)\left(V_{\tau(j+1)}(r^*)-q(0)\right)}\,\mathrm{e}^{-M\left(V_{\tau(j+1)}(r)-q(0)\right)}.
\end{aligned}$$

Then

$$
\begin{aligned}
&\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-n(V_{\tau(j+1)}(r)-q(0))}\,\mathrm{d}r\\
&\le \mathrm{e}^{-(n-M)V_{\tau(j+1)}(r^*)}\,\mathrm{e}^{(n-M)q(0)}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r,\\
&= \mathrm{e}^{-(n-M)\left[q(r^*_{\tau(j+1)})-\frac{2j+2}{n}\ln(r^*_{\tau(j+1)})\right]}\,\mathrm{e}^{(n-M)q(0)}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r,\\
&= \mathrm{e}^{-(n-M)q(r^*_{\tau(j+1)})}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{n-M}{n}}\,\mathrm{e}^{(n-M)q(0)}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r,\\
&= \mathrm{e}^{-(n-M)\left[\frac{q''(0)}{2}(r^*_{\tau(j+1)})^2+O(r^*_{\tau(j+1)})^3\right]}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{n-M}{n}}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r,\\
&\le \mathrm{e}^{-\frac{(n-M)}{2}q''(0)(r^*_{\tau(j+1)})^2)}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{n-M}{n}}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r,\\
&= \mathrm{e}^{-c_1\frac{(n-M)}{2}r^2_{\tau(j+1)}\ln(n)^2}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{n-M}{n}}\int_{r^*_{\tau(j+1)}}^{\infty} 2\,\mathrm{e}^{-MV_{\tau(j+1)}(r)}\,\mathrm{d}r := \epsilon_n(j)
\end{aligned}
$$

for some $c_1 > 0$ and M as given in the other proof such that the integral converges. Since $r_\tau = O(\tau^{\frac{1}{2}})$, there is a constant $c_2 > 0$ such that as n goes to infinity

$$
\begin{aligned}
\epsilon_n(j) &= O\left(\mathrm{e}^{-c_1\frac{(n-M)}{2}\tau(j+1)\ln(n)^2}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{n-M}{n}}\right),\\
&= O\left(\mathrm{e}^{-c_2(j+1)\ln(n)^2}\,{r^*_{\tau(j+1)}}^{(2j+2)\frac{n-M}{n}}\right),\\
&= O\left(\mathrm{e}^{-c_2(j+1)(\ln n)^2}\left(\frac{j+1}{n}(\ln(n))^2\right)^{(j+1)\frac{n-M}{n}}\right).
\end{aligned}
$$

Therefore we obtain

$$
\begin{aligned}
h_j &= \int_0^{\infty} 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d}r = \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-nq(r)}\,\mathrm{d} + \mathrm{e}^{-nq(0)}\,\epsilon_n(j),\\
&= \int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-n\left(q(0)+\frac{1}{2}q''(0)r^2\right)+O(n(r^*_{\tau(j+1)})^3)}\,\mathrm{d}r + \mathrm{e}^{-nq(0)}\,\epsilon_n(j),\\
&= \mathrm{e}^{-nq(0)}\left[\int_0^{r^*_{\tau(j+1)}} 2r^{2j+1}\,\mathrm{e}^{-\frac{1}{2}q''(0)r^2n}\,\mathrm{d}r\left(1+O(n(r^*_{\tau(j+1)})^3)\right)+\epsilon_n(j)\right],\\
&= \mathrm{e}^{-nq(0)}\left[n^{-(j+1)}\Gamma(j+1)\left(\frac{1}{2}q''(0)\right)^{-(j+1)}\left(1+O(n^{-\frac{1}{2}}(j+1/2)^{\frac{3}{2}}(\ln n)^3)\right)+\epsilon_n(j)\right]
\end{aligned}
$$

where Γ is the usual gamma function with integral representation

$$
\Gamma(j+1) = \int_0^{\infty} t^j\,\mathrm{e}^{-t}\,\mathrm{d}t
$$

and we used $t = 1/2q''(0)r^2n$.

□

5.3.2 Asymptotics of the Summation

So now we need to return to Theorem 2.3 and calculate each term for $\sum_{j=0}^{n-1} \ln h_j$ using the expansion in terms of Theorem 5.5 and Theorem 5.6 for the expression

$$\sum_{j=0}^{n} \ln(h_j) = \sum_{j=0}^{m_n-1} \ln(h_j) + \sum_{j=m_n}^{n-1} \ln(h_j).$$

Theorem 5.7. *We have the following asymptotics on n for the summation of* $\ln(h_j)$ *from* 0 *to* $n-1$*,*

$$\sum_{j=0}^{n-1} \ln(h_j) = -n^2 I[\mu_Q] - \frac{1}{2} n \ln(n) - \frac{n}{2}\left(E_Q[\mu_Q] - \ln(2\pi)\right) - \frac{\ln(n)}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-12}(\ln n)^3)$$

where $I_Q[\mu_Q]$*,* $F_V[D_{r_1}]$ *and* $E_Q[\mu_Q]$ *are as in Theorem 5.4.*

To prove this theorem, we must give the asymptotics of both summations. This means we need both Lemma 5.8, referring to the sum from powers 0 to $m_n - 1$, and Lemma 5.9, referring to the sum from m_n to $n-1$. We start by the lower power terms.

Lemma 5.8. *We have the following asymptotic for the summation of* $\ln(h_j)$ *from* 0 *to* $m_n - 1$*,*

$$\sum_{j=0}^{m_n-1} \ln(h_j) = -m_n n q(0) - \frac{m_n(m_n+1)}{2}\left(\ln(n) + \ln\left(\frac{1}{2}q''(0)\right)\right) + \frac{m_n^2 \ln(m_n)}{2} - \frac{3}{4}m_n^2 + \frac{\ln(2\pi)m_n}{2} - \frac{\ln(m_n)}{12} + \zeta'(-1) + O(m_n^{-2} + n^{-\frac{1}{2}(1-5\epsilon)}(\ln(n))^3).$$

where ϵ *is as in the definition for* m_n*.*

Proof. The first term in the right is straightforwardly determined, we need only to write, using Theorem 5.6,

$$\begin{aligned}\sum_{j=0}^{m_n-1} \ln(h_j) &= -m_n n q(0) - \frac{m_n(m_n+1)}{2}\left(\ln(n) + \ln\left(\frac{1}{2}q^{(2)}(0)\right)\right) \\ &\qquad + \ln(G(m_n+1)) + O(n^{-\frac{1}{2}(1-5\epsilon)}(\ln(n))^3), \\ &= -m_n n q(0) - \frac{m_n(m_n+1)}{2}\left(\ln(n) + \ln\left(\frac{1}{2}q^{(2)}(0)\right)\right) + \frac{m_n^2 \ln(m_n)}{2} - \frac{3}{4}m_n^2 \\ &\qquad + \frac{\ln(2\pi)m_n}{2} - \frac{\ln(m_n)}{12} + \zeta'(-1) + O(m_n^{-2} + n^{-\frac{1}{2}(1-5\epsilon)}(\ln(n))^3)\end{aligned}$$

where G is the Barnes G-function defined recursively as

$$G(z+1) = \Gamma(z)G(z),$$

and we have used the asymptotic for the G-Barnes function

$$\ln(G(z+1)) = \frac{z^2\ln(z)}{2} - \frac{3}{4}z^2 + \frac{\ln(2\pi)z}{2} - \frac{\ln(z)}{12} + \zeta'(-1) + O\left(\frac{1}{z^2}\right).$$

□

Lemma 5.9. *We have the following asymptotic for the summation of* $\ln(h_j)$ *from* m_n *to* $n-1$,

$$\sum_{j=m_n}^{n-1} \ln(h_j) = -n^2 I[\mu_Q] + \frac{(n-m_n)}{2}\ln\frac{2\pi}{n} + nm_n q(0) + \frac{3}{4}m_n^2 - \frac{n}{2}E_Q[\mu_Q]$$
$$-\frac{m_n}{2}(m_n+1)\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2}\left(m_n^2 - \frac{1}{6}\right)\ln\left(\frac{m_n}{n}\right) + F_V[D_{r_1}] + O(n^{-12}(\ln n)^3),$$

where $I_Q[\mu_Q]$, $F_V[D_{r_1}]$ *and* $E_Q[\mu_Q]$ *are as in Theorem 5.4.*

The second summation is a bit more detailed, we use the previously defined asymptotics in Theorem 5.5,

$$\ln(h_j) = -nV_{\tau(j)}(r_{\tau(j)}) + \frac{1}{2}\ln\left(\frac{8\pi r_{\tau(j)}^2}{nV_{\tau(j)}^{(2)}(r_{\tau(j)})}\right) + \frac{1}{n}\mathfrak{B}(r_{\tau(j)}) + O(j^{-\frac{3}{2}}(\ln(n))^{\alpha})$$

and need to determine the asymptotic of three terms in order to get the full summation,

$$-n\sum_{j=m_n}^{n-1} V_{\tau(j)}(r_{\tau(j)}), \quad \sum_{j=m_n}^{n-1}\ln\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right) \quad \text{and} \quad \sum_{j=m_n}^{n-1}\ln(r_{\tau(j)}).$$

This is done in Proposition 5.3, Proposition 5.4 and Proposition 5.5.

Proposition 5.3. *The following asymptotic holds under the same context as Lemma 5.9.*

$$-n\sum_{j=m_n}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = -n^2 I_Q[\mu_Q] + nU_{\mu_Q}(0) + \frac{1}{6}\ln(r_1) + nm_n q(0) + \frac{3}{4}m_n^2$$
$$-\frac{1}{2}\left(m_n^2 - m_n + \frac{1}{6}\right)\ln\left(\frac{4m_n}{nV_0^{(2)}(0)}\right) - \frac{m_n}{2} + O\left(n^{-\frac{1}{2}(3-5\epsilon)}\right).$$

Here, $I_Q[\mu_Q]$ *is as in Theorem 5.4,* ϵ *is as in the definition for* m_n, *and*

$$2U_{\mu_Q}(0) := q(r_1) - 2\ln(r_1) - q(0).$$

Proof. We proceed as before and use the Euler-Maclaurin formula to write

$$\sum_{j=m_n}^{n-1} V_{\tau(j)}(r_{\tau(j)}) = \int_{m_n}^{n} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\left(V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(m_n)}(r_{\tau(m_n)})\right) + \frac{1}{12}\left[\partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t V_{\tau(t)}(r_{\tau(t)})|_{t=m_n}\right] + O(n^{-3}).$$

Then, we are able to rewrite, setting $s = r_{\tau(j)}$,

$$\begin{aligned}\int_{m_n}^{n} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= \int_{m_n}^{n} (q(r_{\tau(t)}) - 2\tau(t)\ln(r_{\tau(t)}))\mathrm{d}t,\\ &= \frac{1}{2}\int_{r_{\tau(m_n)}}^{r_1} (q(s) - sq'(s)\ln(s))sV^{(2)}(s)n\mathrm{d}s,\\ &= n\left[\frac{1}{2}\int_{r_{\tau(m_n)}}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \frac{1}{2}\int_{r_{\tau(m_n)}}^{r_1} s^2q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s\right].\end{aligned}$$

Using that $r_1q'(r_1) = 2$, apply integration by parts on the second term and get

$$\begin{aligned}&\frac{1}{2}\int_{r_{\tau(m_n)}}^{r_1} s^2q'(s)\ln(s)V^{(2)}(s)\mathrm{d}s\\ &= \frac{1}{2}\int_{r_{\tau(m_n)}}^{r_1} s^2q'(s)\ln(s)\frac{1}{s}(sq'(s))'\mathrm{d}s,\\ &= \frac{1}{4}\int_{r_{\tau(m_n)}}^{r_1} ((sq'(s))^2)'\ln(s)\mathrm{d}s,\\ &= \frac{1}{4}(sq'(s))^2\ln(s)|_{r_{\tau(m_n)}}^{r_1} - \frac{1}{4}\int_{r_{\tau(m_n)}}^{r_1} (sq'(s))^2\frac{1}{s}\mathrm{d}s,\\ &= \ln(r_1) - \frac{1}{4}\int_{r_{\tau(m_n)}}^{r_1} sq'(s)^2\mathrm{d}s - \frac{m_n^2}{n^2}\ln(r_{\tau(m_n)}),\\ &= \ln(r_1) - \frac{1}{4}(sq'(s)q(s))|_{r_{\tau(m_n)}}^{r_1} + \frac{1}{4}\int_{r_{\tau(m_n)}}^{r_1} q(s)(sq'(s))'\mathrm{d}s - \frac{m_n^2}{n^2}\ln(r_{\tau(m_n)}),\\ &= \ln(r_1) - \frac{1}{2}q(r_1) + \frac{1}{4}\int_{r_{\tau(m_n)}}^{r_1} q(s)(sq'(s))'\mathrm{d}s - \frac{m_n^2}{n^2}\ln(r_{\tau(m_n)}) + \frac{m_n}{2n}q(r_{\tau(m_n)}).\end{aligned}$$

Leaving us with

$$\begin{aligned}&n\left[\frac{1}{4}\int_{r_{\tau(m_n)}}^{r_1} q(s)sV^{(2)}(s)\mathrm{d}s - \ln(r_1) + \frac{1}{2}q(r_1) - \frac{m_n^2}{n^2}\ln(r_{\tau(m_n)}) + \frac{m_n}{2n}q(r_{\tau(m_n)})\right]\\ &:= nI_Q[\mu_Q] - \frac{1}{4}\int_{r_0}^{r_{\tau(m_n)}} q(s)sV^{(2)}(s)\mathrm{d}s - \frac{m_n^2}{n}\ln(r_{\tau(m_n)}) + \frac{m_n}{2}q(r_{\tau(m_n)}).\end{aligned}$$

where we expand some terms using the following results given by Proposition 5.2

$$\ln(r_{\tau(m_n)}) = \frac{1}{2}\ln\left(\frac{4\tau(m_n)}{V_0^{(2)}(0)}\right) + O(\tau(m_n)^{\frac{1}{2}}) = \frac{1}{2}\ln\left(\frac{4\tau(m_n)}{V_0^{(2)}(0)}\right) + O(n^{-\frac{1}{2}(1-\epsilon)}), \tag{5.16}$$

$$q(r_{\tau(m_n)}) = q(0) + \frac{1}{2}q''(0)(r_{\tau(m_n)})^2 + O((\tau(m_n)^{\frac{3}{2}})) = q(0) + \tau(m_n) + O(n^{-\frac{3}{2}(1-\epsilon)}), \tag{5.17}$$

and, using integration by parts we solve the integral

$$\begin{aligned}
\frac{1}{4}\int_{r_0}^{r_{\tau(m_n)}} q(s)sV^{(2)}(s)\mathrm{d}s &= \frac{1}{4}\int_{r_0}^{r_{\tau(m_n)}} q(s)(sq'(s))'\mathrm{d}s,\\
&= \frac{1}{2}\tau(m_n)q(r_{\tau(m_n)}) - \frac{1}{4}\int_0^{r_{\tau(m_n)}} s(q'(s))^2\mathrm{d}s,\\
&= \frac{1}{2}\tau(m_n)q(r_{\tau(m_n)}) - \frac{1}{4}(q''(0))^2\int_0^{r_{\tau(m_n)}} s^3\mathrm{d}s + O(r_{\tau(m_n)}^5),\\
&= \frac{1}{2}\tau(m_n)q(r_{\tau(m_n)}) - \frac{1}{4}\tau(m_n)^2 + O(n^{-5/2(1-\epsilon)}).
\end{aligned}$$

Therefore, for this first term, gathering the above and expanding using Proposition 5.2 and (5.17),

$$\begin{aligned}
\int_{m_n}^{n} V_{\tau(t)}(r_{\tau(t)})\mathrm{d}t &= nI_Q[\mu_Q] - \frac{1}{2}\tau(m_n)q(r_{\tau(m_n)}) - \frac{1}{4}\tau(m_n)^2 + O(n^{-5/2(1-\epsilon)})\\
&\quad - \frac{m_n^2}{2n}\ln\left(\frac{4\tau(m_n)}{V_0^{(2)}(0)}\right) + O(n^{-\frac{3}{2}(1-\epsilon)}) + \frac{m_n}{2}\left(q(0)+\tau(m_n)\right) + O(n^{-\frac{1}{2}(1-\epsilon)}),\\
&= nI_Q[\mu_Q] - m_nq(0) - \frac{3}{4}\frac{m_n^2}{n} + \frac{m_n^2}{2n}\ln\left(\frac{4m_n}{nV_0^{(2)}(0)}\right) + O\left(n^{-\frac{1}{2}(3-5\epsilon)}\right).
\end{aligned}$$

For the other terms in the Euler-Maclaurin formula we have

$$\begin{aligned}
V_{\tau(n)}(r_{\tau(n)}) - V_{\tau(m_n)}(r_{\tau(m_n)}) &= q(r_1) - 2\ln(r_1) - q(r_{\tau(m_n)}) + 2\tau(m_n)\ln(r_{\tau(m_n)}),\\
&= q(r_1) - 2\ln(r_1) - q(0) - \frac{m_n}{n} + \frac{m_n}{n}\ln\left(\frac{4m_n}{nV_0^{(2)}(0)}\right) + O(n^{-3/2(1-\epsilon)}),\\
&:= 2U_{\mu_Q}(0) - \frac{m_n}{n} + \frac{m_n}{n}\ln\left(\frac{4m_n}{nV_0^{(2)}(0)}\right) + O(n^{-3/2(1-\epsilon)}),
\end{aligned}$$

and,

$$\begin{aligned}
\partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=n} - \partial_t(V_{\tau(t)}(r_{\tau(t)}))|_{t=m_n} &= \frac{2}{n}(\ln(r_{\tau(m_n)}) - \ln(r_1)),\\
&= \frac{1}{n}\ln\left(\frac{4m_n}{nV_0^{(2)}(0)}\right) - \frac{2}{n}\ln(r_1) + O(n^{-1/2(3-\epsilon)}).
\end{aligned}$$

Finally, gathering terms, we get the lemma. □

Proposition 5.4. *We have following asymptotic*

$$\sum_{j=m_n}^{n-1}\ln\left(\frac{V_{\tau(j)}^{(2)}(r_{\tau(j)})}{4}\right) = nE_Q(\mu_Q) - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(r_0)}\right) - m_n\ln\left(\frac{V_0^{(2)}(0)}{4}\right) + O(n^{-1/2(1-\epsilon)}),$$

where $E_Q[\mu_Q]$ is as in Theorem 5.4 and ϵ is as in the definition for m_n.

Proof. Again, we start with the Euler-Maclaurin formula

$$\sum_{j=m_n}^{n-1} \ln\left(\frac{V^{(2)}_{\tau(j)}(r_{\tau(j)})}{4}\right) = \int_{m_n}^{n} \ln\left(\frac{V^{(2)}_{\tau(t)}(r_{\tau(t)})}{4}\right)\mathrm{d}t - \frac{1}{2}\ln\left(\frac{V^{(2)}_{1}(r_1)}{V^{(2)}_{\tau(m_n)}(r_{\tau(m_n)})}\right)+$$
$$\frac{1}{12}\left[\partial_t \ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t \ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=m_n}\right] + O(n^{-3}).$$

We first tackle the integral on the right hand side. Changing variables $s = r_{\tau(t)}$ we have

$$\begin{aligned}
\int_{m_n}^{n} \ln\left(\frac{V^{(2)}_{\tau(t)}(r_{\tau(t)})}{4}\right)\mathrm{d}t &= \frac{n}{2}\int_{r_{\tau(m_n)}}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)(sq'(s))'\mathrm{d}s,\\
&= \frac{n}{2}\int_{r_{\tau(m_n)}}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)sV^{(2)}(s)\mathrm{d}s,\\
&= \frac{n}{2}\left[\int_{r_0}^{r_1} \ln\left(\frac{V^{(2)}(s)}{4}\right)sV^{(2)}(s)\mathrm{d}s - \int_{r_0}^{r_{\tau(m_n)}} \ln\left(\frac{V^{(2)}(s)}{4}\right)sV^{(2)}(s)\mathrm{d}s\right],\\
&= \frac{n}{2}\left[2E_Q[\mu_Q] - \int_{r_0}^{r_{\tau(m_n)}} \ln\left(\frac{V^{(2)}(s)}{4}\right)sV^{(2)}(s)\mathrm{d}s\right],
\end{aligned}$$

where one also compute

$$\begin{aligned}
&\int_{r_0}^{r_{\tau(m_n)}} \ln\left(\frac{V^{(2)}(s)}{4}\right)sV^{(2)}(s)\mathrm{d}s\\
&= \int_{r_0}^{r_{\tau(m_n)}} \ln\left(\frac{V^{(2)}_0(0)}{4} + \frac{V^{(2)}_0(0)}{4}s + O(s^2)\right)s\left(V^{(2)}_0(0) + V^{(2)}_0(0)s + O(s^2)\right)\mathrm{d}s,\\
&= \int_{r_0}^{r_{\tau(m_n)}} \ln\left(\frac{V^{(2)}_0(0)}{4}\right)sV^{(2)}_0(0)\mathrm{d}s + O(r^3_{\tau(m_n)}),\\
&= \ln\left(\frac{V^{(2)}_0(0)}{4}\right)r^2_{\tau(m_n)}V^{(2)}_0(0) + O(r^3_{\tau(m_n)}),\\
&= \frac{2m_n}{n}\ln\left(\frac{V^{(2)}_0(0)}{4}\right) + O(n^{-3(1-\epsilon)})
\end{aligned}$$

noting the definition for V and Proposition 5.2. Furthermore, we know

$$\begin{aligned}
\partial_t \ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)}) &= \partial_V\left(\ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\frac{1}{4}\partial_r\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)\partial_\tau\left(r_{\tau(t)}\right),\\
&= \frac{2\partial_r V^{(2)}(r_{\tau(t)})}{nr_{\tau(t)}\left(V^{(2)}_{\tau(t)}(r_{\tau(t)})\right)^2},\\
&= O\left(n/m_n\right)^{-1/2}.
\end{aligned}$$

Therefore, for the third term on the right hand side

$$\begin{aligned}
\partial_t \ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=n} - \partial_t \ln\frac{1}{4}V^{(2)}_{\tau(t)}(r_{\tau(t)})|_{t=m_n} &= \frac{2\partial_r V^{(2)}(r_1)}{nr_1\left(V^{(2)}_1(r_1)\right)^2} - O\left(n/m_n\right)^{-1/2},\\
&= O\left(n/m_n\right)^{-1/2} = O(n^{-1/2(1-\epsilon)}).
\end{aligned}$$

Then, simply expanding the series on $V^{(2)}_{\tau(m_n)}(r_{\tau(m_n)})$, we write,

$$\sum_{j=0}^{n-1} \ln V^{(2)}_{\tau(j)}(r_{\tau(j)}) = nE_Q(\mu_Q) - \frac{1}{2}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)}\right) - m_n \ln\left(\frac{V_0^{(2)}(0)}{4}\right) + O(n^{-1/2(1-\epsilon)}).$$

□

Proposition 5.5. *The following asymptotic holds under the same context as Lemma 5.9.*

$$\sum_{j=m_n}^{n-1} \ln(r_{\tau(j)}) = n\left[-U_{\mu_Q}(0) + \frac{1}{2}\tau(m_n)\right] - \frac{1}{2}\ln(r_1) - \frac{2m_n - 1}{4}\ln\left(\frac{4\tau(m_n)}{V_0^{(2)}(0)}\right) + O(n^{-\epsilon})$$

where $U_{\mu_Q}(0)$ is as in Proposition 5.3.

Proof. As usual, we start with the Euler-Maclaurin formula.

$$\sum_{j=m_n}^{n-1} \ln(r_{\tau(m_n)}) = \int_{m_n}^{n} \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_{\tau(m_n)}}\right) + \frac{1}{12}\left[\partial_t \ln r_{\tau(t)}|_{t=n} - \partial_t \ln r_{\tau(t)}|_{t=m_n}\right]$$

where we directly note that

$$\partial_t \ln r_{\tau(t)}|_{t=m_n} = \partial_r \ln r_{\tau(t)} \partial_\tau r_{\tau(t)} \partial_t \tau(t)|_{t=m_n} = \left.\frac{2}{r^2_{\tau(t)} V^{(2)}_{\tau(t)}(r_{\tau(t)})n}\right|_{t=m_n} = O(m_n^{-1}),$$

$$\partial_t \ln r_{\tau(t)}|_{t=n} = \left.\frac{2}{r^2_{\tau(t)} V^{(2)}_{\tau(t)}(r_{\tau(t)})n}\right|_{t=n} = O(n^{-1}).$$

Now, for the other terms we start with the first term, the integral, to get using integration by parts

$$\begin{aligned}
\int_{m_n}^{n} \ln(r_{\tau(t)})\mathrm{d}t &= \frac{n}{2}\int_{r_{\tau(m_n)}}^{r_1} \ln(s)(s'q(s))'\mathrm{d}s, \\
&= \frac{n}{2}\left[\ln(s)sq'(s)\Big|_{r_{\tau(m_n)}}^{r_1} - \int_{r_{\tau(m_n)}}^{r_1} q'(s)\mathrm{d}s\right], \\
&= \frac{n}{2}\left[\ln(r_1)r_1 q'(r_1) - \ln(r_{\tau(m_n)})r_{\tau(m_n)} q'(r_{\tau(m_n)}) - q(r_1) + q(r_{\tau(m_n)})\right], \\
&= \frac{n}{2}\left[2\ln(r_1) - 2\tau(m_n)\ln(r_{\tau(m_n)}) - q(r_1) + q(r_{\tau(m_n)})\right].
\end{aligned}$$

Ultimately, using (5.16) and (5.17),

$$\begin{aligned}
\int_{m_n}^{n} \ln(r_{\tau(t)})\mathrm{d}t - \frac{1}{2}\ln\left(\frac{r_1}{r_{\tau(m_n)}}\right) + O(m_n^{-1}) &= n\left[\ln(r_1) - \frac{q(r_1) - q(0)}{2} + \frac{1}{2}\tau(m_n)\right] \\
&\quad - \frac{1}{2}\ln(r_1) - \frac{2m_n - 1}{4}\ln\left(\frac{4\tau(m_n)}{V_0^{(2)}(0)}\right) + O(n^{-\epsilon}).
\end{aligned}$$

□

Finally, we turn our attention to gathering these propositions and proving the original lemma.

Proof. (Lemma 5.9) We start by stating from (5.13), with the new limits,

$$\frac{1}{n}\sum_{j=m_n}^{n-1}\mathfrak{B}(r_{\tau(j)}) = -\frac{3}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})}\right) - \frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_n)}V_{\tau(m_n)}^{(3)}(r_{\tau(m_n)})}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})}\right) - \frac{1}{12}\int_{r_{\tau(m_n)}}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r + O(n^{-2}).$$

We should have by the asymptotic for h_j in Theorem 5.5, referring to $j > m_n$

$$\sum_{j=m_n}^{n-1}\ln(h_j) = \sum_{j=m_n}^{n-1}\left(-nV_{\tau(j)}(r_{\tau(j)}) + \ln(r_{\tau(j)}) - \frac{1}{2}\ln\frac{1}{4}V_{\tau(j)}^{(2)}(r_{\tau(j)}) + \frac{1}{n}\mathfrak{B}(r_{\tau(j)})\right) + \frac{(n-m_n)}{2}\ln\frac{2\pi}{n} + O(m_n^{-1/2}(\ln n)^{\alpha}) + O(n^{\frac{1}{2}(1-5\epsilon)}).$$

By the lemmas just proved, we write, as $n \to \infty$,

$$\begin{aligned}\sum_{j=m_n}^{n-1}\ln(h_j) = &-n^2I[\mu_Q] + \frac{(n-m_n)}{2}\ln\frac{2\pi}{n} + nm_nq(0) + \frac{3}{4}m_n^2 - \frac{n}{2}E_Q[\mu_Q]\\ &-\frac{m_n}{2}(m_n+1)\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2}\left(m_n^2 - \frac{1}{6}\right)\ln\left(\frac{m_n}{n}\right) + \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1)\\ &+\frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})}\right) - \frac{1}{32}\left(\frac{r_1V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{r_{\tau(m_n)}V_{\tau(m_n)}^{(3)}(r_{\tau(m_n)})}{V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)})}\right)\\ &-\frac{1}{12}\int_{r_{\tau(m_n)}}^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2}r\mathrm{d}r + O(n^{-12}(\ln n)^3).\end{aligned}$$

Using Proposition 5.2 we write that

$$r_{\tau(m_n)} = r_0 + O(\tau(m_n)^{\frac{1}{2}}) = O(n^{\frac{1}{2}(\epsilon-1)})$$

and

$$V_{\tau(m_n)}^{(2)}(r_{\tau(m_n)}) = V_0^{(2)}(r_0) + O(n^{(\epsilon-1)}).$$

Then, we rewrite

$$\begin{aligned}\sum_{j=m_n}^{n-1}\ln(h_j) = &-n^2I[\mu_Q] + \frac{(n-m_n)}{2}\ln\frac{2\pi}{n} + nm_nq(0) + \frac{3}{4}m_n^2 - \frac{n}{2}E_Q[\mu_Q]\\ &-\frac{m_n}{2}(m_n+1)\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{2}\left(m_n^2 - \frac{1}{6}\right)\ln\left(\frac{m_n}{n}\right) + F_V[D_{r_1}] + O(n^{-12}(\ln n)^3).\end{aligned}$$

where

$$F_V[D_{r_1}] := \frac{1}{6}\ln\left(\frac{4}{V_0^{(2)}(0)}\right) - \frac{1}{3}\ln(r_1) + \frac{5}{32}\ln\left(\frac{V_1^{(2)}(r_1)}{V_0^{(2)}(0)}\right) - \frac{1}{32}\frac{r_1 V_1^{(3)}(r_1)}{V_1^{(2)}(r_1)} - \frac{1}{12}\int_0^{r_1}\frac{V^{(3)}(r)^2}{V^{(2)}(r)^2} r\mathrm{d}r.$$

□

And here finally we return to the full summation.

Proof. (Theorem 5.7) Gathering Lemma 5.8 and Lemma 5.9, we get

$$\sum_{j=0}^{n-1}\ln(h_j) = -n^2 I[\mu_Q] - \frac{1}{2}n\ln(n) - \frac{n}{2}\left(E_Q[\mu_Q] - \ln(2\pi)\right) - \frac{\ln(n)}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-12}(\ln n)^3).$$

Incredibly, all the terms depending on m_n cancels out. □

We finally prove Theorem 5.4.

Proof. (Theorem 5.4) We restate the previous result

$$\sum_{j=0}^{n-1}\ln(h_j) = -n^2 I_Q[\mu_Q] - \frac{1}{2}n\ln n - \frac{n}{2}\left(E_Q[\mu_Q] - \ln(2\pi)\right) - \frac{\ln n}{12} + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-1/12}(\ln n)^3).$$

We get the asymptotic expansion for the Free energy noting that all terms involving m_n vanish and using that

$$\ln(n!) = n\ln(n) - n + \frac{1}{2}\ln(n) + \frac{1}{2}\ln(2\pi) + O(n^{-1})$$

that is,

$$\begin{aligned}\ln \mathcal{Z}_n^Q &= \ln n! + \sum_{j=0}^{n-1}\ln h_j \\ &= -n^2 I_Q[\mu_Q] + \frac{1}{2}n\ln n + n\left(\frac{\ln(2\pi)}{2} - 1 - \frac{1}{2}E_Q[\mu_Q]\right) - \frac{5}{12}\ln n\frac{\ln(2\pi)}{2} \\ &\quad + \zeta'(-1) + F_V[D_{r_1}] + O(n^{-1/12}(\ln n)^3).\end{aligned}$$

□

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